

10 Years' UPSC PYQ Trend Analysis **GS Prelims (2015-2024)** **GEOGRAPHY**



AHMEDABAD



BENGALURU



BHOPAL



CHANDIGARH



DELHI



GUWAHATI



HYDERABAD



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Table of Content



UPSC Syllabus	4
Topic Wise no. of Questions	4
Question Source Type Analysis.	5
Important Observations Based On 10 Year's Prelims Questions.	6
Sources To Be Referred	6
Agriculture	7
Climatology	15
Earth & Solar System	33
Geomorphology.	36
Indian Drainage System	44
Indian Economic Geography (Minerals).	61
Indian Physiography.	72
Map Base Questions	82
Oceanography	94
Soil	100
Vegetation	103
Water Resources	108
World Drainage System.	109
World Economic Geography.	113
Biogeography	117
Annexure	118

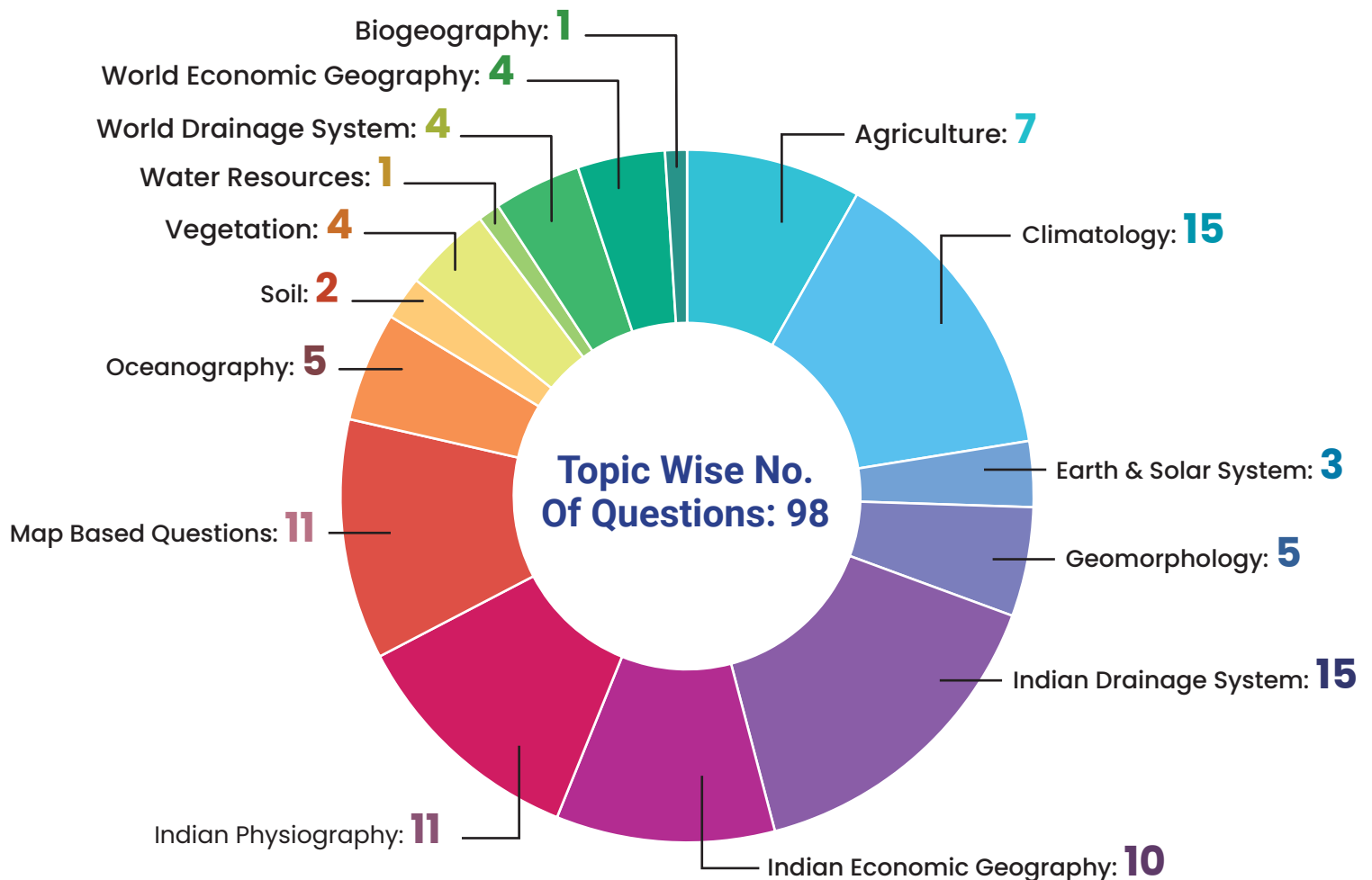




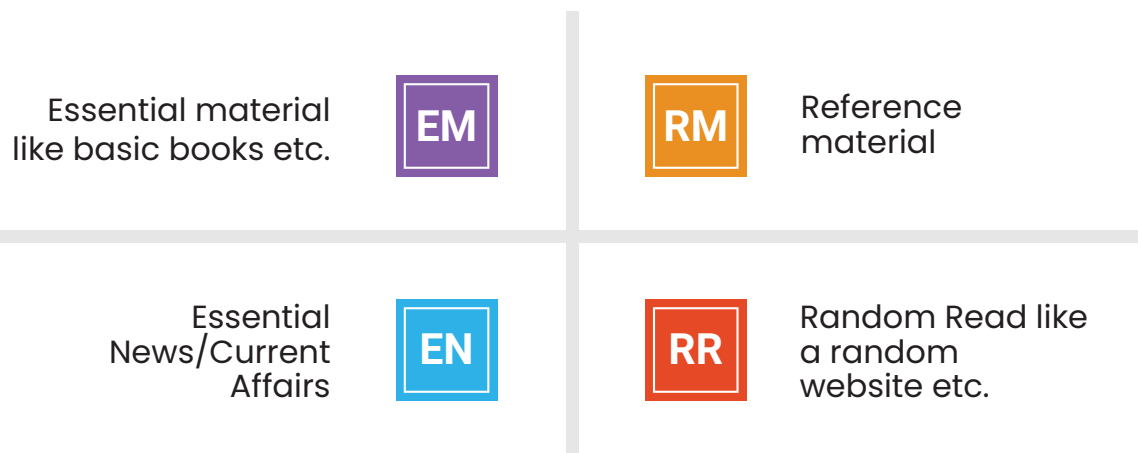
UPSC Prelims Geography Syllabus:

Indian and World Geography- Physical, Social, Economic Geography of India and the World.

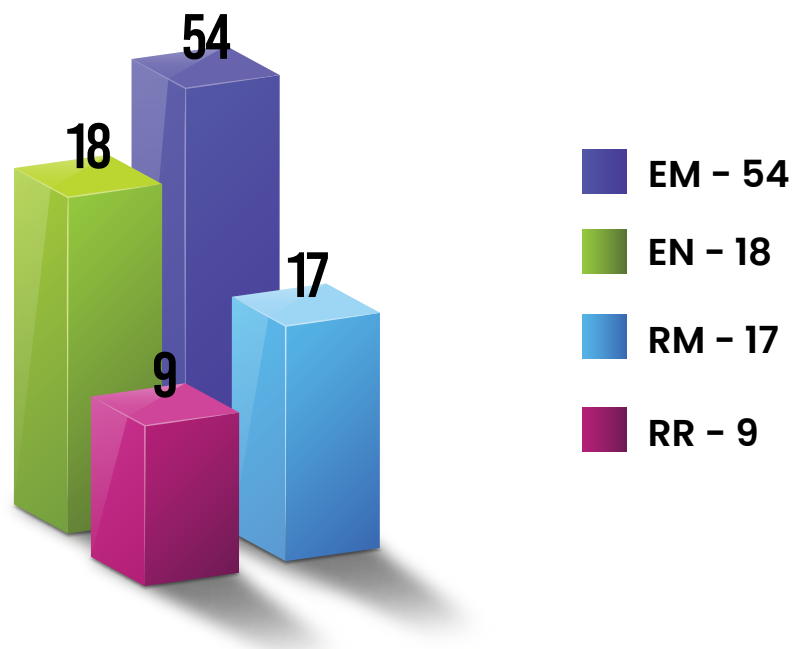
Total No. Of Geography Questions: 98



Source Type:



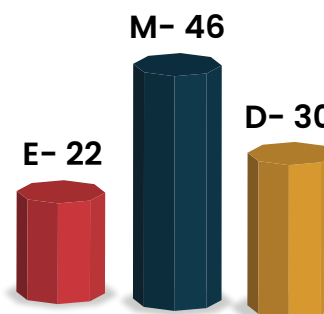
Questions Source Type Analysis:



Difficulty Level:

E: Easy
M: Medium
D: Difficult

Difficulty Level Analysis



Nature of Question:

F: Fundamental, Conventional and conceptual question which is easily available in commonly recommended books. If a current affairs source is mentioned then it indicates that you had one more reason to prepare this.

FA: Fundamental Applied question is an analytical question which requires information + application of mind. Answers to these questions are from commonly recommended books but not so obvious to find.

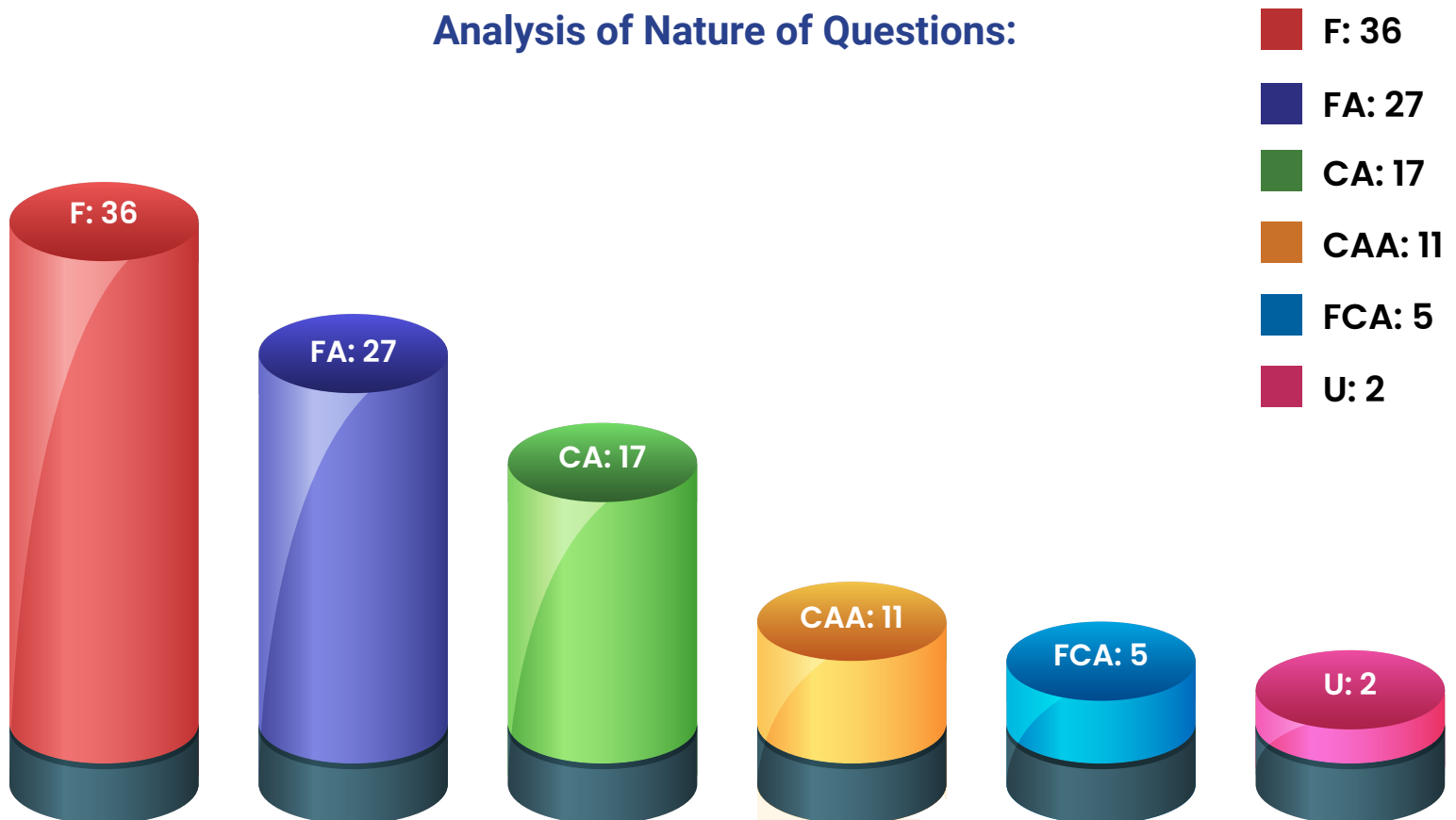
CA: Current Affair question which can be answered almost completely using a given source alone.

CAA: Current Affair Applied question needs information from more than one source to answer AND/OR application of mind (common sense and overall understanding) of aspirant. This is why sources for these questions may not be answering the question completely.

FCA: Fundamental + Current affair. This is a F or FA question supplemented with current affairs or a current affairs question that needs background information for complete answer. Source to these questions might not answer the questions in entirety.

U: Unconventional Question means the question that is distantly related to the syllabus provided by UPSC. It is neither present in even reference sources (in addition to recommended books) nor prominently in news.

Analysis of Nature of Questions:



Important Observations Based On 10 Year's Prelims Questions.

- Mostly questions in this section have come from static and standard sources which makes the difficulty level overall to moderate level.
- Very few questions have come from unconventional sources.
- There is a link in some questions from newspaper articles and current affairs.

Sources To Be Referred:

Essential Material:

- NCERTs (Class VI – Class XII)
- Certificate Physical and Human Geography by G.C. Leong
- Atlas

Reference Material:

- News Paper
- Annual Reports of Ministry of Agriculture
- Economic Survey

Agriculture

Total No. of questions: 07

Kharif Crop	Pulses	Cotton	Sugarcane	Tea	Agricultural Land	Irrigation Method
01	01	01	01	01	01	01

2019

Kharif crops

With reference to the cultivation of Kharif crops in India in the last five years, consider the following statements:

1. Area under rice cultivation is the highest.
2. Area under the cultivation of jowar is more than that of oilseeds.
3. Area of cotton cultivation is more than that of sugarcane.
4. Area under sugarcane cultivation has steadily decreased.

Which of the statements given above is/are correct?

- (a) 1 and 3 only
- (b) 2, 3 and 4 only
- (c) 2 and 4 only
- (d) 1, 2, 3 and 4

Answer: A



Explanation:

► About area of cotton cultivation

Crops	Yield (kg/hectare)		
	2014-15	2015-16	2016-17*
Rice	2391	2400	2550
Coarse cereals	1703	1579	1784
Oilseeds	1075	968	1225
Sugarcane	71512	70720	69886

Crops	Production (Million Tonnes)		
	2014-15	2015-16	2016-17*
Rice	441.10	104.41	110.15
Coarse cereals	42.86	38.52	44.19
Oilseeds	27.51	25.25	32.10
Sugarcane	362.33	348.45	306.72

Crops	Area (lakh hectare)		
	2014-15	2015-16	2016-17*
Rice	441.10	434.99	431.94
Coarse cereals	251.70	243.89	247.71
Oilseeds	255.96	260.87	262.06
Sugarcane	50.66	49.27	43.89

► Hence option (a) is the correct answer.

Source

➤ Annual report of Ministry of Agriculture and Farmers Welfare



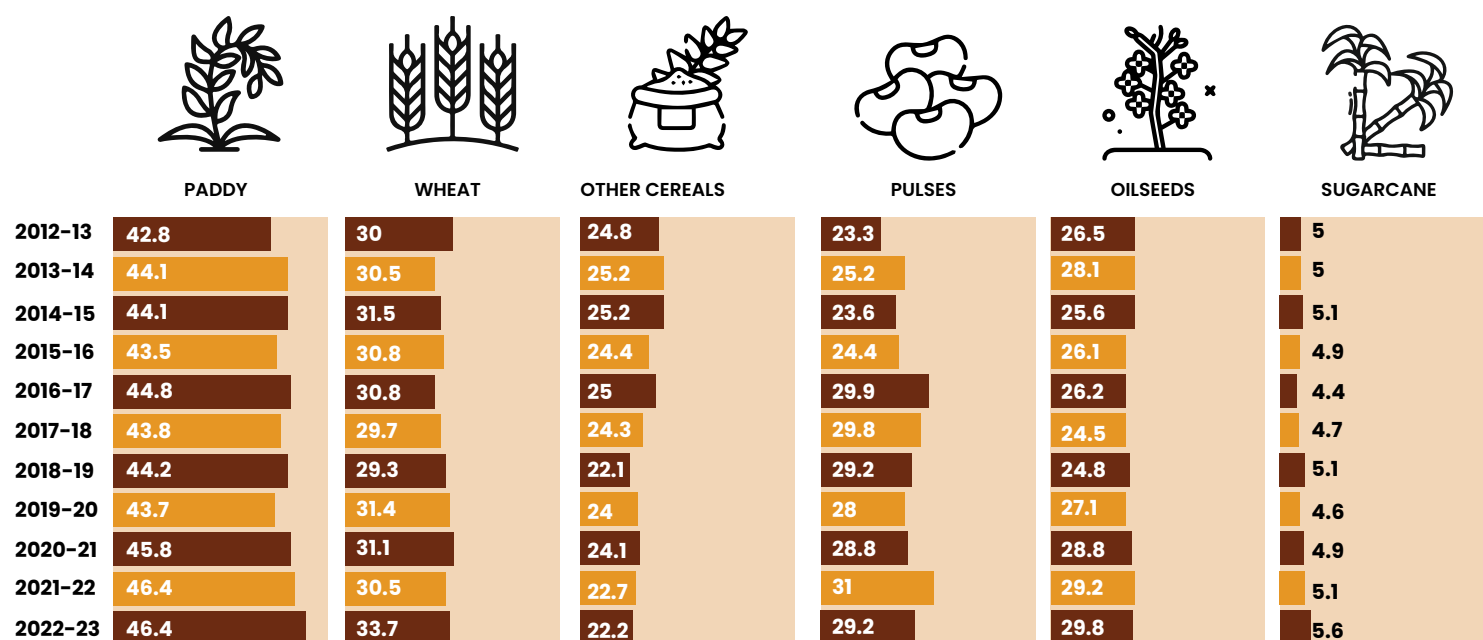
Motivation: Crops and their acreage is often discussed in the news.

Elimination Technique

Question can be solved using the following smart guesses and logic.

- Paddy production is highest (approximately 1.5 times of the wheat production) among all crops produced in India, using this logic, either option (a) or (d) has to be correct.
- Also, India is the fourth largest oilseeds producer in the world, whereas jowar is a semi-arid region crop. Using this logic, we can eliminate option 2 as well.

Area under major crops in India (million hectares)



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2020

Pulses Cultivation

With reference to pulse production in India, consider the following statements:

1. Black gram can be cultivated as both kharif and rabi crops.
2. Green-gram alone accounts for nearly half of pulse production.
3. In the last three decades, while the production of kharif pulses has increased the production of rabi pulses has decreased.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 and 3 only
- (c) 2 only
- (d) 1, 2 and 3

Answer: A



Explanation:

- **Statement 1 is correct:** Many crops are cultivated in both kharif and rabi seasons. Though pulses are grown in both Kharif and Rabi seasons, Rabi pulses contribute more than 60 percent of the total production. Black gram can be cultivated as both kharif and rabi crops. It is consumed in a variety of ways across the North to South in preparation of different regular and popular dishes like vada, idli, dosa etc. During kharif, it is cultivated throughout the country. It is best suited to rice fallows during rabi in southern and south-eastern parts of India.
- **Statement 2 is not correct:** Bengal Gram (Desi ChickPea / Desi Chana), Pigeon Peas (Arhar / Toor / Red Gram), Green Beans (Moong Beans), ChickPeas (Kabuli Chana), Black Matpe (Urad / Mah / Black Gram), Red Kidney Beans (Rajma), Black Eyed Peas (Lobiya), Lentils (Masoor), White Peas (Matar) are major pulses grown and consumed in India. Gram (Chickpeas) is the most dominant pulse having a share of around 40 per cent in the total production followed by Tur/Arhar at 15 to 20 percent and Urad/Black Matpe and Moong at around 8-10 per cent each.
- **Statement 3 is not correct:** The country's pulses production stood at 234 lakh tonnes in the 2018-19 crop year, down from the record level of 254 lakh tonnes in 2017-18. The production of pulses for both Rabi and Kharif seasons has increased in the last three decades. Production of Kharif pulses in 2000-01 was 44.48 lakh tonnes which increased to 55.30 lakh tonnes by 2016-17. The production of Rabi pulses has also increased from 66.27 lakh tonnes in 2000-01 to 108.18 lakh tonnes in 2015-16.

Source

- Annual Reports of Ministry of Agriculture and Farmers
- Economic Survey
- Geography NCERT Class X and XII



Motivation: Pulses production is frequently discussed in the news.

2020

Cotton Cultivation

"The crop is subtropical in nature. A hard frost is injurious to it. It requires at least 210 frost-free days and 50 to 100 centimetres of rainfall for its growth. A light well-drained soil capable of retaining moisture is ideally suited for the cultivation of the crop." Which one of the following is that crop?

- (a) Cotton
- (b) Jute
- (c) Sugarcane
- (d) Tea

Answer: A



Explanation:

- **Option (a) is the correct Answer:** Cotton is grown on a variety of soils across the world. Deep, fertile soil with adequate humus and high water holding capacity and good internal drainage is best suited for growing cotton. Cotton can be grown in places wherever, at least 180-200 frost free days are available. Optimum temperature for germination 20-30°C. Germination will be delayed if the temperature is <18°C.
- **The climatic conditions in the cotton growing regions of India show considerable variations. High temperature of about 45 degree celsius during sowing and seedling emergence and low temperature accompanied by occasional frost coinciding with the picking period and moderate rainfall ranging from 300-700 mm are the features of the north zone. In the Southern and Central zones, the climate is equal.**

Source

- G.C. Leong and Geography NCERTs



Motivation: Fundamental questions on Indian agriculture are often covered by the UPSC.

Elimination Technique:

Jute and Sugarcane are water intensive (require water more than 100cm). Hence option (b) and (c) can be eliminated.

2021

Sugarcane Cultivation

Among the following, which one is the least water-efficient crop?

- (a) Sugarcane
- (b) Sunflower
- (c) Pearl millet
- (d) Red gram

Answer: A

Source

➤ News Paper



Motivation: Sugarcane cultivation was in the news due to drought in Marathwada region.



Explanation:

- It takes about 210 litres of water to produce 1 kg of sugar cane. Water footprint of sunflowers is 7–9 litres per kg. Millets are sown in dry areas with minimal irrigation. Red gram requires 35 to 40 cm water, during its entire growth period. Optimum moisture is required during budding, flowering, and pod formation stages. As Red gram is a rain-fed crop grown in assured rainfall areas, usually it does not need any irrigation. **Hence option (a) is the correct Answer.**

2022

Tea Cultivation

Consider the following States:

1. Andhra Pradesh
2. Kerala
3. Himachal Pradesh
4. Tripura

How many of the above are generally known as tea-producing States?

- (a) Only one State
- (b) Only two States
- (c) Only three States
- (d) All four States

Answer: D

Source

- › Part of the answer is given in Geography NCERT of Class X, Chapter-4

RR Source

- › Official website of Tea Board of India



Explanation:

- › Today the major tea growing states are Assam, West Bengal, Tripura, Tamil Nadu, Kerala and Karnataka. Himachal Pradesh and Uttarakhand are also traditional tea growing states, albeit to a less significant extent.
- › In Himachal Pradesh, tea is grown in the Mandi and Kangra districts over an area of 2,063 hectares. Kangra, known as "the valley of gods," is famous for its distinct flavoured tea. Tripura is categorised as a traditional tea growing State with about 54 Tea Estates, 21 Tea processing factories, and more than 2500 small tea growers.
- › The Araku Valley in Visakhapatnam district of Andhra Pradesh is known for its tea plantations. Hence option (d) is the correct answer.

2023

Comparison of India and China in agriculture sector

Consider the following statements:

1. India has more arable areas than China.
2. The proportion of irrigated area is more in India as compared to China.
3. The average productivity per hectare in Indian agriculture is higher than that in China.

How many of the above statements are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

Answer: A



Explanation:

- › India and China are the most populous countries in the world. However, with limited arable land about 120 million hectares in China, and 156 million hectares in India, both face the challenge of producing enough food, fodder, and fiber for their population. Both have adopted modern technologies in agriculture, starting with High Yielding Variety (HYV) seeds in the mid-1960s, increased irrigation cover, and use of more chemical fertilizers to get more food from this limited land. Hence statement 1 is correct.

- Worldwide, irrigated agriculture accounts for about four-fifths of global water withdrawals. The share of irrigated land ranges widely, from 4 percent of the total area cropped in Africa to 42 percent in South Asia. The leading countries are India and China with about 30 percent and 52 percent of all croplands irrigated, respectively. **Hence statement 2 is not correct.**
- Better seeds that result from higher R&D expenditures generally demand more fertilizer. It is worth noting that China's fertilizer consumption in 2016 was 503 kg/ha of arable area compared to just 166 kg/ha for India, as per World Bank estimates. No wonder, China's productivity in most crops is 50 to 100% higher than India's.
- Agricultural yield is the quantity of a crop produced on one unit of land. In India agricultural yield of food grains has increased by more than four times since 1950-51, and was 3479 kg/hectare in 2021. Whereas the average yield of food grains was 6321 kg/hectare. **Hence statement 3 is not correct.**

Source

- Newspapers and World Bank data



Motivation: This question was inspired due to various studies done on low agricultural productivity in India.

Note:

Information regarding various crops and the conditions required for their growth is mentioned in Annexure A of this document.

2024

Irrigation Method

Which of the following is/are the advantage/advantages of practising drip irrigation?

1. Reduction in weed
2. Reduction in soil salinity
3. Reduction in soil erosion

Select the correct answer using the code given below.

- (a) 1 and 2 only
- (b) 3 only
- (c) 1 and 3 only
- (d) None of the above is an advantage of practising drip irrigation

Answer: C



Explanation:

- Advantages of Drip Irrigation:
 - Maximum use of available water.
 - No water being available to weeds.
 - Maximum crop yield.
 - High efficiency in the use of fertilizers.
 - **Less weed growth** and restricts population of potential hosts.
 - Low labour and relatively low operation cost.
 - **No soil erosion.**
 - Improved infiltration in soil of low intake.
 - Ready adjustment to sophisticated automatic control.
 - No runoff of fertilizers into ground water.
 - Less evaporation losses of water as compared to surface irrigation.
 - Improves seed germination.
 - Decreased tillage operations.
- Disadvantages of Drip Irrigation:
 - In spite of the fact that drip irrigation has so many potential benefits, there are a certain limitation also, there are as follow:
 - Sensitivity to clogging
 - Moisture distribution problem
 - Salinity hazards
 - High cost compared to furrow.
 - High skill is required for design, install and operation.

Source

- <http://www.agriinfo.in/default.aspx?page=topic&superid=8&topicid=2243>



Motivation: Drip irrigation is often discussed in the news.

Climatology

Total No. of questions: 15

2015

“Each day is more or less the same, the morning is clear and bright with a sea breeze; as the Sun climbs high in the sky, heat mounts up, dark clouds form, then rain comes with thunder and lightning. But rain is soon over”. Which of the following regions is described in the above passage?

- (a) Savannah
- (b) Equatorial
- (c) Monsoon
- (d) Mediterranean

Answer: B



Explanation:

- Wet equatorial climate, is characterized by consistently high temperatures (around 30 °C [86 °F]), with plentiful precipitation (150–1,000 cm [59–394 inches]) on most afternoons., heavy cloud cover, and high humidity, with very little annual temperature variation. Wet equatorial regions lie within about 12° latitude of the Equator.

Source

- <https://www.britannica.com/science/wet-equatorial-climate#:~:text=wet%20equatorial%20climate%2C%20major%20climate,very%20little%20annual%20temperature%20variation.>



Motivation: Basic concepts on climatology are often covered by UPSC.

2015

In the South Atlantic and South-Eastern Pacific regions in tropical latitudes, cyclone does not originate. What is the reason?

- (a) Sea surface temperatures are low
- (b) Inter-Tropical Convergence Zone seldom occurs
- (c) Coriolis force is too weak
- (d) Absence of land in those regions

Answer: B



Explanation:

- ITCZ stays at or near the equator and does not shift southward over the south Atlantic or South Pacific region. Therefore, these regions do not have the ITCZ over them which is a major cause of the lack of cyclones. Lower sea temperatures are also a reason.
- Tropical cyclones originate and intensify over warm tropical oceans. The conditions favourable for the formation and intensification of tropical storms are:
 - (i) Large sea surface with a temperature higher than 27°C ;
 - (ii) Presence of the Coriolis force;
 - (iii) Small variations in the vertical wind speed;
 - (iv) A pre-existing weak- low-pressure area or low-level-cyclonic circulation;
 - (v) Upper divergence above the sea level system.
- The most proximate reasons for the lack of activity in the South Atlantic are sea surface temperatures that tend to run a shade cooler than ideal for tropical cyclone formation even in the southern summer, climatologically high values of vertical wind shear across that basin throughout the year, and a lack of pre-existing centres of rotation (vorticity) in that area.

Source

- NCERT Fundamental Geography Ch 10 Page 94



Motivation: Cyclone and jet streams are often seen in news.

2015

Consider the following statements:

1. The winds which blow between 30°N and 60°S latitude throughout the year are known as westerlies.
2. The moist air masses that cause winter rains in North-Western region of India are part of westerlies.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Answer: B



Explanation:

- The westerly winds, also known as the westerlies, occur at two regions on Earth: between 30 and 60 degrees latitude in the northern hemisphere and between 30 and 60 degrees latitude in the southern hemisphere. Western Disturbance occurs in India, Pakistan, Bangladesh and Nepal to describe an extratropical storm originating in the Mediterranean, that brings sudden winter rain and snow to the northwestern parts of the Indian subcontinent. This is a non-monsoonal precipitation pattern driven by the Westerlies.

Source

- NCERT Class 11th India Physical Environment Page 35-36



Motivation: Western disturbance and Westerlies are often seen in news.

2017

Indian Ocean Dipole

With reference to 'Indian Ocean Dipole (IOD)' sometimes mentioned in the news while forecasting Indian monsoon, which of the following statements is/are correct?

1. IOD phenomenon is characterized by a difference in sea surface temperature between tropical Western Indian Ocean and tropical Eastern Pacific Ocean.
2. An IOD phenomenon can influence an El Nino's impact on the monsoon.

Select the correct Answer using the code given below:

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Answer: B

Source

➤ Standard newspapers



Motivation: IOD and El Nino are also seen in the news frequently.

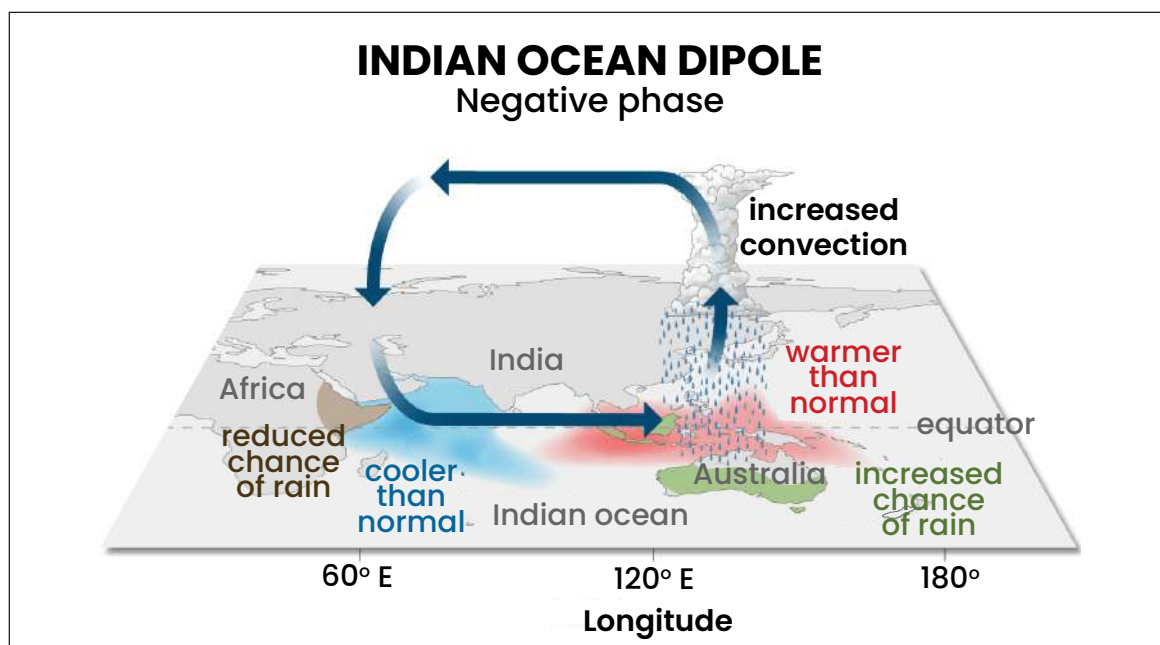
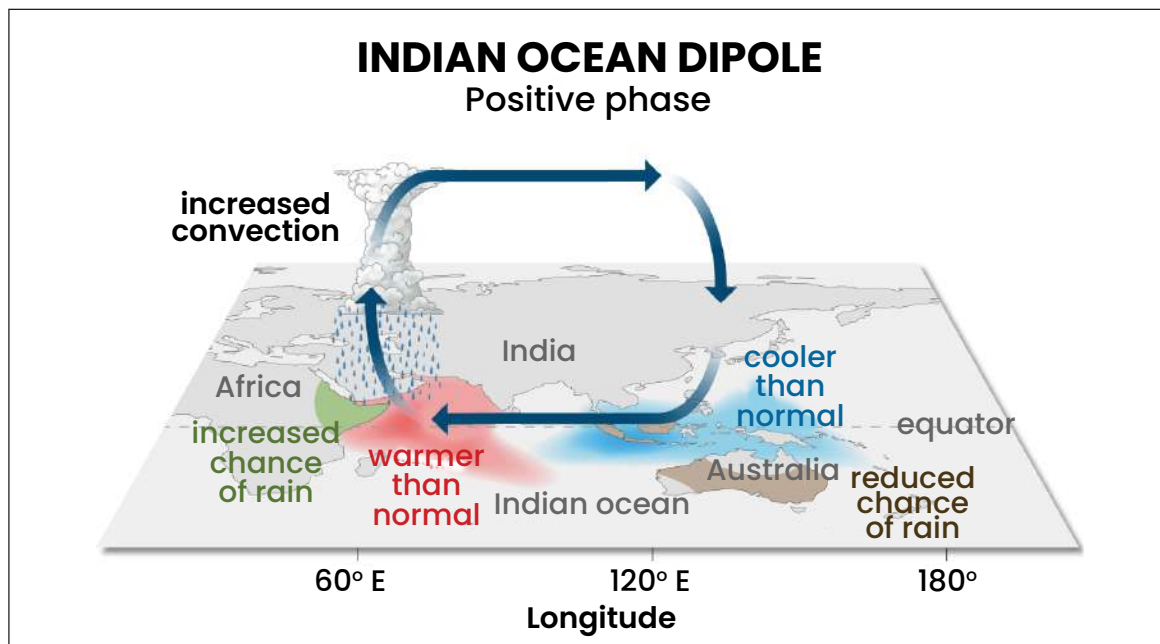


Explanation:

- The Indian Ocean Dipole is an irregular oscillation of sea-surface temperatures in which the western Indian Ocean becomes alternately warmer and then colder than the eastern part of the ocean. In scientific terms, the IOD is a coupled ocean and atmosphere phenomenon, similar to ENSO but in the equatorial Indian Ocean. **Hence statement 1 is not correct.**
- The IOD affects the climate of Australia and other countries that surround the Indian Ocean Basin, and is a significant contributor to rainfall variability in this region.
 - An IOD can either aggravate or weaken the impact of El Nino on Indian monsoon. If there is a positive IOD, it can bring good rains to India despite an El Nino year. **Hence statement 2 is correct.**
- **Positive event linked to it:**
 - Warmer sea surface temperatures in the western Indian Ocean relative to the east
 - Easterly wind anomalies across the Indian Ocean and less cloudiness to Australia's northwest
 - Less rainfall over southern Australia and the Top End.

➤ **Negative event linked to it:**

- Cooler sea surface temperatures in the western Indian Ocean relative to the east
- Winds become more westerly, bringing increased cloudiness to Australia's northwest
- More rainfall in the Top End and southern Australia



Note:

Other similar phenomena to be prepared on these lines are Pacific Decadal Oscillation, Atlantic Meridional Overturning Circulation, Arctic Oscillation, Antarctic Oscillation, Madden-Julian Oscillation (MJO), North Atlantic Oscillation (NAO) etc.

More information on El Nino, La Nina and formation of tropical cyclones is mentioned in Annexure B of this document.

2019

Clouds and Dew Formation

Why are dewdrops not formed on a cloudy night?

- (a) Clouds absorb the radiation released from the Earth's surface.
- (b) Clouds reflect back the Earth's radiation.
- (c) The Earth's surface would have low temperatures on cloudy nights.
- (d) Clouds deflect the blowing wind to ground level.

Answer: B



Explanation:

- When the moisture is deposited in the form of water droplets on cooler surfaces of solid objects (rather than nuclei in air above the surface) such as stones, grass blades and plant leaves, it is known as dew.
- The ideal conditions for its formation are clear sky, calm air, high relative humidity, and cold and long nights.
- For the formation of dew, it is necessary that the dew point is above the freezing point.
- Dew is not formed on a cloudy night:
 - on cloudy nights because cloudy nights are warmer than clear nights
 - in shady places because air cannot move there freely
 - on windy nights because wind carries away the moist air
- The atmosphere, heated by the absorption of Earth radiation, in turn **radiates heat back to the Earth's surface increasing the Earth's surface temperature.**
- **On cloudy night clouds reflect the heat back to the ground hence ground never gets cold enough for the dew to be formed.**

Source

- NCERT, Class XI Fundamentals of Geography



Motivation: Basic concepts on climatology are often covered by UPSC.

Additional information

- Condensation can take place when the dew point is
 - lower than the freezing point,
 - higher than the freezing point.
- While frost, snow and some clouds (cirrus clouds) are produced when the temperature is lower than the freezing point.
- Dew, fog and clouds result even when the temperature is higher than the freezing point.

Consider the following statements:

1. Jet streams occur in the Northern Hemisphere only.
2. Only some cyclones develop an eye.
3. The temperature inside the eye of a cyclone is nearly 10°C lesser than that of the surroundings.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 and 3 only
- (c) 2 only
- (d) 1 and 3 only

Answer: C



Explanation:

➤ **Statement 1 is not correct:**

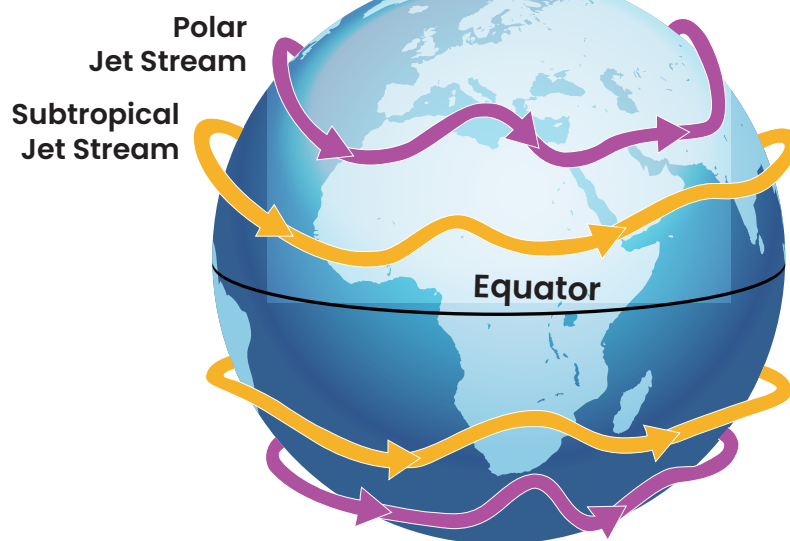
Jet streams occur in both the Northern and Southern Hemispheres. The actual appearance of jet streams result from the complex interaction between many variables – such as the location of high and low pressure systems, warm and cold air, and seasonal changes.

➤ **Statement 2 is correct:**

The hurricane's centre is a relatively calm, generally clear area of sinking air and light winds that usually do not exceed 15

mph (24 km/h) and is typically 20–40 miles (32–64 km) across. An eye will usually develop when the maximum sustained wind speeds go above 74 mph (119 km/h) and is the calmest part of the storm. **As the storm system rotates faster and faster, an eye forms in the centre.**

➤ **Statement 3 is not correct:** The eye is the region of lowest surface pressure and warmest temperatures aloft – the eye temperature may be 10°C warmer or more at an altitude of 12 km than the surrounding environment, but only $0\text{--}2^{\circ}\text{C}$ warmer at the surface in the tropical cyclone.



Major Jet Streams of Earth

Source

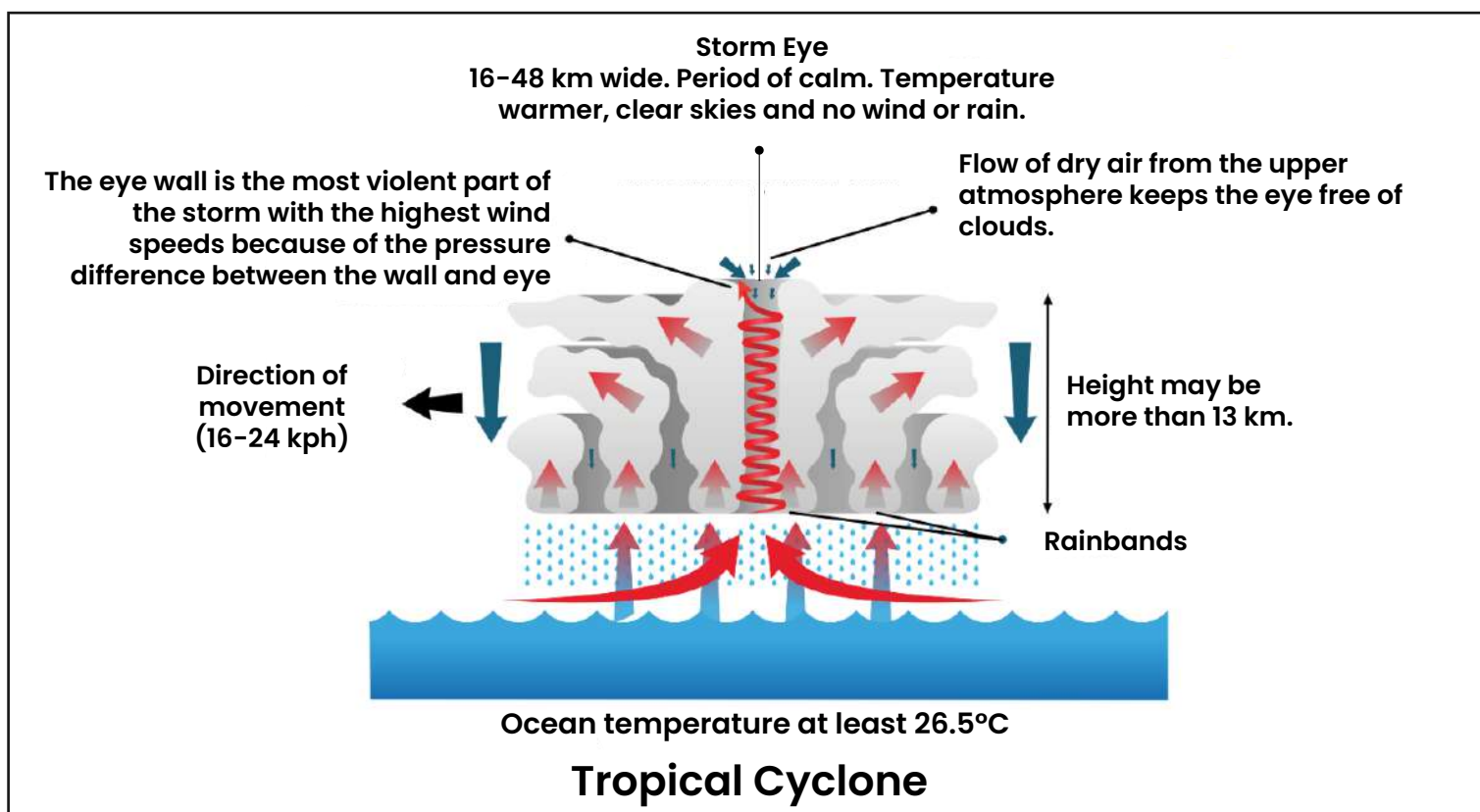
- NCERT, Class XI Fundamentals of Geography
- NCERT, Class IX Contemporary India



Motivation: Basic concepts on climatology are often covered by UPSC.

Elimination Technique

- If the temperature inside the eye region is lower (by any degree, let alone 10), the pressure will have to be higher (than the rest of the cyclone). We know that the eye is the region of lowest pressure. So the eye must have the highest temperature of all parts of cyclone



2021

Consider the following statements:

1. In the tropical zone, the western sections of the oceans are warmer than the eastern sections owing to the influence of trade winds.
2. In the temperate zone, westerlies make the eastern sections of oceans warmer than the western sections.

Which of the statements given above is/are correct?

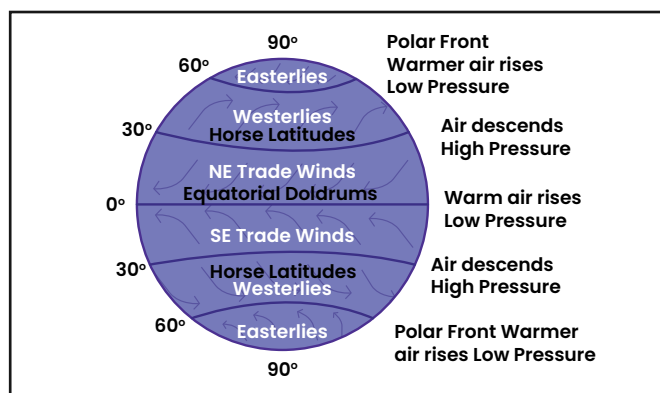
- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Answer: C



Explanation:

- Mid-ocean surface temperatures vary with latitude in response to the balance between incoming solar radiation and outgoing longwave radiation. There is an excess of incoming solar radiation at latitudes less than approximately 45° and an excess of radiation loss at latitudes higher than approximately 45° .
- At low latitudes the trade winds act to move water away from the lee coasts of the landmasses to produce areas of coastal upwelling of water from depth and reduce surface temperatures.
- **Statement 1 is correct:** Warmer water is transported westward in the ocean by the Northeast trade winds in the Northern hemisphere of the tropical zone (as the figure given below shows). So, in tropical zones, the western section of ocean is warmer than eastern sections due to trade winds.
- **Statement 2 is correct:** Similarly, the Westerlies play an important role in carrying the warm, equatorial waters and winds to the western coasts of continents (that is the eastern section of the Oceans in temperate zone). Thus, in temperate zones, westerlies make the eastern section of the ocean warmer than the western sections.



Source

- NCERT, Class XI, Fundamentals of Physical Geography



Motivation: Basic concepts on climatology are often covered by UPSC.

Heat Budget of Earth

Consider the following statements:

1. High clouds primarily reflect solar radiation and cool the surface of the Earth.
2. Low clouds have a high absorption of infrared radiation emanating from the Earth's surface and thus cause warming effect.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Answer: D

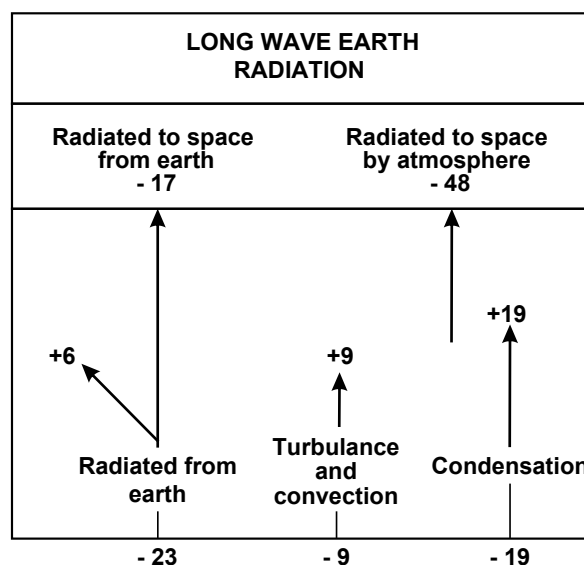
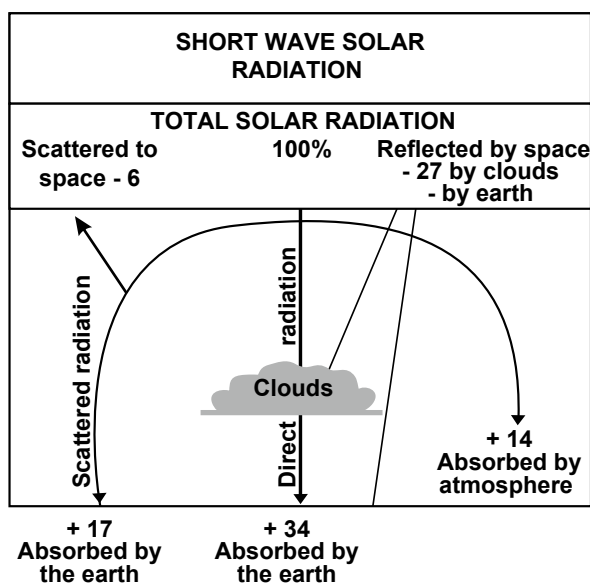


Explanation:

- **Statement 1 is not correct:** Low, thick clouds primarily reflect solar radiation and cool the surface of the Earth.
- **Statement 2 is not correct:** High, thin clouds primarily transmit incoming solar radiation; at the same time, they trap some of the outgoing infrared radiation emitted by the Earth and radiate it back downward, thereby warming the surface of the Earth.

Source

- NCERT, Class XI, Fundamentals of Physical Geography



Heat Budget of the Earth

With reference to the Earth's atmosphere, which one of the following statements is correct?

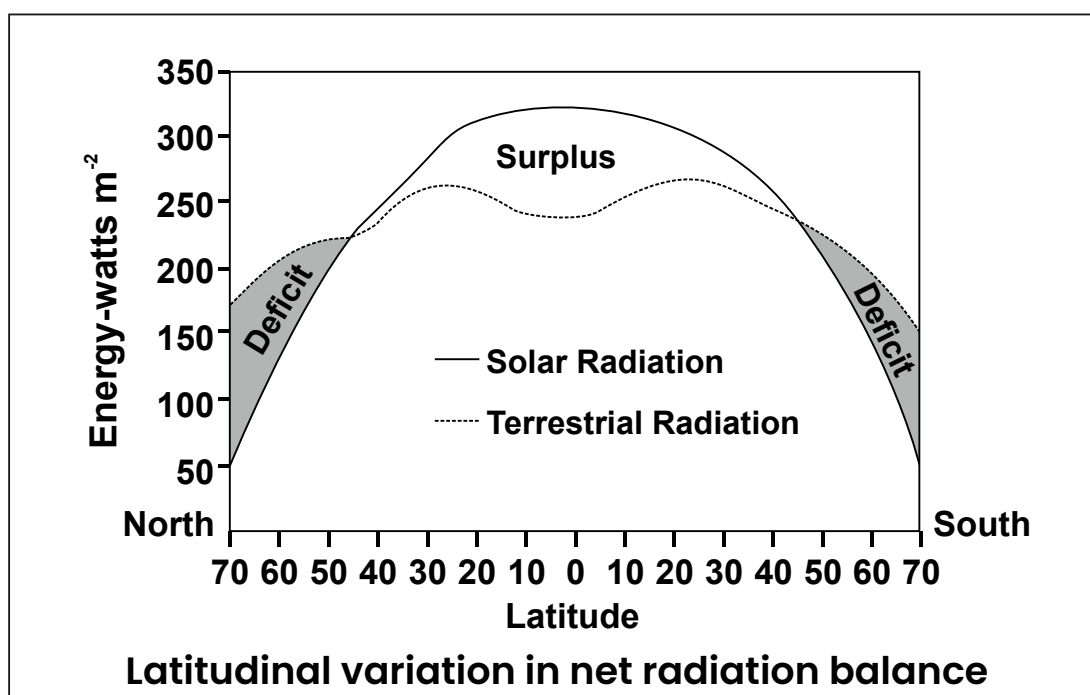
- (a) The total amount of insolation received at the equator is roughly about 10 times of that received at the poles.
- (b) Infrared rays constitute roughly two-thirds of insolation.
- (c) Infrared waves are largely absorbed by water vapour that is concentrated in the lower atmosphere.
- (d) Infrared waves are a part of visible spectrum of electromagnetic waves of solar radiation.

Answer: C



Explanation:

- The earth's surface receives most of its energy in short wavelengths. The energy received by the earth is known as incoming solar radiation which in short is termed as insolation. The insolation received at the surface varies from about 320 Watt/m² in the tropics to about 70 Watt/m² in the poles. Therefore, the total amount of insolation received at the equator is roughly about 5 times of that received at the poles. **Hence option (a) is not correct.**
- Insolation is frequently referred to as shortwave radiation; it falls primarily within the ultraviolet and visible portions of the electromagnetic spectrum and consists predominantly of wavelengths of 0.39 to 0.76 micrometres. Radiation emitted from Earth is called long-wave radiation; it falls within the infrared portion of the spectrum and has typical wavelengths of 4 to 30 micrometres. **Hence option (b) is not correct.**



- The atmosphere is largely transparent to shortwave solar radiation. The incoming solar radiation passes through the atmosphere before striking the earth's surface. Within the troposphere water vapor, ozone and other gasses absorb much of the near infrared radiation.
- Water vapor is the most abundant greenhouse gas in the Earth's atmosphere, and it has strong absorption properties in the infrared portion of the electromagnetic spectrum. As infrared radiation passes through the atmosphere, water vapor molecules can absorb and re-emit a significant portion of the longwave radiation. **Hence option (c) is the correct Answer.**
- Infrared waves are not part of the visible spectrum of electromagnetic waves in solar radiation. The visible spectrum refers to the range of electromagnetic waves that are visible to the human eye, which includes wavelengths between approximately 400 to 700 nanometers. Infrared waves have wavelengths longer than approximately 700 nanometers. Most of the radiation emitted by a moderately heated surface is infrared. **Hence option (d) is not correct.**

Source

- NCERT, Class XI, Fundamentals of Physical Geography



Motivation: Basic concepts on climatology are often covered by UPSC.

2024

Consider the following statements:

Statement-I: The atmosphere is heated more by incoming solar radiation than by terrestrial radiation.

Statement-II: Carbon dioxide and other greenhouse gases in the atmosphere are good absorbers of long-wave radiation.

Which one of the following is correct in respect of the above statements?

- (a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I
- (b) Both Statement-I and Statement-II are correct, but Statement-II does not explain Statement-I
- (c) Statement-I is correct, but Statement-II is incorrect
- (d) Statement-I is incorrect, but Statement-II is correct

Answer: D

Source

- NCERT, Class XI: Fundamental of Physical Geography, Page No. 68-69



Motivation: Basic concepts on climatology are often covered by UPSC.



Explanation:

- The atmosphere is largely transparent to short wave solar radiation. The incoming solar radiation passes through the atmosphere before striking the earth's surface. Within the troposphere water vapour, ozone and other gases absorb much of the near infrared radiation.
- The insolation received by the earth is in short waves forms and heats up its surface. The earth after being heated itself becomes a radiating body and it radiates energy to the atmosphere in long wave form. This energy heats up the atmosphere from below. This process is known as terrestrial radiation.
- The long wave radiation is absorbed by the atmospheric gases particularly by carbon dioxide and the other greenhouse gases. Thus, the atmosphere is indirectly heated by the earth's radiation.
- **Hence, Statement-I is incorrect, but Statement-II is correct.**

2024

Consider the following statements:

Statement-I:

Thickness of the troposphere at the equator is much greater as compared to poles.

Statement-II:

At the equator, heat is transported to great heights by strong convectional currents.

Which one of the following is correct in respect of the above statements?

- (a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I
- (b) Both Statement-I and Statement-II are correct, but Statement-II does not explain Statement-I
- (c) Statement-I is correct, but Statement-II is incorrect
- (d) Statement-I is incorrect, but Statement-II is correct

Answer: A



Explanation:

- The troposphere is the lowermost layer of the atmosphere. Its average height is 13 km and extends roughly to a height of 8 km near the poles and about 18 km at the equator.
- Thickness of the troposphere is greatest at the equator because heat is transported to great heights by strong convectional currents. **Hence option (a) is the correct answer.**
- This layer contains dust particles and water vapour. All changes in climate and weather take place in this layer. The temperature in this layer decreases at the rate of 1°C for every 165m of height. This is the most important layer for all biological activity. The zone separating the troposphere from stratosphere is known as the tropopause. The air temperature at the tropopause is about minus 80°C over the equator and about minus 45°C over the poles. The temperature here is nearly constant, and hence, it is called the tropopause.

Source

➤ NCERT, Class XI: Fundamentals of Physical Geography, Page No. 65



Motivation: Basic concepts on climatology are often covered by UPSC.

All India Test Series Test Code: 4141

Consider the following statements regarding the troposphere:

1. The temperature in the troposphere changes at a positive lapse rate.
2. All weather phenomena like cyclones, rainfall, fog and hailstorms, etc. are confined to this layer.
3. The temperature at the tropopause is nearly constant.

How many statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

2024

Which of the following is/are correct inference/inferences from isothermal maps in the month of January?

1. The isotherms deviate to the north over the ocean and to the south over the continent.
2. The presence of cold ocean currents Gulf Stream and North Atlantic Drift make the North Atlantic Ocean colder and the isotherms bend towards the north.

Select the answer using the code given below:

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Answer: A

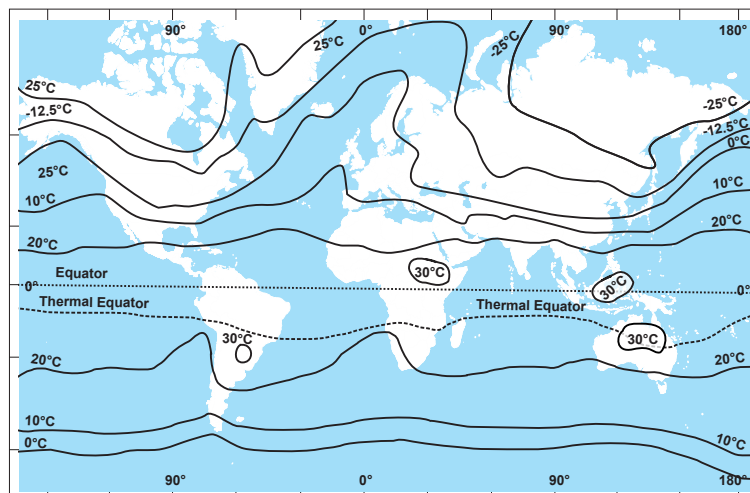
Source

➤ NCERT, Class XI: Fundamentals of Physical Geography, Page No. 71



Explanation:

- The Isotherms are lines joining places having equal temperature.
 - Figure given below shows the distribution of surface air temperature in the month of January. In general, the effect of the latitude on temperature is well pronounced on the map as the isotherms are generally parallel to the latitude. The deviation from this general trend is more pronounced in January than in July, especially in the northern hemisphere. In the northern hemisphere the land surface area is much larger than in the southern hemisphere. Hence, the effects of land mass and the ocean currents are well pronounced.
 - In January the isotherms deviate to the north over the ocean and to the south over the continent. This can be seen on the North Atlantic Ocean. The presence of warm ocean currents, Gulf Stream and North Atlantic drift, make the Northern Atlantic Ocean warmer and the isotherms bend towards the north. Over the land the temperature decreases sharply and the isotherms bend towards south in Europe. It is much pronounced in the Siberian plain. Hence statement 1 is correct and statement 2 is not correct.
 - The mean January temperature along 60° E longitude is minus 20° C both at 80° N and 50° N latitudes. The mean monthly temperature for January is over 27° C, in equatorial oceans over 24° C in the tropics and 2° C - 0° C in the middle latitudes and -18° C to -48° C in the Eurasian continental interior.
 - The effect of the ocean is well pronounced in the southern hemisphere. Here the isotherms are more or less parallel to the latitudes and the variation in temperature is more gradual than in the northern hemisphere. The isotherm of 20° C, 10° C, and 0° C runs parallel to 35° S, 45° S and 60° S latitudes respectively.



The distribution of surface air temperature in the month of January

All India 2024 Prelims Test Series: Test 4141

Consider the following statements regarding Isotherms:

1. The spacing of isotherms depicts the temperature gradient across a portion of the Earth's surface.
2. Widely spaced isotherms indicate a small change in temperature over distance and closely spaced isotherms indicate large changes in temperature.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

With reference to "water vapour", which of the following statements is/are correct?

1. It is a gas, the amount of which decreases with altitude.
2. Its percentage is maximum at the poles.

Select the answer using the code given below:

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Answer: A



Explanation:

- Water vapour is also a variable gas in the atmosphere, which decreases with altitude. In the warm and wet tropics, it may account for four per cent of the air by volume, while in the dry and cold areas of desert and polar regions, it may be less than one per cent of the air. Hence statements 1 is correct and 2 is not correct.
- Water vapour also decreases from the equator towards the poles. It also absorbs parts of the insolation from the sun and preserves the earth's radiated heat. It thus, acts like a blanket allowing the earth neither to become too cold nor too hot. Water vapour also contributes to the stability and instability in the air.

Source

- NCERT, Class XI: Fundamentals of Physical Geography, Page No.64-73



Motivation: Conceptual Understanding of the aspirants is often tested by UPSC.

2024

Consider the following description:

1. Annual and daily range of temperatures is low.
2. Precipitation occurs throughout the year.
3. Precipitation varies between 50 cm – 250 cm.

What is this type of climate?

- (a) Equatorial climate
- (b) China type climate
- (c) Humid subtropical climate
- (d) Marine West Coast climate

Answer: D



Explanation:

- The marine west coast climate (British type) is found along a relatively narrow strip of coastal Oregon, Washington, British Columbia, and southern Alaska in North America. It is likewise found along coastal Chile in South America. The marine west coast climate extends further inland into northwest Europe than in North America due to the orientation of mountains. This climate is also found on the southeast coast of Australia and New Zealand.
- The marine west coast is noted for its mild summers and winters and, as a result, a small annual temperature range. Its location on the west coast of a continent in the midlatitudes places the climate in the path of the Westerlies. In this situation, the climate receives a constant influx of oceanic air throughout the year. The mild temperatures are a direct result of the moderating influence of ocean bodies on air temperatures. This is especially true for those situations where a warm ocean current borders the continent, like the North Atlantic Drift's effect on northwestern Europe. Temperature ranges increase as one moves away from the coast.
- The marine west coast climate is dominated by cyclonic activity embedded in the Westerlies. Frequent cyclonic storms bring prolonged periods of rain, drizzle and fog to these west coast locations. In some locations, it is not uncommon to receive as much as 250 cm (100 in) of precipitation in a year. **Hence option (d) is the correct answer.**

Source

- Certificate Geography by GC Leong Chapter 22



Motivation: Basic concepts on climatology are often covered by UPSC.

With reference to "Coriolis force", which of the following statements is/are correct?

1. It increases with increase in wind velocity.
2. It is maximum at the poles and is absent at the equator.

Select the answer using the code given below:

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Answer: C



Explanation:

- The rotation of the earth about its axis affects the direction of the wind. This force is called the Coriolis force after the French physicist who described it in 1844. It deflects the wind to the right direction in the northern hemisphere and to the left in the southern hemisphere. The deflection is more when the wind velocity is high. **Hence statement 1 is correct.**
- The Coriolis force is directly proportional to the angle of latitude. It is maximum at the poles and is absent at the equator. **Hence statement 2 is correct.**
- The Coriolis force acts perpendicular to the pressure gradient force. The pressure gradient force is perpendicular to an isobar. The higher the pressure gradient force, the more is the velocity of the wind and the larger is the deflection in the direction of wind. As a result of these two forces operating perpendicular to each other, in the low-pressure areas the wind blows around it.
- At the equator, the Coriolis force is zero and the wind blows perpendicular to the isobars. The low pressure gets filled instead of getting intensified. That is the reason why tropical cyclones are not formed near the equator.

Source

- NCERT, Class XI: Fundamentals of Physical Geography, Page No. 78-79



Motivation: Basic concepts on climatology are often covered by UPSC.

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Earth & Solar System,

Total No. of Questions: 03

2019

On 21st June, the Sun

- (a) does not set below the horizon at the Arctic Circle
- (b) does not set below the horizon at Antarctic Circle
- (c) shines vertically overhead at noon on the Equator
- (d) shines vertically overhead at the Tropic of Capricorn

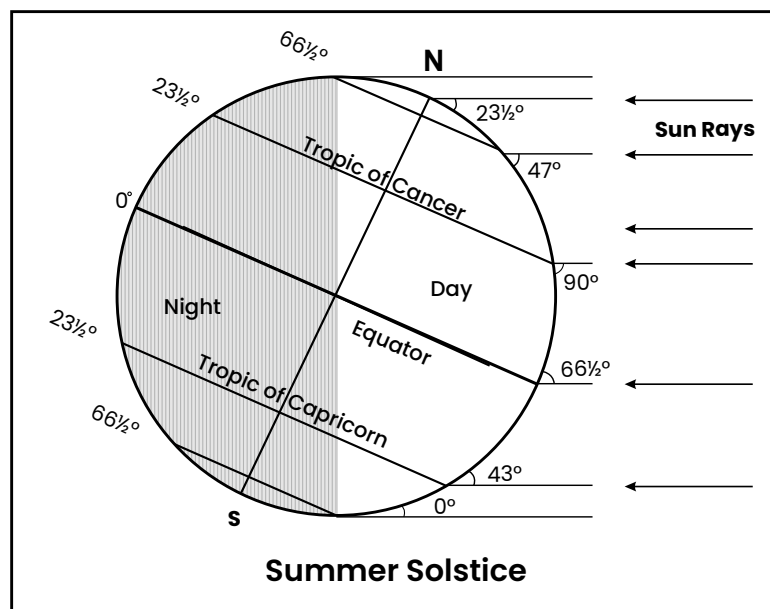
Answer: A



Explanation:

- On 21st June, the Sun is directly over the Tropic of Cancer. Geometrically, it means the Sun is normal (90°) to Tropic of Cancer and makes an angle of 47° to the Arctic Circle. As such, the Sun is above the horizon for all areas north of Arctic Circle. The entire area in the Arctic circle experiences a day. The North Pole is inclined towards the sun and **the places beyond the Arctic Circle experience continuous daylight** for about six months. Since a large portion of the Northern Hemisphere is getting light from the sun, it is summer in the regions north of the equator. The longest day and the shortest night at these places occur on 21st June.

- At this time in the Southern Hemisphere all these conditions are reversed. It is winter season there. The nights are longer than the days. This position of the earth is called the Summer Solstice.



Source

- Certificate Physical and Human Geography by G C Leong
- NCERT, Class XI – Fundamentals of Physical Geography



Motivation: Conceptual understanding of the aspirants is often tested by UPSC in this section.

2022

In the northern hemisphere, the longest day of the year normally occurs in the:

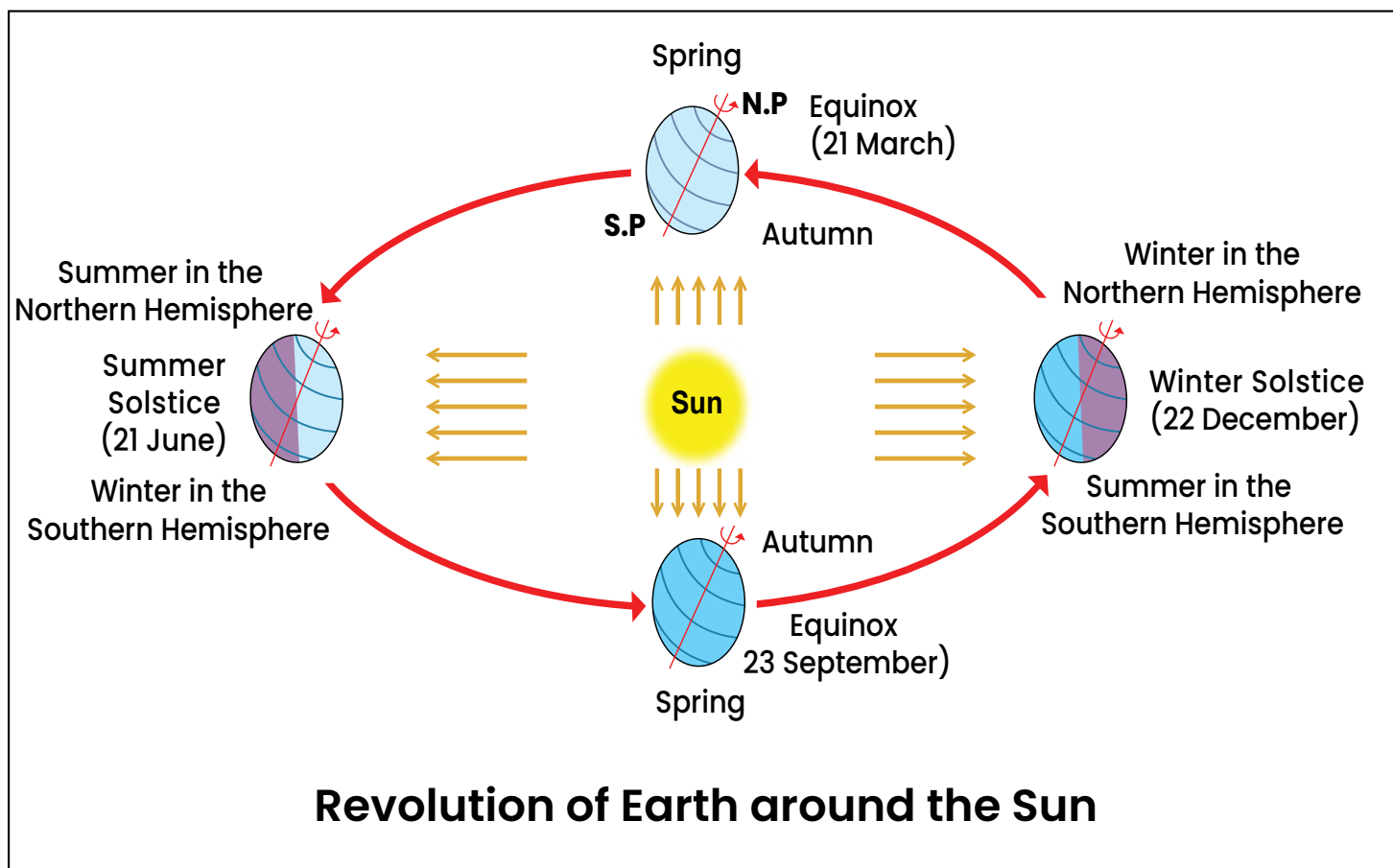
- (a) First half of the month of June
- (b) Second half of the month of June
- (c) First half of the month of July
- (d) Second half of the month of July

Answer: B



Explanation:

- The summer solstice, also known as festival solstice or midsummer, occurs when one of Earth's poles has its maximum tilt toward the Sun. It happens twice yearly, once in each hemisphere. During the Northern Hemisphere's summer solstice—which always falls around June 21—the Southern Hemisphere gets its winter solstice. Hence, In the northern hemisphere, the longest day of the year normally occurs in the Second half of the month of June. **Hence option (b) is the correct answer.**



Source

- Certificate Physical and Human Geography by G C Leong
➤ NCERT, Class VI Geography

2024

On June 21 every year, which of the following latitude(s) experience(s) a sunlight of more than 12 hours?

1. Equator
2. Tropic of Cancer
3. Tropic of Capricorn
4. Arctic Circle

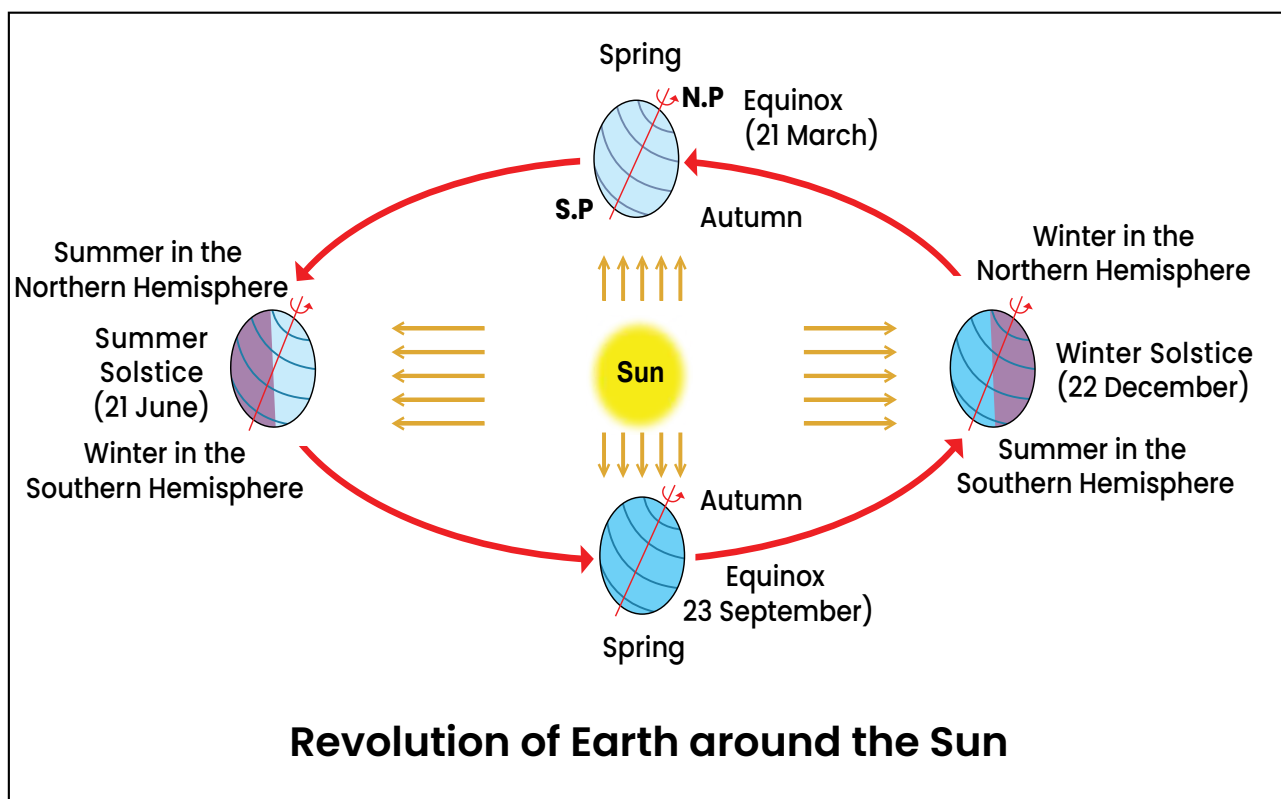
Select the correct answer using the code given below:

- (a) 1 only
- (b) 2 only
- (c) 3 and 4
- (d) 2 and 4 only

Answer: D



Explanation:



Source

➤ NCERT, Class VI GC Leong Page No. 6



Motivation: Conceptual Understanding of the aspirants is often tested by UPSC.

Geomorphology

Total No.of Questions: 05

2018

Formation of Earth

Consider the following statements:

1. The Earth's magnetic field has reversed every few hundred thousand years.
2. When the Earth was created more than 4000 million years ago, there was 54% oxygen and no carbon dioxide.
3. When living organisms originated, they modified the early atmosphere of the Earth.

Which of the statements given above is/are correct?

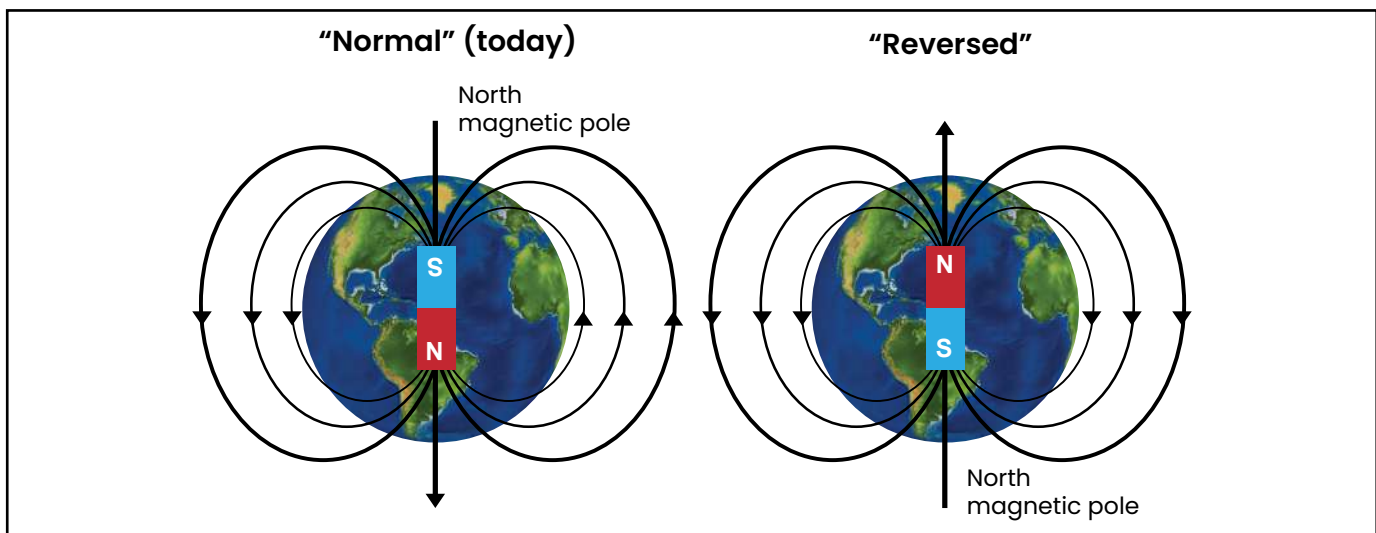
- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Answer: C



Explanation:

- **Statement 1 is correct.** Earth's magnetic field has flipped its polarity many times over the millennia. Magnetic Reversals are the rule, not the exception. Earth has settled in the last 20 million years into a pattern of a magnetic pole reversal about every 200,000 to 300,000 years.
- In Normal Polarity, Earth's North Magnetic Pole is the South Pole of its Magnetic Field.
 - In Reverse Polarity, Earth's North Magnetic Pole is the North Pole of its Magnetic Field.



- **Statement 2 is not correct.** When Earth formed 4.6 billion years ago, it had almost no atmosphere. As Earth cooled, an atmosphere formed mainly from gases spewed from volcanoes. It included hydrogen sulphide, methane, and 10 to 200 times as much carbon dioxide as today's atmosphere.
- The Earth's early atmosphere wasn't rich in oxygen. Oxygen started to build up in the atmosphere around 2.5 billion years ago, due to photosynthesis by early life forms.
- **Statement 3 is correct.** Living organisms include plants and microbes. Life started to have a major impact on the environment once photosynthetic organisms evolved. While photosynthetic life reduced the carbon dioxide content of the atmosphere, it also started to produce oxygen. So, when living beings originated, they did modify the atmosphere.

Source

- NCERT, Class XI, Fundamentals of physical geography

2023

Earthquake Waves

Consider the following statements:

1. In a seismograph, P waves are recorded earlier than S waves.
2. In P waves, the individual particles vibrate to and fro in the direction of wave propagation whereas in S waves, the particles vibrate up and down at right angles to the direction of wave propagation.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Answer: C



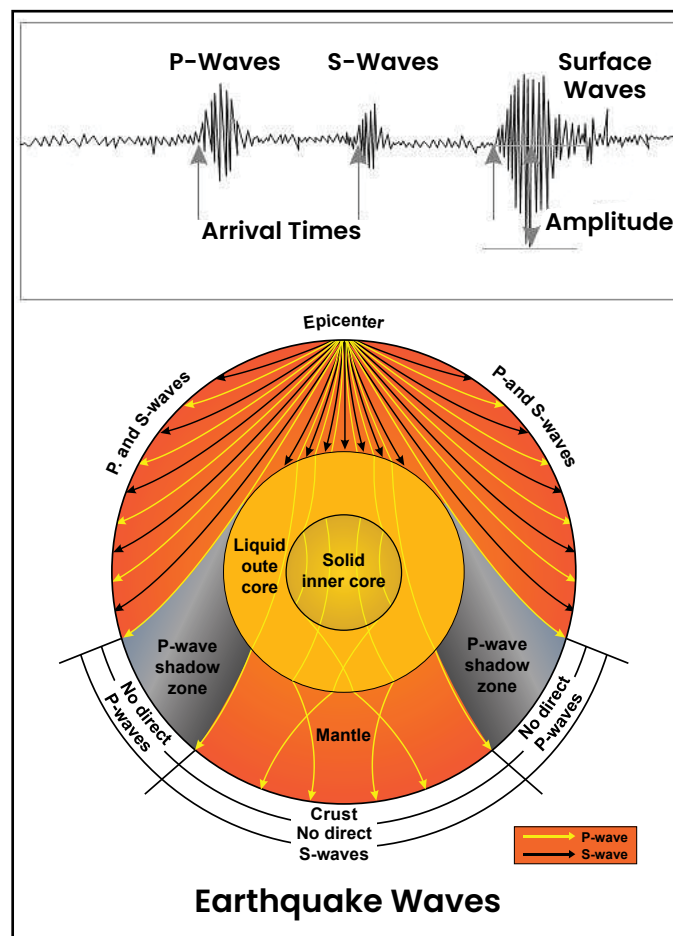
Explanation:

- Earthquake waves are basically of two types: body waves and surface waves. Body waves are generated due to the release of energy at the focus and move in all directions travelling through the body of the earth. Hence, the name body waves. The body waves interact with the surface rocks and generate new sets of waves called surface waves. These waves move along the surface. The velocity of waves changes as they travel through materials with different densities. The denser the material, the higher is the velocity. Their direction also changes as they reflect or refract when coming across materials with different densities.

- P-waves move faster and are the first to arrive at the surface. These are also called 'primary waves'. The P-waves are similar to sound waves. They travel through gaseous, liquid and solid materials. S-waves arrive at the surface with some time lag. These are called secondary waves and they can travel only through solid materials. **Hence statement 1 is correct.**
- P-waves vibrate parallel to the direction of the wave. This exerts pressure on the material in the direction of the propagation. As a result, it creates density differences in the material leading to stretching and squeezing of the material. The direction of vibrations of S-waves is perpendicular to the wave direction in the vertical plane. Hence, they create troughs and crests in the material through which they pass. Surface waves are considered to be the most damaging waves. **Hence statement 2 is correct.**

Source

- NCERT Class XI: Fundamentals of Physical geography



2024

Consider the following:

1. Pyroclastic debris
2. Ash and dust
3. Nitrogen compounds
4. Sulphur compounds

How many of the above are products of volcanic eruptions?

- (a) Only one
- (b) Only two
- (c) Only three
- (d) All four

Answer: D



Explanation:

- A volcano is a place where gases, ashes and/or molten rock material – lava – escape to the ground. A volcano is called an active volcano if the materials mentioned are being released or have been released out in the recent past. The layer below the solid crust is mantle. It has higher density than that of the crust. The mantle contains a weaker zone called asthenosphere. It is from this that the molten rock materials find their way to the surface.
- The material in the upper mantle portion is called magma. Once it starts moving towards the crust or it reaches the surface, it is referred to as lava.
- The material that reaches the ground includes lava flows, pyroclastic debris, volcanic bombs, ash and dust and gases such as nitrogen compounds, sulphur compounds and minor amounts of chlorine, hydrogen and argon. **Hence option (d) is the correct answer.**

Source

- NCERT, Class XI: Fundamentals of Physical Geography, Page No. 23



Motivation: Volcanic eruptions are often seen in news.

Sandhan Vision IAS Prelims Test Series Initiative

Which of the following statements is correct regarding volcanic eruptions?

- (a) Composite Volcanoes are the largest of all the volcanoes on the earth.
- (b) Deccan Traps formations in the Indian peninsula is an example of caldera volcanoes.
- (c) Batholiths are a large mass of magmatic material that cools on the earth surface.
- (d) Volcanic eruptions do not typically occur at transform boundaries.

Consider the following statements:

Statement-I: Rainfall is one of the reasons for weathering of rocks.

Statement II: Rain water contains carbon dioxide in solution.

Statement-III: Rain water contains atmospheric oxygen.

Which one of the following is correct in respect of the above statements?

- (a) Both Statement-II and Statement-III are correct and both of them explain Statement-I
- (b) Both Statement-II and Statement-III are correct, but only one of them explains Statement-I
- (c) Only one of the Statements II and III is correct and that explains Statement-I
- (d) Neither Statement-II nor Statement-III is correct

Answer: A



Explanation:

- Chemical weathering is the basic process by which denudation proceeds. It is the extremely slow and gradual decomposition of rocks due to exposure to air and water. Air and water contain chemical elements, which though they may be in small quantities, are sufficient to set up chemical reactions in the surface layers of exposed rocks. Such reactions may weaken or entirely dissolve certain constituents of the rock, thus loosening the other crystals and weakening the whole surface. There are three major chemical weathering processes:
 - **Solution:** Many minerals are dissolved by water, especially when, as with rain-water, it contains enough carbon dioxide to make it a weak acid. Solution is the most potent weathering process in limestone regions because the rainwater attacks and dissolves the calcium carbonate of which the rock is chiefly formed. The dissolved calcium carbonate is carried away by the water, joints and cracks in the rock are quickly widened and whole systems of caves and passages are worn out. Limestone, however, is by no means the only rock to suffer from solution. All rocks are subject to solution to some extent, though the process is much slower than with limestone. Hence Statement II is correct and explains Statement I.
 - **Oxidation:** Dissolved oxygen is present in all rainwater and surface supplies due to contact with the atmosphere. Oxidation is the reaction of oxygen in air or water with minerals in the rock. For example, most rocks contain a certain amount of iron, which when it comes in contact with air is changed to iron oxide, familiar brownish crust or rust. Iron oxide crumbles easily and is far more easily eroded than the original iron. It is thus removed, loosening the overall structure of the rocks and weakening them. Hence Statement III is correct and explains Statement I.
 - **Decomposition by Organic acids:** Within the soil which covers most rocks are bacteria which thrive on decaying plant or animal material. These bacteria produce acids which, when dissolved in water, help to speed up the weathering of the underlying rocks.
- Hence option (a) is the correct answer.

➤ NCERT, Class XI: Fundamentals of Physical Geography, Page No. 40–41 Certificate Geography by GC Leong, Page No.: 28– 29

Sandhan Vision IAS Prelims Test Series Initiative

Consider the following statements about the weathering:

1. A very little or no motion of materials takes place in weathering.
2. Carbonation is a common process of breaking down feldspars and carbonate minerals.
3. Physical weathering processes are caused by thermal expansion and pressure release.

Which of the statements given above are correct?

- (a) 1 and 3 only
- (b) 1, 2 and 3
- (c) 2 and 3 only
- (d) 1 and 2 only

Carbonation is the reaction of carbonate and bicarbonate with minerals and is a common process helping the breaking down of feldspars and carbonate minerals. Carbon dioxide from the atmosphere and soil air is absorbed by water, to form carbonic acid that acts as a weak acid. Calcium carbonates and magnesium carbonates are dissolved in carbonic acid and are removed in a solution without leaving any residue resulting in cave formation.

Consider the following information:

Region	Name of the mountain range	Type of mountain
1. Central Asia	Vosges	Fold mountain
2. Europe	Alps	Block mountain
3. North America	Appalachians	Fold mountain
4. South America	Andes	Fold mountain

In how many of the above rows is the given information correctly matched?

- (a) Only one
- (b) Only two
- (c) Only three
- (d) All four

Answer: B

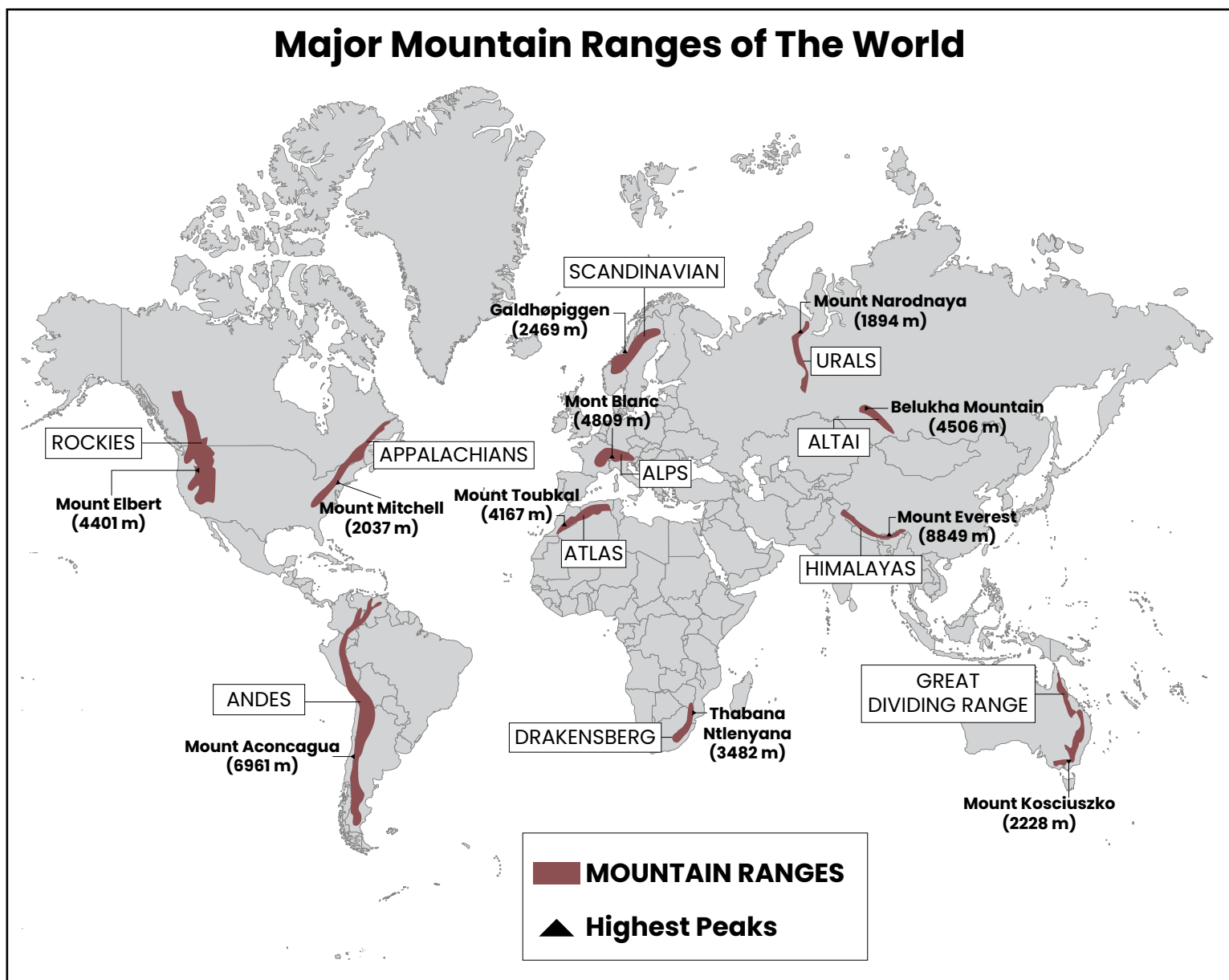


Explanation:

- The Andes are a fold mountain range in South America.
- Appalachian Mountains are a set of old fold mountains that form the eastern boundary of North America.
- Alps are not a block mountain, but rather a fold mountain range in Europe
- Vosges Mountains are in Europe (not Central Asia), and are fold mountains

Type of Mountain	Description	Examples
Fold Mountains	Formed by the folding of rock layers due to tectonic forces.	Himalayas, Alps, Andes
Fault-block Mountains	Result from the uplift of blocks of rock along fault lines.	Sierra Nevada, Harz Mountains
Volcanic Mountains	Formed by the accumulation of solidified volcanic materials.	Mount Kilimanjaro, Mount Fuji, Cascade Range
Dome Mountains	Characterized by upwarped and circular-shaped rock layers.	Black Hills, Vosges Mountains

Plateau Mountains	Elevated flat-topped landforms that underwent uplift and erosion.	Colorado Plateau, Deccan Plateau
Residual Mountains	Remnants of older mountain systems that have undergone erosion.	Appalachian Mountains, Scottish Highlands
Coastal Mountains	Found near coastlines, influenced by tectonic activity or erosion.	Coast Mountains, Great Dividing Range



Source

➤ Certificate Geography by GC Leong, Page No. 15-16

Indian Drainage System

Total No. of Questions: 15

Rivers

10

Lakes, Wetlands and Water Reservoirs

05

2015

Godavari River System

Consider the following rivers:

1. Vamsadhara
2. Indravati
3. Pranhita
4. Pennar

Which of the above are tributaries of Godavari?

- (a) 1, 2 and 3
- (b) 2, 3 and 4
- (c) 1, 2 and 4
- (d) 2 and 3 only

Answer: D

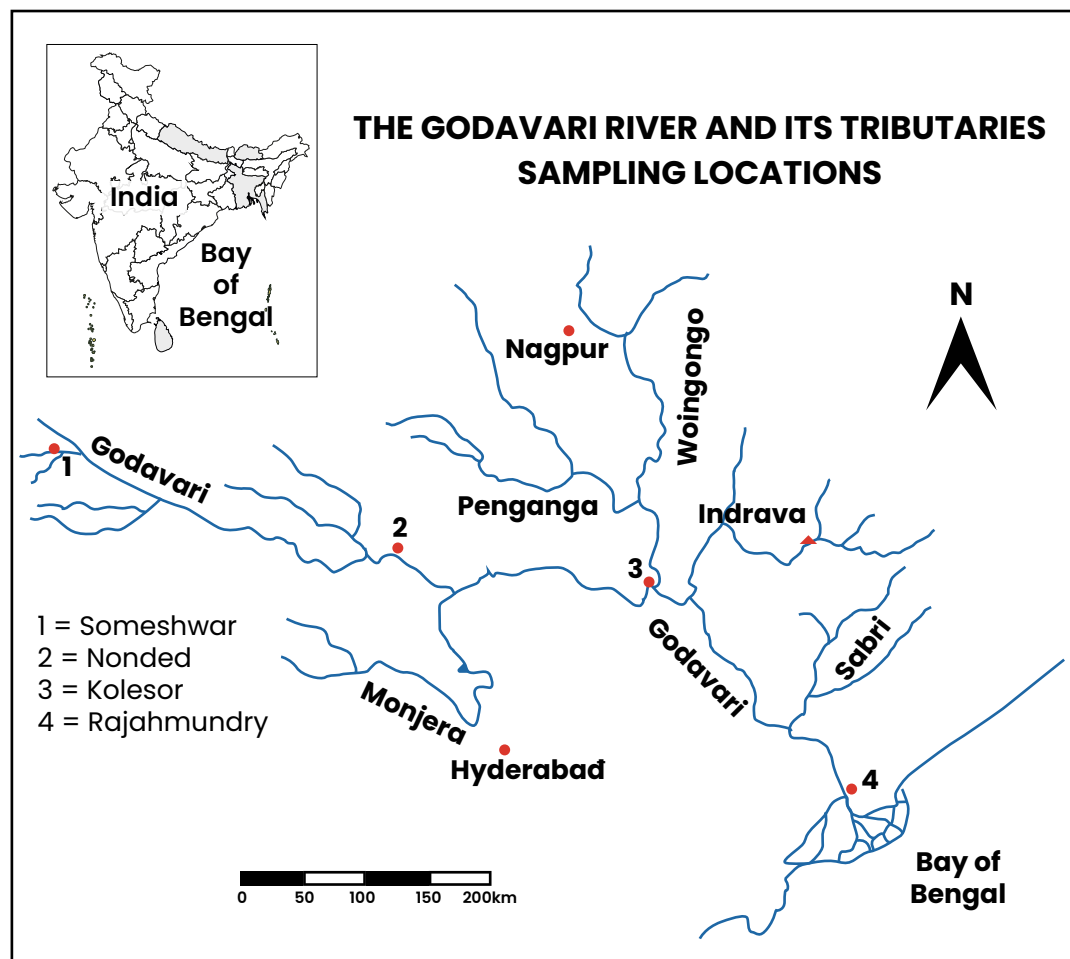


Explanation:

- The Penganga, the Indravati, the Pranhita, and the Manjra are principal tributaries of the Godavari.

Source

- NCERT Class 11th India Physical Environment Page 27



2016

Brahmaputra River System

Which of the following is/are tributary/tributaries of Brahmaputra?

1. Dibang
2. Kameng
3. Lohit

Select the correct answer using the code given below:

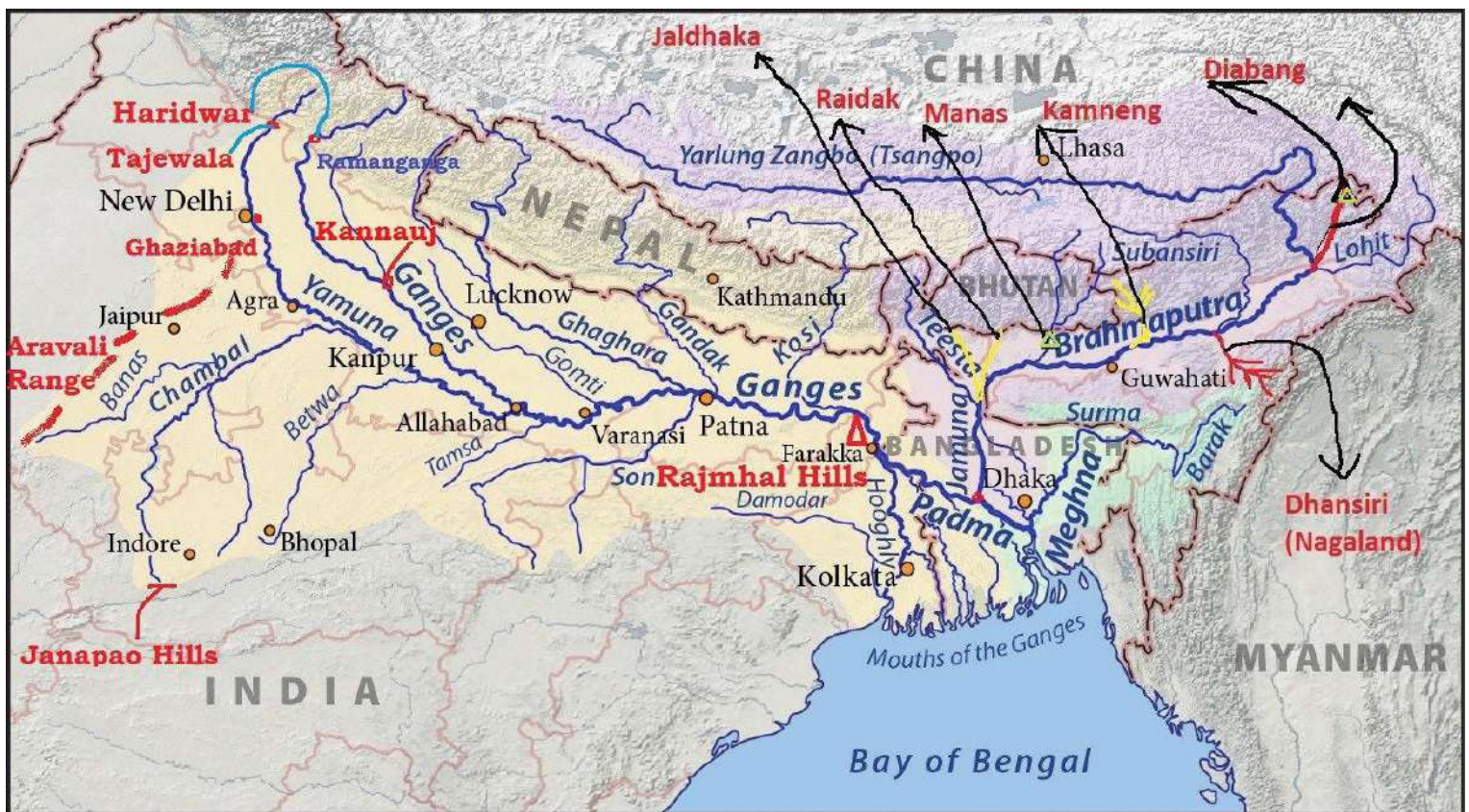
- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Answer: D



Explanation:

➤ All the given rivers Dibang, Kameng and Lohit are the tributaries of Brahmaputra.



Source

➤ NCERT 11, India physical environment, page 26

2016

River linking

Recently, linking of which of the following rivers was undertaken?

- (a) Cauvery and Tungabhadra
- (b) Godavari and Krishna
- (c) Mahanadi and Son
- (d) Narmada and Tapi

Answer: B



Explanation:

- Andhra Pradesh took the historic step of interlinking its two major rivers Godavari and Krishna, a development that is being seen as a boon for farmers of Krishna delta who are facing water scarcity. Government decided to divert Godavari water from the Pattiseema Lift Irrigation Scheme.

Source

- <http://www.thehindu.com/news/national/andhra-pradesh/interlinking-godavari-krishna-my-most-satisfying-exercise/article8051320.ece>
- <http://indianexpress.com/article/explained/godavari-and-krishna-rivers-interlink-when-two-rivers-meet/>



Motivation: Pattiseema Lift Irrigation Scheme has been often seen in news

2017

Teesta River

With reference to river Teesta, consider the following statements:

1. The source of river Teesta is the same as that of Brahmaputra but it flows through Sikkim.
2. River Rangeet originates in Sikkim and it is a tributary of river Teesta.
3. River Teesta flows into Bay of Bengal on the border of India and Bangladesh.

Which of the statements given above is/are correct?

- (a) 1 and 3 only
- (b) 2 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

Answer: B



Explanation:

- **Statement 1 is not correct.** The Teesta River originates from the Pahunri (or Teesta Kangse) glacier and flows southward through the Sikkim Himalaya. where the Rangpo River joins, and where it forms the border between Sikkim and West Bengal up to Teesta Bazaar. Just before the Teesta Bridge, where the roads from Kalimpong and Darjeeling join, the river is met by its main tributary, the Rangeet River. The Brahmaputra River originates from the Angsi Glacier in Burang County, Tibet, in the northern side of the Himalayas.
- The Rangit river and its tributaries originate in the Talung glacier in West Sikkim and after flowing for about 60 km, joins Teesta below Malli near the border of Sikkim with West Bengal. River Rangit is a major tributary of river Teesta from the Western Sikkim. **Hence Statement 2 is correct.**
- It changes course southwards, flowing into West Bengal. The river then merges up with the Brahmaputra River after it bifurcates the city of Jalpaiguri and flows just touching Cooch Behar district at Mekhliganj and moves to Fulchori in Bangladesh. Thus it does not flow directly into the Bay of Bengal. **Hence statement 3 is not correct.**



Source

- School Atlas



Motivation: River Teesta has been often seen in news with respect to India-Bangladesh international relations. Its waters are still disputed between the two nations.

Elimination Technique

- Teesta river is the tributary of the Brahmaputra river. Hence statement 3 can be eliminated to arrive at the correct answer option (b).

2018

Artificial Lakes

Which one of the following is an artificial lake?

- (a) Kodaikanal (Tamil Nadu)
- (b) Kolleru (Andhra Pradesh)
- (c) Nainital (Uttarakhand)
- (d) Renuka (Himachal Pradesh)

Answer: A



Explanation:

- Kodaikanal Lake, also known as Kodai Lake is a manmade lake located in the Kodaikanal city in Dindigul district in Tamilnadu, India. Sir Vere Henry Levinge, the then Collector of Madurai, was instrumental in creating the lake in 1863, amidst the Kodaikanal town which was developed by the British and early missionaries from the USA. **Hence option (a) is the correct answer.**

Source

- Standard ATLAS



Motivation: : Kodaikanal lake and its maintenance by private firms was in news

2019

Water Reservoirs

What is common to the places known as Aliyar, Isapur and Kangsabati?

- (a) Recently discovered uranium deposits
- (b) Tropical rain forests
- (c) Underground cave systems
- (d) Water reservoirs

Answer: D



Explanation:

- Aliyar is a village located near Pollachi Town in Coimbatore district in Tamil Nadu, India. The famous Aliyar Reservoir is located in this village.
- Isapur Dam is an earthfill dam on Penganga river near Pusad in the state of Maharashtra.
- The Kangsabati Reservoir Project was started in 1956 as part of the Indian Second Five-year Plan to provide water to 3,484.77 square kilometres (1,345.48 sq mi) of land in the districts of Midnapur(erstwhile), Bankura, and Hooghly. It involves irrigation land using water from the Kangsabati River, as well as the Shilabati and the Bhoiobbanki rivers.

Source

➤ Newspaper and Standard Atlas



Motivation: Aliyar, Isapur and Kangsabati were in news in the context of water sharing.

Note:

A map denoting the important Dams of India and their location is given in Annexure E of this document.

2019

Glaciers

Consider the following pairs:

Glacier	: River
1. Bandarpunch	: Yamuna
2. Bara Shigri	: Chenab
3. Milam	: Mandakini
4. Siachen	: Nubra
5. Zemu	: Manas

Which of the pairs given above are correctly matched?

- (a) 1, 2 and 4
- (b) 1, 3 and 4
- (c) 2 and 5
- (d) 3 and 5

Answer: A



Explanation:

➤ Bandarpunch is located at the western edge of the High Himalayan Range. It is part of the Sankari Range and lies within the Govind Pashu Vihar National Park and Sanctuary. It is a major watershed for the headwaters of the Yamuna River, whose source lies above Yamunotri, on the west end of the massif below White Peak. **Hence pair 1 is correctly matched.**

- Bara-Sigri glacier, which is the second longest glacier in Himalaya after Gangotri, is located in the Chandra Valley of Lahaul. The glacier feeds the Chenab River. **Hence pair 2 is correctly matched.**
 - Milam Glacier is a major glacier of the Kumaon Himalaya. It is a part of the Pithoragarh district of Uttarakhand, India, about 15 kilometers (9 mi) northeast of Nanda Devi. Mandakini is a tributary of the Alaknanda River which originates from the Chorabari Glacier near Kedarnath in Uttarakhand, India. **Hence pair 3 is not correctly matched.**
- Siachen Glacier is located in the eastern Karakoram range in the Himalayas. The glacier's melting waters are the main source of the Nubra River in the Indian region of Ladakh, which drains into the Shyok River. **Hence pair 4 is correctly matched.**
- Zemu Glacier is the largest glacier in the Eastern Himalaya. It is located at the base of Kanchenjunga in the Himalayan region of Sikkim, India. Manas River is a transboundary river in the Himalayan foothills between southern Bhutan and India. **Hence pair 5 is not correctly matched.** Important Glaciers of Himalayas are given in **Annexure F**.

Source

- Part of the answer is given in Bander punch and Milam glaciers mentioned in Class XI NCERT.



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GTB Nagar Metro (Mukherjee Nagar): 27 DEC, 6 PM | 6 JAN, 8 AM

हिन्दी माध्यम | DELHI: 6 जनवरी, 11 AM | 13 दिसंबर, 8 AM

AHMEDABAD: 4 JAN	BENGALURU: 5 DEC	BHOPAL: 5 DEC	HYDERABAD: 9 DEC	JAIPUR: 20 JAN
JODHPUR: 3 DEC	LUCKNOW: 5 DEC	PUNE: 20 JAN	ADMISSION OPEN	CHANDIGARH

2020

Cauvery Basin

Which of the following Protected Areas are located in the Cauvery basin?

1. Nagarhole National Park
2. Papikonda National Park
3. Sathyamangalam Tiger Reserve
4. Wayanad Wildlife Sanctuary

Select the correct Answer using the code given below:

- (a) 1 and 2 only
- (b) 3 and 4 only
- (c) 1, 3 and 4 only
- (d) 1, 2, 3 and 4

Answer: C



Explanation:

- **Option 1 is correct:** Nagarhole National Park, also known as Rajiv Gandhi National Park, is a wildlife reserve in the South Indian state of Karnataka. Nagarhole National Park is in the Cauvery Basin.
- **Option 2 is not correct:** Papikonda National Park is located near Rajamahendravaram in the Papi Hills in East Godavari and West Godavari districts of Andhra Pradesh, and covers an area of 1,012.86 km² (391.07 sq mi). It is an Important Bird and Biodiversity Area and home to some endangered species of flora and fauna. River Godavari flows through Papikonda National Park.
- **Option 3 is correct:** Sathyamangalam Tiger Reserve is a protected area and tiger reserve in the Eastern Ghats in the Erode District of the Indian state of Tamil Nadu. It is located at the confluence of two distinct geographical regions of biodiversity landscape; Western Ghat and Eastern Ghat. Sathyamangalam Tiger Reserve, the Gateway to Eastern Ghats, is a significant ecosystem and a wildlife corridor in the Nilgiri Biosphere Reserve between the Western Ghats and the rest of the Eastern Ghats and a genetic link between the five other protected areas which it adjoins, including the Biligiriranga Swamy Temple Wildlife Sanctuary, Sigur Plateau, Mudumalai National Park, Bandipur National Park and the Cauvery Wildlife Sanctuary. Sathyamangalam Tiger Reserve is in the Cauvery Basin.
- **Option 4 is correct:** The Wayanad Wildlife Sanctuary is an animal sanctuary in Wayanad, Kerala, India. Kabini river (a tributary of Cauvery river) flows through the sanctuary.

Source

- **Standard Reference Atlas**



Motivation: Papikonda National Park and others were in the news.

2021

Indus River System

With reference to the Indus river system, of the following four rivers, three of them put into one of them which joins the Indus direct. Among the following, which one is such river that joins the Indus direct?

- (a) Chenab
- (b) Jhelum
- (c) Ravi
- (d) Sutlej

Answer: D



Explanation:

- The Indus originates in the northern slopes of the Kailash range in Tibet near Lake Manasarovar. It follows a northwesterly course through Tibet. It enters Indian territory in Ladakh. The main tributaries of the Indus in India are Jhelum, Chenab, Ravi, Beas and Sutlej. Jhelum and Ravi join Chenab, Beas joins Sutlej, and then Sutlej and Chenab join to form Panjnad, 10 miles north of Uch Sharif in Muzaffargarh district. The combined stream runs southwest for approximately 44 miles and joins the Indus River at Mithankot. **Hence option (d) is the correct answer.**

Source

- NCERT, Class IX: Contemporary India-I
- NCERT, Class XI: India Physical Environment



2021

Saline Lakes

With reference to India, Didwana, Kuchaman, Sargol and Khatu are the names of

- (a) glaciers
- (b) mangrove areas
- (c) Ramsar sites
- (d) saline lakes

Answer: D



Explanation:

- **Playas are small lakes** with flat floors and undrained basins in which water collects after rains and evaporates quickly are called **playas**.
- **Didwana** is a playa. Other such playas are the **Kuchaman, the Sargol and the Khatu lakes**.
- Didwana, Kuchaman, Sargol and Khatu are names of towns with or near saline lakes. Didwana is a town in the Nagaur district of Rajasthan. It also has a famous Didwana salt-water lake of Rajasthan. Kuchaman lake occupies an area of about 8.5 sq.km in Nagaur District of Rajasthan. **Hence option (d) is the correct answer.**

Source

- Standard Atlas



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2021

Rivers of eastern Ghats

Consider the following rivers:

1. Brahmani
2. Nagavali
3. Subarnarekha
4. Vamsadhara

Which of the above rise from the Eastern Ghats?

- (a) 1 and 2
- (b) 2 and 4
- (c) 3 and 4
- (d) 1 and 3

Answer: B



Explanation:

- Nagavali River forms a prime river of north eastern Andhra Pradesh and southern Odisha. The origin of Nagavali River lies in the eastern slopes of the Eastern Ghats near Lakhbahal, located at an altitude of 1,300 metres in the Kalahandi district of the Indian state Odisha. **Hence option 2 is correct.**
- Vamsadhara river rises in the Eastern Ghats on the border of Kalahandi district and Rayagada district of Odisha. It runs for a distance of about 254 kilometres, where it joins the Bay of Bengal at Kalingapatnam, Andhra Pradesh. **Hence option 4 is correct.**
- Brahmani is a major seasonal river in the Odisha state of eastern India. The Brahmani is formed by the confluence of the Sankh and South Koel rivers near the major industrial town of Rourkela. The Sankh has its origins near the Jharkhand-Chhattisgarh border. The South Koel too arises in Jharkhand, near Lohardaga, on the other side of a watershed that also gives rise to the Damodar River. Both of these sources are in the Chota Nagpur Plateau. **Hence, option 1 is not correct.**
- Subarnarekha River flows through the Indian states of Jharkhand, West Bengal and Odisha. After originating near Ranchi it traverses a long distance through Ranchi and East Singhbhum. Thereafter, it flows for shorter distances through West Bengal and Odisha before joining the Bay of Bengal. **Hence, option 3 is not correct.**

Source

- School Atlas

2022

Wetlands

Consider the following pairs:

Wetland/Lake	-	Location
1. Hokera Wetland	-	Punjab
2. Renuka Wetland	-	Himachal Pradesh
3. Rudrasagar Lake	-	Tripura
4. Sasthamkotta Lake	-	Tamil Nadu

How many pairs given above are correctly matched?

- (a) Only one pair
- (b) Only two pairs
- (c) Only three pairs
- (d) All four pairs

Answer: B



Explanation:

- **Pair 1 is not correctly matched:** It is located at the northwest Himalayan biogeographic province of Kashmir, back of the snow-draped Pir Panchal.
- **Pair 2 is correctly matched:** Renuka Wetland is located in Himachal Pradesh. It is a natural wetland with freshwater springs and inland subterranean karst formations.
- **Pair 3 is correctly matched:** Rudrasagar Lake, also known as Twijilikma, is a lake located in Melaghar, Tripura, India. The Government of India's Ministry of Environment and Forest has identified Rudrasagar as one of the wetlands of National Importance for conservation and sustainable use based on its bio-diversity and socio economic importance.
- **Pair 4 is not correctly matched:** Sasthamcotta Lake or Sasthamkotta Lake, also categorized as a wetland, is the largest fresh water lake in Kerala. Sasthamkotta lake is also known as the Queen of Lakes.

Source

- Standard Atlas and Newspaper

Ramsar Sites in India as on December, 2024

JAMMU & KASHMIR

1. Hokera Wetland
2. Surinsar-Mansar Lakes
3. Wular Lake
4. Hygam Wetland Conservation Reserve
5. Shallbugh Wetland Conservation Reserve

PUNJAB

1. Beas Conservation Reserve
2. Hanke Lake
3. Kanjli Lake
4. Keshopur-Mani Community Reserve
5. Nangal Wildlife Sanctuary
6. Ropar Lake

HARYANA

1. Bhindawas Wildlife Sanctuary
2. Sultanpur National Park

RAJASTHAN

1. Keoladeo National Park
2. Sambhar Lake

GUJARAT

1. Nalsarovar Bird Sanctuary
2. Thol Lake Wildlife Sanctuary
3. Wadhvana Wetland
4. Khijadia Bird Sanctuary

MADHYA PRADESH

1. Bhoj Wetlands
2. Sirpur wetland
2. Sakhya Sagar Lake
4. Yashwant Sagar
5. Tawa Reservoir

MAHARASHTRA

1. Lonar Lake
2. Nandur Madhameshwar
3. Thane Creek

GOA

1. Nanda Lake

KARNATAKA

1. Ranganathittu Bird Sanctuary
2. Ankasamudra Bird Conservation Reserve
3. Aghanashini Estuary
4. Magadi Kere Conservation Reserve

KERALA

1. Asthamudi Wetland
2. Sasthamkotta Lake
3. Vembanad Kol Wetland

LADAKH

1. Tso Kar Wetland Complex
2. Tsomorini Lake

HIMACHAL PRADESH

1. Chandertal Wetland
2. Pong Dam Lake
3. Renuka Wetland

UTTARAKHAND

1. Asan Conservation Reserve

UTTAR PRADESH

1. Nawabganj Bird Sanctuary
2. Parvati Agra Bird Sanctuary
3. Saman Bird Sanctuary
4. Samaspur Bad Sanctuary
5. Sandi Bird Sanctuary 5
6. Sarsai Nawar Jheel
7. Sur Sarovar
8. Upper Ganga River
9. Haiderpur Wetland
10. Bakhira Bird Sanctuary

ASSAM

1. Deepar Beel

MANIPUR

1. Loktak Lake

MIZORAM

1. Pala wetland

TRIPURA

1. Rudrasagar Lake

WEST BENGAL

1. East Kolkata Wetlands
2. Sunderbans Wetland

BIHAR

1. Kanwar Taal or Kabar Taal Lake
2. Nagi Bird Sanctuary
3. Nakti Bird Sanctuary

ORISSA

1. Bhitarkanika Mangroves
2. Chilka Lake
3. Saltkosia Gorge
4. Tampara Lake
5. Hirakud Reservoir
6. Ansupa Lake

ANDHRA PRADESH

1. Kolleru Lake

TAMIL NADU

1. Point Calimere Wildlife and Bird Sanctuary
2. Karikili Bird Sanctuary
3. Pallikarandai Marsh Reserve Forest
4. Pichhavaram Mangrove
5. Gulf of Manar Marine Biosphere Reserve
6. Vembannur Wetland Complex
7. Chitrangudi Bird Sanctuary
8. Karaivetti Bird Sanctuary
9. Longwood Shola Reserve Forest

10. Suchindram Theroor Wetland Complex
11. Vaduvur Bird Sanctuary
12. Kanjirankulam Bird Sanctuary
13. Vellore Bird Sanctuary
14. Udhayamarthandapuram Bird Sanctuary
15. Koonthankutam Bird Sanctuary
16. Vedanthangal Bird Sanctuary
17. Nanjarayan Bird Sanctuary
18. Kazhuvelli Sanctuary

2022

Consider the following pairs:

Reservoirs

States

- | | | |
|-----------------|---|----------------|
| 1. Ghataprabha | - | Telangana |
| 2. Gandhi Sagar | - | Madhya Pradesh |
| 3. Indira Sagar | - | Andhra Pradesh |
| 4. Maithon | - | Chhattisgarh |

How many pairs given above are **not** correctly matched?

- (a) Only one pair
- (b) Only two pairs
- (c) Only three pairs
- (d) All four pairs

Answer: C



Explanation:

- **Pair 1 is not correctly matched:** Raja Lakhamagouda dam, also known as Hidkal dam, is a dam constructed across the Ghataprabha River in the Krishna River basin. It is situated at Hidkal village in Hukkeri Taluk of Belagavi district in North Karnataka.
- **Pair 2 is correctly matched:** Gandhisagar Dam is located in the Mandasaur, districts of the state of Madhya Pradesh. The Dam is constructed on the Chambal River.
- **Pair 3 is not correctly matched:** The Indira Sagar Dam is the largest dam in India, in terms of volume of water stored in the reservoir. It is located on the Narmada River at the town of Narmada Nagar, Punasa in the Khandwa district of Madhya Pradesh in India.
- **Pair 4 is not correctly matched:** Maithon is located on the banks of river Barakar. The Maithon Dam is located about 48 kms from the Coal City of Dhanbad, Jharkhand.

Source

- Standard Reference Atlas

Consider the following statements:

1. Jhelum River passes through Wular Lake.
2. Krishna River directly feeds Kolleru Lake
3. Meandering of Gandak River formed Kanwar Lake.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

Answer: A



Explanation:

- Wular Lake is the 2nd largest fresh-water lake of Asia, situated on the foothills of Haramuk Mountain. It is spread in a total area of 200 square km covering almost 24 km in length while its breadth is 10 km. The lake lies between the towns of Sopur and Bandippore, in Sangrama, near Baramulla Road. Main source of water for Wular Lake is River Jhelum. This lake also has a small island in its center called the 'Zaina Lank'. This island was constructed by King Zainul-Abi-Din. Wular Lake is also said to be a remnant of Satisar Lake that existed in ancient times. The premises of this lake also form a popular sunset point. **Hence statement 1 is correct.**
- Kolleru Lake is a natural eutrophic lake, situated between the two major river basins of the Godavari and the Krishna, fed by two seasonal rivers Budameru and Tamleru and a number of drains and channels, which functions as a natural flood balancing reservoir between the deltas of the two rivers. **Hence statement 2 is not correct.**
- Once a haven for migratory birds, the Kanwar lake in Bihar, Asia's largest freshwater oxbow lake, is today a dying wetland ecosystem. Kanwar jheel, as it is locally called, is located 22 km north-west of Begusarai town. It is a residual oxbow lake, formed due to meandering of Burhi Gandak river, a tributary of Ganga, in the geological past. It is the largest of a series of shallow permanent and ephemeral wetlands formed in the depression between River Burhi Gandak and palaeochannel of River Bagmati. Kanwar lake was declared a notified area under the Wildlife (Protection) Act of 1972. To check poaching of birds, it was declared a protected zone by the Bihar state government in 1986; the government of India declared it a bird sanctuary in 1989. The authorities had notified 15,000 acres in the area as wetland, which makes it six times bigger than Keoladeo National Park in Bharatpur, Rajasthan—another well-known bird sanctuary of India. **Hence statement 3 is not correct.**
- River Burhi Gandak rises in the West Champaran district from the spring of Someshwar hills. The river is known as Sikrchan in its upper reaches. It flows through the West Champaran district in south east direction and is joined by several tributaries like Masan, Harbora, Tilawe, Siriswa, Koria, Pasaha, Tiar Hahwa with their catchment in Someshwar hills. After the confluence of the Tiber river Sikrchan is known as Burhi. After receiving tributary Bainty Balan river flows in easterly direction and falls into the River Ganga about 7 km east from the district Khagaria. The Burhi Gandak Basin is bounded on the north by the Himalayas, on the south by the River

Ganga, on the east by the Kosi Basin and on the west by the Gandak Basin. The basin extends over the district West Champaran, Muzaffarpur, Samastipur and Khagaria of Bihar state.

- The river Gandhak originates at an altitude of 7620 m in the north of Dhaulagiri in Tibet near Nepal border in the shown peaks of the Himalayas. The river enters Indian territory near Tribeni town in Nepal. Thereafter the river flows in a southerly direction and forms a boundary between Uttar Pradesh and Bihar. It flows through the west Champaran, East Champaran, Muzaffarpur, Gopalganj, Siwan, Saren and Vaishali districts of Bihar and Gorakhpur and Deoria district of Uttar Pradesh before joining Ganga at Hajipur near Patna.

Source

- School Atlas and News Paper



Motivation: Wular, Kolleru and Kanwar lakes were in news due to various conservation efforts.

Note:

Important Lakes of India and their location is given in Annexure D of this document.

2024

Consider the following information

Waterfall

1. Dhuandhar
2. Hundru
3. Gersoppa

Region

1. Malwa
2. Chota Nagpur
3. Western Ghats

River

1. Narmada
2. Subarnarekha
3. Netravati

In how many of the above rows is the given information correctly matched?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

Answer: A



Explanation:

- **Information given in row 1 is not correctly matched:** The Dhuandhar Falls is a waterfall in Jabalpur district in the Indian state of Madhya Pradesh, it is created by river Narmada. Jabalpur is not located in the Malwa region. It is located in the Mahakoshal region. The Vindhya Range forms the northern boundary of the Mahakoshal region; north of the Vindhya Range lie the regions of Malwa to the northwest, Bundelkhand to the north, and Bagelkhand to the northeast. Chhattisgarh state lies to the east, and the Vidarbha region of Maharashtra state lies to the south across the Satpura Range.
- **Information given in row 2 is correctly matched:** The Hundru Falls Ranchi, Jharkhand is created on the course of the Subarnarekha River, where it falls from a height of 320 feet creating the highest waterfalls of the state. It is the 34th highest waterfall in India. The Subarnarekha originates from the Ranchi Plateau in Jharkhand Forming the boundary between West Bengal and Odisha in its lower course. It joins the Bay of Bengal forming an estuary between the Ganga and Mahanadi deltas. Its total length is 395 km.
- **Information given in row 3 is not correctly matched:** Gersoppa waterfall is created by the Sharavathi River falling from a height of 253 m (830ft), making it the second-highest plunge waterfall in India. Located near Sagara, Shimoga District of Karnataka state. It is also called by alternative names of Gerusoppa falls, Gersoppa Falls and Jogada Gundi. The Sharavathi River basin is situated in the central part of the Western Ghats. It originates at Ambuthirtha and flows towards west for about 132 km before joining the Arabian Sea at Honnavar.

Source

- <https://ranchi.nic.in/tourist-place/hundru-waterfall/#:~:text=The%20Hundru%20Falls%20Ranchi%20is%20created%20on%20the%20course%20of,water%20falls%20of%20the%20state.>
- <https://karnatakaturism.org/must-visit-waterfalls-in-karnataka>

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Consider the following pairs:

Waterfall

1. Dhuandhar
2. Shivasamudram
3. Kanchenjunga water falls

River

- Narmada River
- Cauvery River
- Tista River

How many pairs given above are correctly matched?

- (a) Only one pair
- (b) Only two pairs
- (c) All three pairs
- (d) None of the pairs

Indian Economic Geography (Minerals)

Total No. of questions: 10

2015

In India, the steel production industry requires the import of

- (a) Saltpetre
- (b) Rock phosphate
- (c) Coking coal
- (d) All of the above

Answer: C



Explanation:

- Saltpetre and Rock phosphate are not utilized by majority Steel companies in India. Further, India is a producer of saltpetre and usually does not import this product in large quantity.

2016

In which of the following regions of India are shale gas resources found?

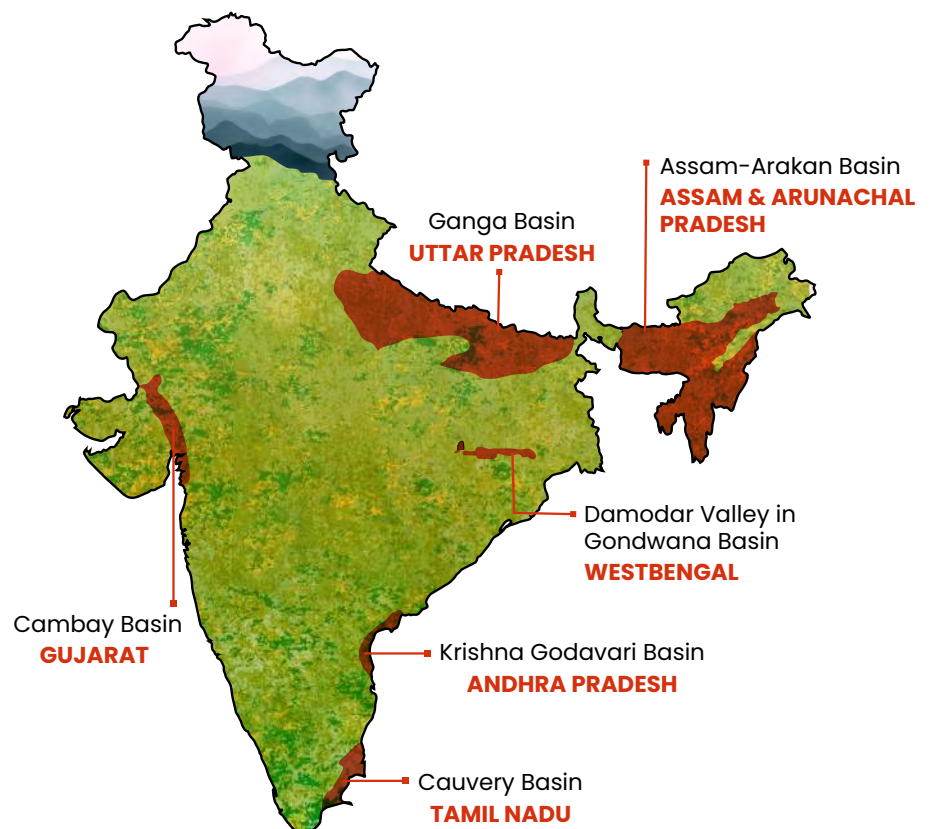
1. Cambay Basin
2. Cauvery Basin
3. Krishna-Godavari Basin

Select the correct answer using the code given below.

- (a) 1 and 2 only
- (b) 3 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

Answer: D

PROSPECTIVE BASINS FOR PHASE I SHALE OIL AND GAS EXPLORATION





Explanation:

- Shale gas is the natural gas that is trapped within shale formations. ONGC sought permission for drilling 11 exploratory wells for shale oil and shale gas in Cambay basin at Mehsana, Ahmedabad and Bharuch districts of Gujarat, one well in Cauvery basin at Nagapattinam in Tamil Nadu and five wells in KG Basin at East and West Godavari districts of Andhra Pradesh. Other regions in India where shale gas resources are found - Cauvery, Ganga & Assam and Assam - Arakan; Gondwana basin (including Damodar).

Source

- <http://pib.nic.in/newsite/PrintRelease.aspx?relid=121651>
- <http://www.thehindu.com/todays-paper/tp-business/ongc-to-drill-17-exploratory-wells-for-shale-gas-oil/article8517524.ece>

2016

Recently, which of the following States has explored the possibility of constructing an artificial inland port to be connected to sea by a long navigational channel?

- (a) Andhra Pradesh
- (b) Chhattisgarh
- (c) Karnataka
- (d) Rajasthan

Answer: D



Explanation:

- The government of Rajasthan is planning to develop an Inland Shipping Port at Jalore which would be connected to the Arabian Sea by developing a channel along the Kutch Creek.

Source

- <http://pib.nic.in/newsite/PrintRelease.aspx?relid=125015>

2016

Recently, India's first 'National Investment and Manufacturing Zone' was proposed to be set up in

- (a) Andhra Pradesh
- (b) Gujarat
- (c) Maharashtra
- (d) Uttar Pradesh

Answer: A



Explanation:

- Andhra Pradesh is set to house India's first national investment and manufacturing zone after the state assured the Centre of availability of 10 sq km of land in one place in Prakasham district. The imminent final approval for the NIMZ, which is expected to give a fillip to Make in India campaign, comes four years after the concept was mooted to boost manufacturing in the country and two years after the Department of Industrial Policy and Promotion gave an in-principal nod to Andhra Pradesh in this regard.

Source

- <http://www.thehindu.com/news/national/andhra-pradesh/nimz-works-gain-pace-in-prakasam-district/article8030363.ece>
- <http://www.newindianexpress.com/states/telangana/Protest-against-land-acquisition-spreads/2016/06/07/article3470099.ece>



Motivation: Frequently talk in the news.

2018

Non-Coal Mines

Consider the following statements:

1. In India, State Governments do not have the power to auction non-coal mines.
2. Andhra Pradesh and Jharkhand do not have gold mines.
3. Rajasthan has iron ore mines.

Which of the statements given above is/are correct?

- (a) 1 and 2
- (b) 2 only
- (c) 1 and 3
- (d) 3 only

Answer: D



Explanation:

- Unlike coal, the auction of mining licences of non-coal minerals is conducted by the respective state governments. **Hence statement 1 is not correct.**
- Iron ore mines are located in Bhilwara in Rajasthan. **Hence statement 3 is correct.**
- India has gold deposits spread across several states including Andhra Pradesh, Chhattisgarh, Jharkhand, Karnataka, Kerala, Madhya Pradesh, Tamil Nadu and Rajasthan. Governments of both Jharkhand and Andhra Pradesh have auctioned several gold mines in recent years. **Hence statement 2 is not correct.**

Source

- NCERTs, Class VIII, Class X and Class XII Geography

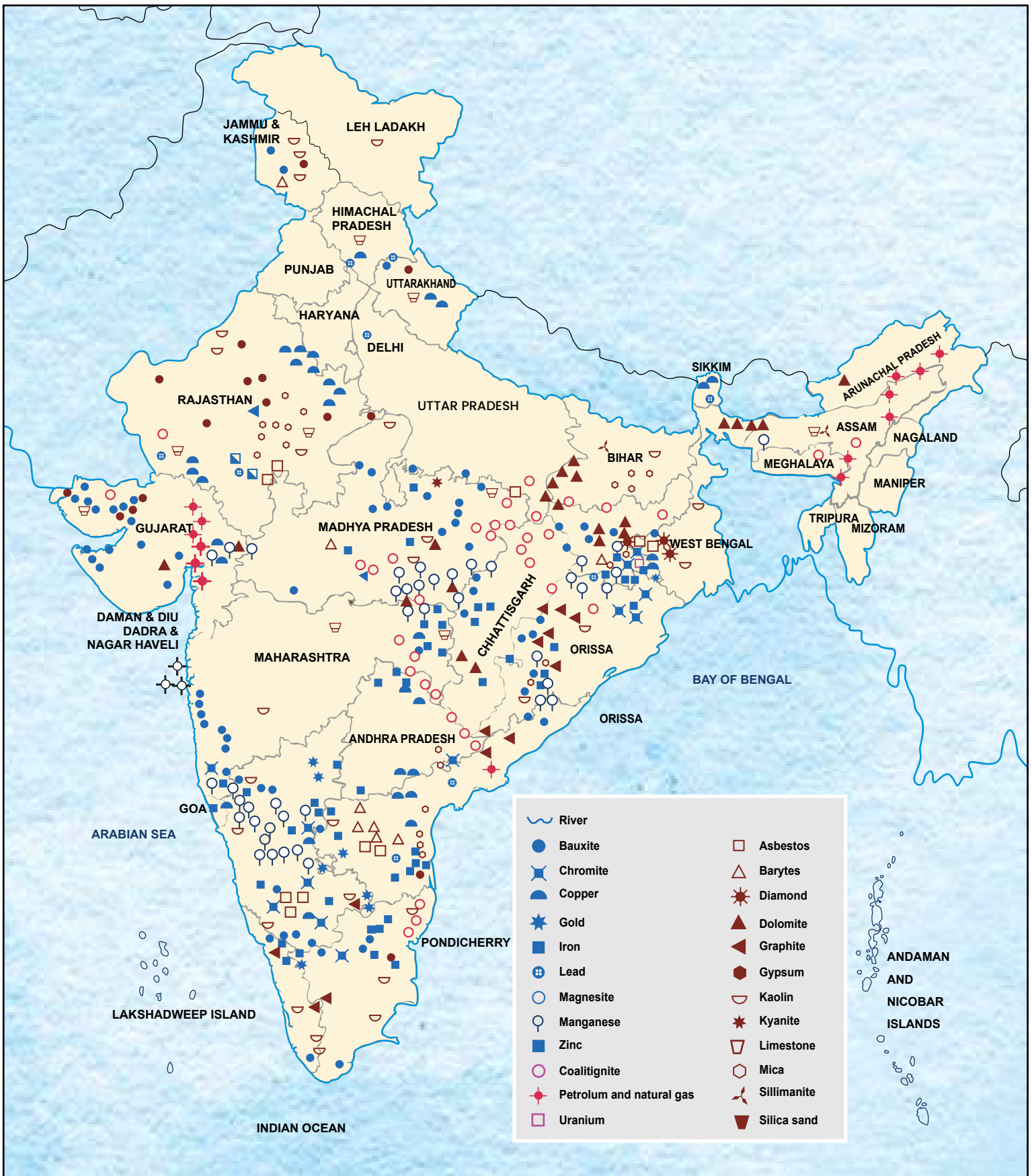


Motivation: Recent news related to auction of non-coal mines and gold mines.

Note:

Major Coal fields of India are mentioned in **Annexure H**.

Additional Information



2019

Minor Minerals

With reference to the management of minor minerals in India, consider the following statements:

1. Sand is a 'minor mineral' according to the prevailing law in the country.
2. State Governments have the power to grant mining leases of minor minerals, but the powers regarding the formation of rules related to the grant of minor minerals lie with the Central Government.
3. State Governments have the power to frame rules to prevent illegal mining of minor minerals.

Which of the statements given above is/are correct?

- (a) 1 and 3 only
- (b) 2 and 3 only
- (c) 3 only
- (d) 1, 2 and 3

Answer: A



Explanation:

- **Statement 1 is correct**– Sand is a minor mineral, as defined under section 3(e) of the Mines and Minerals (Development and Regulation) Act, 1957 (MMDR Act).
- **Statement 2 is not correct**– As per Section 15 of the Mines and Minerals (Development and Regulation) (MMDR) Act, 1957, State Governments have been empowered to frame rules in respect of minor minerals for regulating the grant of quarry leases, mining leases or other mineral concessions in respect of minor minerals and for purposes connected therewith.
- **Statement 3 is correct**– Section 23C of the MMDR Act, 1957 empowers state governments to frame rules to prevent illegal mining, transportation and storage of minerals and for purposes connected therewith. Control of illegal mining is, therefore, under the legislative and administrative jurisdiction of state governments.

Source

- Official website of Indian Bureau of Mines



Motivation: Recent news related to minor minerals and their regulations.

Additional Information

➤ Indian Bureau of Mines, 1948:

1. It is under the Ministry of Mines with responsibilities for conservation and systematic exploitation of mineral resources other than coal, petroleum and natural gas, atomic minerals and minor minerals.
2. IBM performs regulatory functions under MMDRA, 1957.

Minor Minerals	Major Minerals
1. According to MMDR Act, 1957 "Minor Minerals" means building stones, gravel, ordinary clay, ordinary sand other than sand used for prescribed purposes, and any other Central Govt notified mineral. 2. The Center has the power to notify "minor minerals" under MMDR Act, 1957. India has 86 minor minerals out of which 31 were added in 2015. 3. The power to frame law for minor minerals is entirely delegated to State Govts. Thus, the administrative and regulatory jurisdiction of minor minerals falls under State govts. 4. AP tops in the value of minor minerals produced in India followed by Gujarat, Maharashtra, Rajasthan and UP. 1.	1. Major minerals are those specified in the first schedule appended in the MMDR Act 1957 and the common major minerals are Lignite, Coal, Uranium, iron ore, gold etc. 2. There is no official definition for "major minerals" in the MMDR Act. Hence, whatever is not declared as a "minor mineral" may be treated as a major mineral. 3. The power to frame law for major minerals is dealt with by the Ministry of Mines under Center.

2020

Consider the following minerals:

1. Bentonite
2. Chromite
3. Kyanite
4. Sillimanite

In India, which of the above is/are officially designated as major minerals?

- (a) 1 and 2 only
- (b) 4 only
- (c) 1 and 3 only
- (d) 2, 3 and 4 only

Answer: D



Explanation:

- Bentonite is essentially a highly plastic clay containing not less than 85% clay mineral, montmorillonite. It gets its name from the place where its presence and usages were first discovered, Fort Benton, America. As per Govt. of India Notification S.O. 423(E) dated 10th February 2015 bentonite has been declared as 'Minor Mineral'.
- Chromite (Cr) is the single commercially viable ore of chromium which is chemically known as iron chromium oxide ($\text{Fe Cr}_2\text{O}_4$).
- Kyanite, Sillimanite and andalusite are unhydrous aluminum silicate minerals that have the same chemical formula Al_2O_3 but differ in crystal structure and physical properties. **Hence option (d) is the correct answer.**

Source

- Newspaper



Motivation: District Mineral Foundations are frequently seen in the news. It was also recently reported that of the total welfare funds collected, only 17 per cent were spent till 2017-end.

2022

With reference to India, consider the following statements :

1. Monazite is a source of rare earths.
2. Monazite contains thorium.
3. Monazite occurs naturally in the entire Indian coastal sands in India.
4. In India, Government bodies only can process or export monazite.

Which of the statements given above are correct ?

- (a) 1, 2 and 3 only
- (b) 1, 2 and 4 only
- (c) 3 and 4 only
- (d) 1, 2, 3 and 4

Answer: B



Explanation:

- **Statement 1 is correct:** Monazite is a primarily reddish-brown phosphate mineral that contains rare-earth elements. Due to variability in composition, monazite is considered a group of minerals.
- **Statement 2 is correct:** In addition to the rare-earth elements, thorium (Th) can be extracted from monazite ore. Thorium is a slightly radioactive metal, identical to uranium.
- **Statement 3 is not correct:** Atomic Minerals Directorate for Exploration and Research (AMD), a constitute unit of Department of Atomic Energy (DAE) has estimated the presence of 11.93 million tonnes of monazite resources in the beach sand mineral placer deposits along the coastal tracts of India.
- **Statement 4 is correct:** Monazite is a mineral mainly containing rare earths and thorium—a prescribed substance to be handled by the Department of Atomic Energy (DAE). Accordingly, Indian Rare Earths Ltd. (IREL) wholly owned by the Govt. of India, under the administrative control of the Dept. of Atomic Energy (DAE) utilises monazite mainly for production of rare earth compounds, and thorium, as needed in the Department of Atomic Energy.

Source

- Part of the answer is given in Geography NCERT of Class XII, Unit-III, Chapter-5



Motivation: Monazite and thorium reserves are often in the news.

Elimination Technique

Statement 3 is extreme. It is highly unlikely that monazite can be found along the entire coastal sands of India.

Ilmenite and rutile, abundantly available in certain coastal tracts of India, are rich sources of which one of the following?

- (a) Aluminium
- (b) Copper
- (c) Iron
- (d) Titanium

Answer: D



Explanation:

- India is endowed with large resources of heavy minerals which occur mainly along coastal stretches of the country and also in inland places. Heavy mineral sands comprise a group of seven minerals, viz, ilmenite, leucoxene (brown ilmenite), rutile, zircon, sillimanite, garnet and monazite.
- Ilmenite ($\text{FeO} \cdot \text{TiO}_2$) and rutile (TiO_2) are the two chief minerals of titanium. Titanium dioxide occurs in polymorphic forms as rutile, anatase (octahedrite) and brookite. **Hence option (d) is the correct Answer.**
- Ilmenite and rutile along with other heavy minerals are important constituents of beach sand deposits found right from Moti Daman-Umbrat coast (Gujarat) in the west to Odisha coast in the east.

Source

- Part of the answer is given in Geography NCERT Class X, Chapter - 5



Motivation: Beach minerals were in the news. The Union Ministry of Mines has proposed to amend the Mines and Minerals (Development and Regulations) Act, 1957, to remove certain atomic minerals including beach sand minerals from Part B of the act.

2023

Minerals

With reference to coal-based thermal power plants in India, consider the following statements :

1. None of them uses seawater.
2. None of them is set up in water-stressed districts.
3. None of them is privately owned.

How many of the above statements are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

Answer: D



Explanation:

- Thermal power plants are generally constructed near the sea coast to meet their requirement of cooling water. The warm water discharge from the thermal power plant is one of the major environmental concerns in view of the thermal pollution in the sea water. The temperature limit for the warm water discharge from the thermal power plant has to be monitored and controlled. Coastal Gujarat Power Limited (CGPL) is a coal fired power plant which operates (24 × 7) at an “once-through system” based sea water circulation for power generation. The used sea water is then discharged into the sea through an outlet channel.
- In India, several other coal-fired thermal power plants also employ seawater for a variety of functions, including cooling the condenser system. **Hence statement 1 is not correct.**
- The extraction of huge amounts of water by coal-fired thermal power plants may affect the short-term availability of water in India’s river basins, a recent study has pointed out.
- The study, titled Vulnerability assessment of thermal power plants in India under water stress conditions, has looked at the impact of water consumption by coal-fired thermal power plants on India’s growing water crisis.
- The major power grids in India — eastern, western, northern and southern — are dependent on the 14 river basins, with more plants distributed in the Indus, Ganga, Mahanadi, and Godavari basins. The catchments in India are under ‘high water stress’ due to the post-2010 geographical shift in the construction of new coal-fired power plants favouring northern and eastern river basins. **Hence statement 2 is not correct.**
- There are both privately and publicly owned coal-fired thermal power stations in India. Many private corporations have invested in the power industry and run coal-fired thermal power facilities, such as Coastal Gujarat Power Limited, a wholly owned subsidiary of Tata Power. **Hence statement 3 is not correct. Coal-based thermal power plants in India are mentioned in Annexure I.**

Source

- Newspaper

Indian Physiography

Total No. of questions: 11

2015

Consider the following States:

1. Arunachal Pradesh
2. Himachal Pradesh
3. Mizoram

In which of the above States do 'Tropical Wet Evergreen Forests' occur?

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Answer: C



Explanation:

- In India, evergreen forests are found on the eastern and western slopes of the Western Ghats in such states as Tamil Nadu, Karnataka, Kerala and Maharashtra. And also found in Assam, Arunachal Pradesh, Mizoram, Meghalaya, Nagaland, Tripura, West Bengal and Andaman and Nicobar Islands.

Source

- Ncert Class 11th India Physical Environment Page 58

2015

In a particular region in India, the local people train the roots of living tree into robust bridges across streams. As the time passes these bridges become stronger. These unique 'Living Root Bridges' are found in

- (a) Meghalaya
- (b) Himachal Pradesh
- (c) Jharkhand
- (d) Tamil Nadu

Answer: A



Explanation:

- Roots of the rubber tree — *Ficus Elastica* are the pavements for these natural bridges in the thick forests of Khasi Hills district. It is interesting to note that the districts in Meghalaya are named after the tribes in that area, like East Garo Hills district, West Jaitia Hills district and so on. Strong secondary roots of this rubber tree are manipulated to grow horizontally through the tunnels of hollowed betel nut trunks across the fast-flowing rivers.

Source

- <http://www.thehindu.com/todays-paper/tp-features/tp-youngworld/living-root-bridges-of-meghalaya/article4162886.ece>

2017

Indian Physiography

Consider the following statements:

1. In India, the Himalayas are spread over five states only.
2. Western Ghats are spread over five states only.
3. Pulicat Lake is spread over two States only.

Which of the statements given above is/are correct?

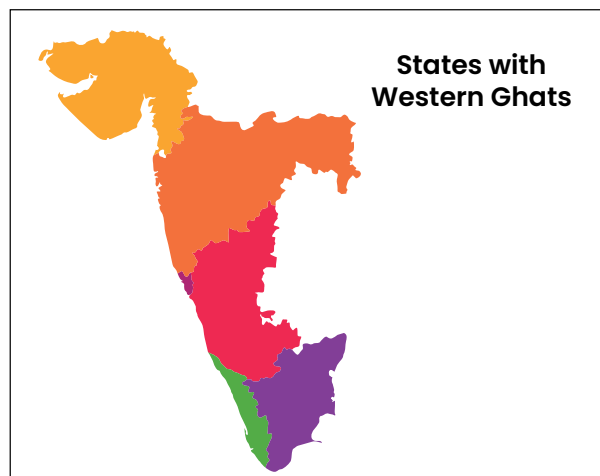
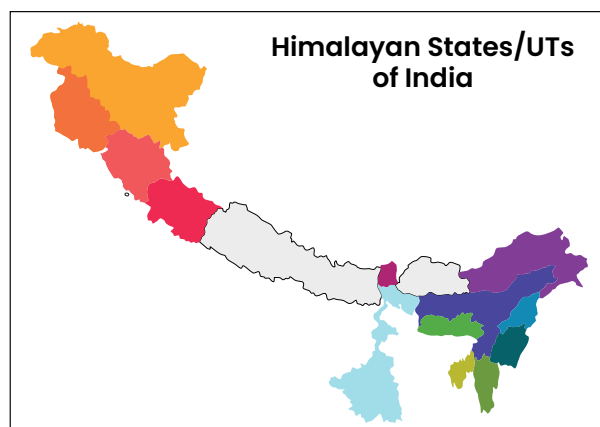
- (a) 1 and 2 only
- (b) 3 only
- (c) 2 and 3 only
- (d) 1 and 3 only

Answer: B



Explanation:

- **Statement 1 is not correct:** In India, the Himalayas are spread over Jammu & Kashmir, Himachal Pradesh, Uttarakhand, Sikkim, Assam, West Bengal, Arunachal Pradesh. Some extensions of Shiwaliks are also present in Punjab and Haryana.
- **Statement 2 is not correct:** Western Ghats are spread over six states, namely- Gujarat, Maharashtra, Goa, Karnataka, Kerala, Tamil Nadu.
- **Statement 3 is correct:** Pulicat, the second largest brackish water lake in India, lies on the border of Andhra Pradesh and Tamil Nadu. Over 40% of it lies in Andhra Pradesh and 60% in Tamil Nadu.



Source

➤ NCERTs, Class IX and XI Geography

2017

Indian Physiography

Which of the following is geographically closest to Great Nicobar?

- (a) Sumatra
- (b) Borneo
- (c) Java
- (d) Sri Lanka

Answer: A



Explanation:

- Distance from Great Nicobar to The island of Sumatra 1,192 km.
- Distance from Great Nicobar to Borneo is 2,398 km.
- Distance from Great Nicobar to Java 1817 km.
- Distance from Great Nicobar to Sri Lanka 1437 km.
- Hence option (a) is the correct answer.

Source

➤ School Atlas



Motivation: Important from the view point of India's territorial extent.

2017

Indian Physiography

From the ecological point of view, which one of the following assumes importance in being a good link between the Eastern Ghats and the Western Ghats?

- (a) Sathyamangalam Tiger Reserve
- (b) Nallamala Forest
- (c) Nagarhole National Park
- (d) Seshachalam Biosphere Reserve

Answer: A



Explanation:

- Sathyamangalam forest range is a significant wildlife corridor in the Nilgiri Biosphere Reserve between the Western Ghats and the rest of the Eastern Ghats. Located in the Erode district of Tamil Nadu, the Sathyamangalam Tiger Reserve is the largest wildlife sanctuary in the state. With a forest area of 1,411.6 square kilometres, the reserve is a part of the Nilgiri Biosphere Reserve and lies between the Western Ghats and the rest of the Eastern Ghats. Situated in a key position between the Western and Eastern Ghats, the Sathyamangalam Tiger Reserve boasts of a wide variety of fauna.
- The reserve also adjoins four other protected areas -- Biligiriranga Swamy Temple Wildlife Sanctuary, Sigur Plateau, Mudumalai National Park and Bandipur National Park. **Hence option (a) is the correct answer.**

Source

- SchoolAtlasandnewspaper



Motivation: The Sathyamangalam Tiger Reserve was in news for an increase in Bear population. Also in December 2016, Sathyamangalam Tiger Reserve ecotourism plan was revived.

Note:

Important Protected Areas of Western Ghats, as depicted on a map, is given in Annexure G of this document.

2017

Indian Physiography

At one of the place in India, if you stand on the seashore and watch the sea, 'you will find that the sea water recedes from the shore line a few kilometres and comes back to the shore, twice a day, and you can actually walk on the sea floor when the water recedes. This unique phenomenon is seen at

- (a) Bhavnagar
- (b) Bheemunipatnam
- (c) Chandipur
- (d) Nagapattinam

Answer: C



Explanation:

- Chandipur is located in Balasore district. This beach is unique in the whole world. No where on earth you can find a beach where the sea water retreats inside the sea from 1km to 5Km every day and it again comes back to the shore slowly during high tide. This happens twice every day. It is also known as Odisha's Hide and Seek Beach. Bhavnagar has the highest tidal range in India. **Hence option (c) is the correct answer.**

Source

- Newspaper



Motivation: It is a phenomenon unique only to Chandipur beach, India. Chandipur-on-sea is also the location of the Indian Army's Integrated Test Range and is often seen in news.

2018

Consider the following statements:

1. The Barren Island volcano is an active volcano located in the Indian territory.
2. Barren Island lies about 140 km east of Great Nicobar.
3. The last time the Barren Island volcano erupted was in 1991 and it has remained inactive since then.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 and 3
- (c) 3 only
- (d) 1 and 3

Answer: A



Explanation:

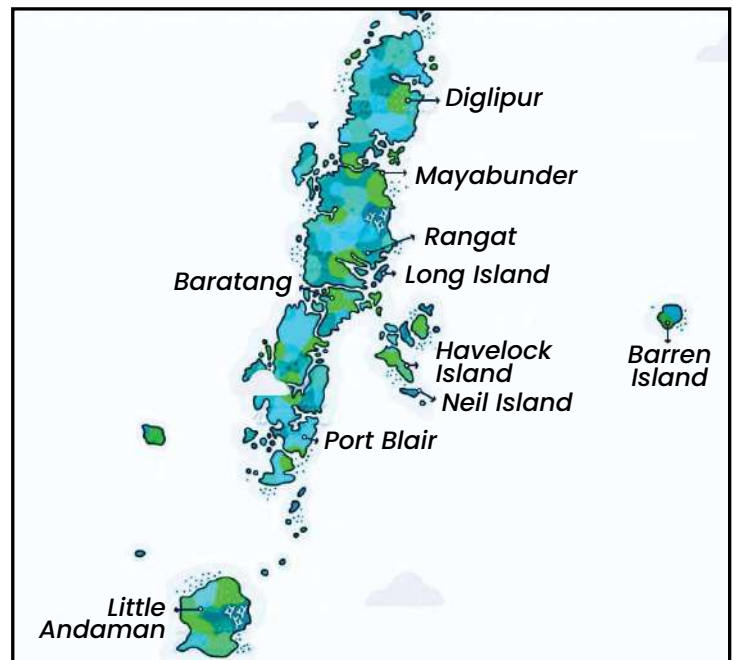
- Barren Island is India's only active volcano. It is located 140kms to the east of Port Blair (lies in South Andaman) while Great Nicobar is southernmost region of Andaman & Nicobar and lies further south of Port Blair.
- The Barren Island volcano had been lying dormant for more than 150 years until it saw a major eruption in 1991. Since then it has shown intermittent activity, including eruptions in 1995, 2005 and 2017. **Hence option (a) is the correct answer.**

Source

- NCERT, Class IX Geography & Newspaper



Motivation: Barren island volcano erupted in January 2017



2022

Indian Physiography

Gandikota canyon of South India was created by which one of the following rivers.

- (a) Cauvery
- (b) Manjira
- (c) Pennar
- (d) Tungabhadra

Answer: C



Explanation:

- Over time, the Pennar River carved the pink granite rock of Erramala Hills and formed Gandikota Canyon. This natural structure is often compared to the Grand Canyon of Arizona, USA. Gandikota is a village and historical fort on the right bank of the Pennar river, in Kadapa district, Andhra Pradesh, India. The fort was the centre of power for various dynasties, such as the Kalyani Chalukyas, Pemmasani Nayakas, and the Golconda Sultanate. **Hence option (c) is the correct answer.**

Source

- School Atlas

2022

Consider the following pairs :

Peak	Mountains
1. Namcha Barwa	Garhwal Himalaya
2. Nanda Devi	Kumaon Himalaya
3. Nokrek	Sikkim Himalaya

Which of the pairs given above is/are correctly matched ?

- (a) 1 and 2
- (b) 2 only
- (c) 1 and 3
- (d) 3 only

Answer: B



Explanation:

- **Pair 1 is not correctly matched:** Namcha Barwa or Namcha Barwa is a mountain peak lying in Tibet in the region of Pemako. It is the easternmost section of the Himalaya in southeastern Tibet and northeastern India. The Garhwal Himalayas are mountain ranges located in the Indian state of Uttarakhand.
- **Pair 2 is correctly matched:** Nanda Devi is the second-highest mountain in India, after Kangchenjunga. It is situated in the Chamoli district of Uttarakhand. Kumaun Himalayas, west-central section of the Himalayas in northern India, extends 200 miles (320 km) from the Sutlej River east to the Kali River. It rises to 7,817 metres at Nanda Devi, the range's highest peak, and to 7,756 metres at Kamet, near the Chinese border.
- **Pair 3 is not correctly matched:** The Nokrek Peak is located in the northeast of India on the Tura Range, which forms part of the Meghalaya Plateau. The entire area is mountainous and Nokrek is the highest peak of the Garo hills, rising up 1,412 metres. Important Mountain Peaks of India and Hills of Peninsular India are shown in Annexure J.

Source

- Standard Reference Atlas

2023

Indian Physiography

Which one of the following is the best example of repeated falls in sea level, giving rise to present-day extensive marshland?

- (a) Bhitarkanika Mangroves
- (b) Marakkanam Salt Pans
- (c) Naupada Swamp
- (d) Rann of Kutch

Answer: D



Explanation:

- Rann of Kutch is a massive expanse of cracked earth, inland from the sea, that promises to take your breath away. The nothingness for miles is both nerve wracking and stunning with small oasis of water bodies and shrub forests doubling up as homes for pink flamingos and wild asses. Tribal hamlets with cylindrical mud bhungas (huts) are the epicenter for Kutchi embroidery, tie and dye, leatherwork, pottery, bell metal craft and the famous Rogan painting by the only surviving expert family. Approximately 200 km east of the Rann, is the Little Rann of Kutch, which houses the 4953-sq-km Wild Ass Sanctuary. It houses the only remaining population of the chestnut-coloured Indian wild ass (khur), as well as blue-bulls, blackbuck and chinkara.
- The area was once a sprawling shallow of the Arabian Sea until a constant geological shift closed off the connection with the sea. Over the years, the region eventually became a seasonal marshy salt desert. During monsoons, the marsh fills up with water and the wetland extends from the Gulf of Kutch on the west through to the Gulf of Cambay on the east. In the summers, the water dries to create a crunchy based bed of white salty land. **Hence option (d) is the correct Answer.**

Source

- Indian Geography by Mazid Hussain

2023

Indian Physiography

Consider the following statements:

1. Amarkantak Hills are at the confluence of Vindhya and Sahyadri Ranges.
2. Biligirirangan Hills constitute the easternmost part of Satpura Range.
3. Seshachalam Hills Constitute the southernmost part of Western Ghats.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

Answer: D



Explanation:

- The Western Ghats, also known as the Sahyadri mountain range, is a mountain range that covers an area of 160,000 km² in a stretch of 1,600 km parallel to the western coast of the Indian peninsula, traversing the states of Gujarat, Maharashtra, Goa, Karnataka, Kerala and Tamil Nadu.
- Amarkantak is situated in Anuppur district of the state of Madhya Pradesh at an altitude of 1065m at the meeting place of the Vindhya and Satpura mountain ranges amidst sylvan surroundings. It has an average elevation of 1048m. The mighty Narmada river as well the Son, the Mahanadi, and Arnadoh which is a major tributary of the Godavari all arise in the Amarkantak plateau. **Hence statement 1 is not correct.**
- The Biligirirangana Hills is a hill range situated in south-western Karnataka, at its border with Tamil Nadu in South India. The area is called Biligiri Ranganatha Swamy Temple Wildlife Sanctuary or simply BRT Wildlife Sanctuary. However, the Satpura Range is a range of hills in central India. The range rises in eastern Gujarat running east through the border of Maharashtra and Madhya Pradesh and ends in Chhattisgarh. **Hence statement 2 is not correct.**
- Seshachalam Hills also known as Tirumala Range are hilly ranges part of the Eastern Ghats in southern Andhra Pradesh state, in southeastern India. The Seshachalam hill ranges are predominantly present in Tirupati district of the Rayalaseema region in Andhra Pradesh. **Hence statement 3 is not correct.**

Source

- School Atlas



Motivation: Fundamental Questions often Asked by UPSC on Indian Physiography.

Elimination Technique:

Seshachalam Biosphere Reserve is in Andhra Pradesh. Biligiri Ranganatha Swamy Temple Tiger Reserve is located in Karnataka. Hence statements 2 and 3 can be eliminated.

Map Base Questions

Total No. of questions: 11

India Map

02

World Map

02

Note:

Some world map based questions are covered under International Relations.

2015

Which one of the following countries of South-West Asia does not open out to the Mediterranean Sea?

- (a) Syria
- (b) Jordan
- (c) Lebanon
- (d) Israel

Answer: B



Explanation:

- Hence, option B is the correct answer

Source

- <http://www.thehindu.com/opinion/editorial/a-looming-refugee-crisis/article7326417.ece>



2015

Which one of the following pairs of States of India indicates the easternmost and westernmost State?

- (a) Assam and Rajasthan
- (b) Arunachal Pradesh and Rajasthan
- (c) Assam and Gujarat
- (d) Arunachal Pradesh and Gujarat

Answer: D



Explanation:

- The mainland of India extends from Kashmir in the north to Kanyakumari in the south and Arunachal Pradesh in the east to Gujarat in the west.

Source

- NCERT Class 11th India Physical Environment Page 2
- Fundamental question from NCERT



2015

Consider the following pairs:

Place of Pilgrimage	Location
1. Srisaillam	Nallamala Hills
2. Omkareshwar	Satmala Hills
3. Pushkar	Mahadeo Hills

Which of the above pairs is/are correctly matched?

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Answer: A



Explanation:

- Srisaillam is reputed for the shrine of Lord Mallikarjuna on the flat top of Nallamala Hills (Andhra Pradesh). Omkareshwar is situated on the Mandhata hills in Vindhya Mountains of Madhya Pradesh. The Pushkar valley is formed between the Aravalli hills.
- Hence, pair 2 and 3 are not correctly matched and pair 1 is correctly matched.

Source

- Fundamental question from NCERT

2016

Consider the following pairs:

Famous Place	Region
1. Bodhgaya:	Baghelkhand
2. Khajuraho:	Bundelkhand
3. Shirdi:	Vidarbha
4. Nasik (Nashik):	Malwa
5. Tirupati:	Rayalaseema

Which of the pairs given above are correctly matched?

- (a) 1, 2 and 4
- (b) 2, 3, 4 and 5
- (c) 2 and 5 only
- (d) 1, 3, 4 and 5

Answer: C



Explanation:

- Bagelkhand or Baghelkhand is a region and also a mountain range in central India that covers the northeastern regions of Madhya Pradesh and a small area of western Uttar Pradesh. Bodhgaya is in Bihar.
- The Khajuraho group of monuments was built during the rule of the Rajput Chandela dynasty. The building activity started almost immediately after the rise of their power, throughout their kingdom to be later known as Bundelkhand.
- Vidarbha is the eastern region of the Indian state of Maharashtra, comprising Nagpur Division and Amravati Division. Shirdi is a town and falls under Ahmednagar District in Maharashtra. It is not in Vidarbha region.
- Tirupati lies in the Rayalaseema region of Andhra Pradesh.
- Nasik Region: The Western Ghats of Sahyadri range stretches from north to south across the western portion of the Nasik district. The larger eastern portion of the district, which lies on the Deccan Plateau.

Source

- ATLAS, NCERT Geography books

2017

The Mediterranean Sea is a border of which of the following countries?

1. Jordan
2. Iraq
3. Lebanon
4. Syria

Select the correct answer using the code given below.

- (a) 1, 2 and 3 only
- (b) 2 and 3 only
- (c) 3 and 4 only
- (d) 1, 3 and 4 only

Answer: C





Explanation:

- The countries with coastlines on the Mediterranean Sea are Albania, Algeria, Bosnia and Herzegovina, Croatia, Cyprus, Egypt, France, Greece, Israel, Italy, Lebanon, Libya, Malta, Morocco, Monaco, Montenegro, Slovenia, Spain, Syria, Tunisia and Turkey. In addition, the Gaza Strip and the British Overseas Territories of Gibraltar and Akrotiri and Dhekelia have coastlines on the sea.
- On the Eastern shore, Turkey, Syria, Lebanon and Israel have their coastline on the Mediterranean Sea. **Hence option (c) is the correct answer.**



Motivation: These countries are often seen in news as they are regions under conflict facing acute refugee crisis.

2017

If you travel by road from Kohima to Kottayam, what is the minimum number of States within India through which you can travel, including the origin and the destination?

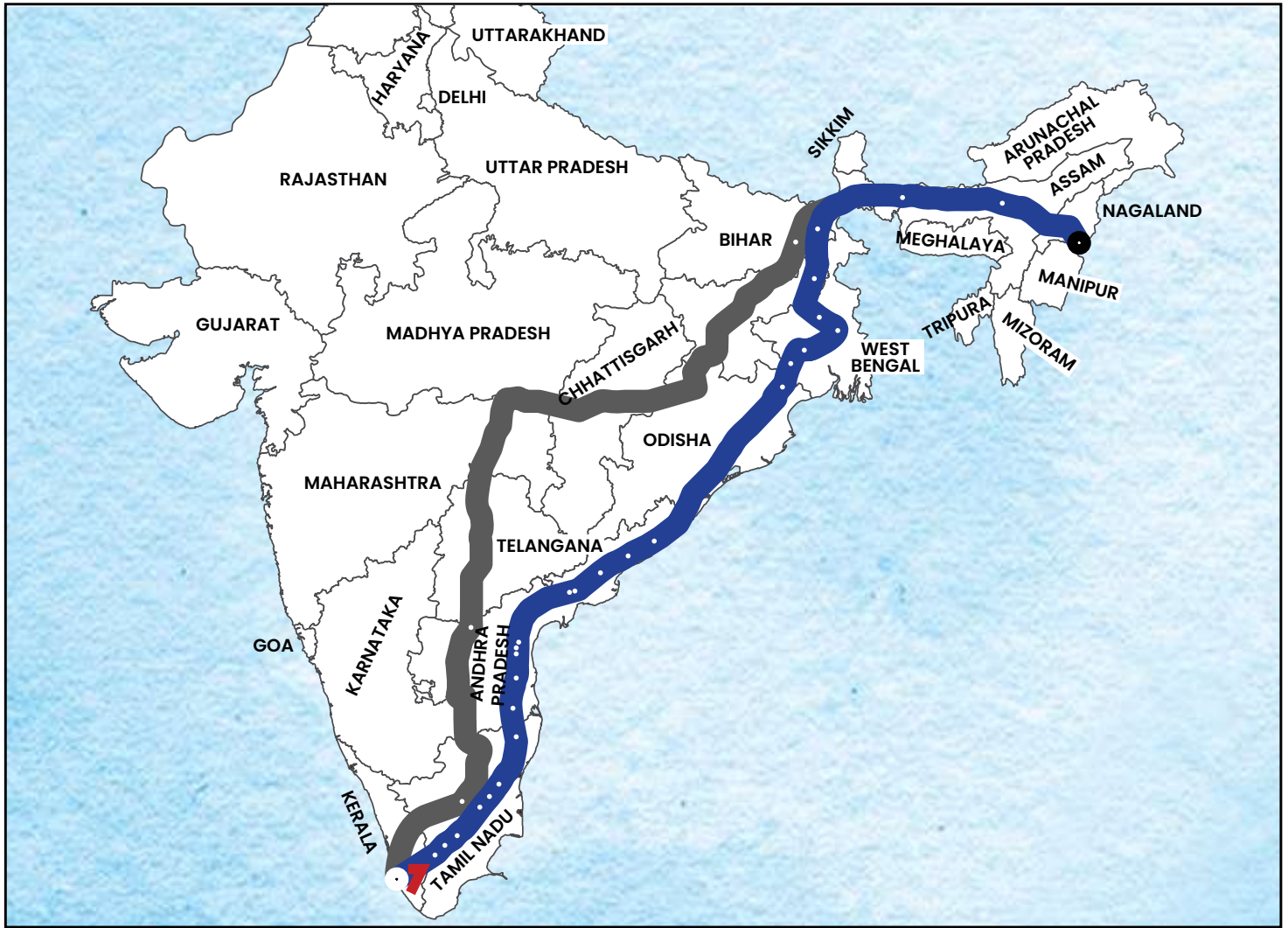
- (a) 6
- (b) 7
- (c) 8
- (d) 9

Answer: B



Explanation:

- The shortest route from Kohima (Nagaland) to Kottayam (Kerala) passes through the following states:
- Nagaland-Assam-West Bengal-Odisha-Andhra Pradesh-Tamil Nadu-Kerala. **Hence option (b) is the correct answer.**



Motivation: Cabinet approved North East Road Network Connectivity Project Phase I.

2018

Among the following cities, which one lies on a longitude closest to that of Delhi?

- (a) Bengaluru
- (b) Hyderabad
- (c) Nagpur
- (d) Pune

Answer: A



Explanation:

- Delhi and Bengaluru are located almost on the same longitude.
- Longitude of Delhi- 77.1025° E
- Longitude of Bengaluru- 77.5946° E
- longitude of Hyderabad- 78.4867° E
- longitude of Nagpur- 79.0882° E
- longitude of pune- 73.8567° E
- **Hence option (a) is the correct answer.**



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Consider the following pairs:

Sea		Bordering country
1. Adriatic Sea	:	Albania
2. Black Sea	:	Croatia
3. Caspian Sea	:	Kazakhstan
4. Mediterranean Sea	:	Morocco
5. Red Sea	:	Syria

Which of the pairs given above are correctly matched?

- (a) 1, 2 and 4 only
- (b) 1, 3 and 4 only
- (c) 2 and 5 only
- (d) 1, 2, 3, 4 and 5

Answer: B



Explanation:

- The Adriatic Sea is a body of water separating the Italian Peninsula from the Balkan peninsula. The countries with coasts on the Adriatic are Albania, Bosnia and Herzegovina, Croatia, Italy, Montenegro and Slovenia. **Hence pair 1 is correctly matched.**
- Black Sea is a large inland sea situated at the southeastern extremity of Europe. It is bordered by Ukraine to the north, Russia to the northeast, Georgia to the east, Turkey to the south, and Bulgaria and Romania to the west. **Hence pair 2 is not correctly matched.**
- Caspian Sea is the world's largest inland body of water, variously classed as the world's largest lake or a full-fledged sea. It is an endorheic basin (a basin without outflows) located between Europe and Asia. It is bounded by Kazakhstan to the northeast, Russia to the northwest, Azerbaijan to the west, Iran to the south, and Turkmenistan to the southeast. **Hence pair 3 is correctly matched.**
- The Mediterranean Sea is an intercontinental sea that stretches from the Atlantic Ocean on the west to Asia on the east and separates Europe from Africa. The countries surrounding the Mediterranean in clockwise order are Spain, France, Monaco, Italy, Slovenia, Croatia, Bosnia and Herzegovina, Montenegro, Albania, Greece, Turkey, Syria, Lebanon, Israel, Egypt, Libya, Tunisia, Algeria, and Morocco. **Hence pair 4 is correctly matched.**
- Red Sea is a narrow strip of water extending southeastward from Suez, Egypt, for about 1,200 miles (1,930 km) to the Bab el-Mandeb Strait, which connects with the Gulf of Aden and thence with the Arabian Sea. The six countries bordering the Red Sea proper are – Saudi Arabia, Yemen, Egypt, Sudan, Eritrea and Djibouti. **Hence pair 5 is not correctly matched.**

Source

- School Atlas



Motivation: Map based basic questions are often covered by UPSC.

2024

The longest border between any two countries in the world is between:

- (a) Canada and the United States of America
- (b) Chile and Argentina
- (c) China and India
- (d) Kazakhstan and Russian Federation

Answer: A



Explanation:

- Canada's border with the United States is the world's longest international border, at 8,890 km. This compares with the 6,846-km boundary between Russia and Kazakhstan and the 5,308-km frontier between Chile and Argentina. India shares 3488 Km of border with China that runs along the Union Territory of Ladakh, States of Himachal Pradesh, Uttarakhand, Sikkim and Arunachal Pradesh.
- Hence option (a) is the correct answer.

Source

- <https://indianexpress.com/article/cities/chandigarh/canada-calling-of-illegal-migrants-from-us-indian-students-in-fake-colleges-and-jagmeet-singh-tok-loss-8479577/>



Motivation: News related to illegal migrants was seen in the news.

With reference to the Himalayan rivers Joining the Ganga downstream of Prayagraj from West to East, which one of the following sequences is correct?

- (a) Ghaghara - Gomati - Gandak - Kosi
- (b) Gomati - Ghaghara - Gandak - Kosi
- (c) Ghaghara - Gomati - Kosi - Gandak
- (d) Gomati - Ghaghara - Kosi - Gandak

Answer: B



Explanation:

➤ The Gomti is an alluvial river that originates from the Gomat Taal, otherwise called Fulhaar Jheel, found near the Madho Tanda in Pilibhit district in Uttar Pradesh, India. After flowing through an incised valley southwards through the districts of Sitapur, Lucknow, Barabanki, Sultanpur, and Jaunpur, it joins River Ganga near Varanasi.

The Ganges-Brahmaputra Basin



- The Ghaghara originates in the glaciers of Mapchachungo. After collecting the waters of its tributaries – Tila, Seti and Beri, it comes out of the mountain, cutting a deep gorge at Shishapani. The river Sarada (Kali or Kali Ganga) joins it in the plain before it finally meets the Ganga at Chhapra.
- The Gandak comprises two streams, namely Kaligandak and Trishulganga. It rises in the Nepal Himalayas between the Dhaulagiri and Mount Everest and drains the central part of Nepal. It enters the Ganga plain in Champaran district of Bihar and joins the Ganga at Sonpur near Patna.
- The Kosi is an antecedent river with its source to the north of Mount Everest in Tibet, where its main stream Arun rises. After crossing the Central Himalayas in Nepal, it is joined by the Son Kosi from the West and the Tamur Kosi from the east. It forms Sapt Kosi after uniting with the river Arun. The Kosi river joins Ganga in the Katihar district of Bihar. **Hence option (b) is the correct answer.**

Source

- All India Abhyaas Test Series Test Code: 4360
- In the map shown, rivers labelled 1, 2, 3, 4, and 5 respectively are
 - (a) Ghaghara, Ganga, Gandak, Yamuna and Kosi
 - (b) Rapti, Gomti, Kosi, Ramganga and Mahananda
 - (c) Rapti, Ghaghara, Gandak, Sharda and Kosi
 - (d) Ghaghara, Sharda, Kosi, Gomti and Teesta

2024

Consider the following countries:

1. Finland
2. Germany
3. Norway
4. Russia

How many of the above countries have a border with the North Sea?

- (a) Only one
- (b) Only two
- (c) Only three
- (d) All four

Answer: B



Explanation:

- The North Sea in northern Europe is part of the Atlantic Ocean. It is bordered by the United Kingdom and Norway to the west, Denmark to the south, Germany, the Netherlands, Belgium, and France to the east, and the Faroe Islands and Norway's Svalbard archipelago to the north. The North Sea connects to the Atlantic Ocean via the English Channel in the southwest and the Baltic Sea in the east via the Kattegat and Skagerrak straits. **Hence option (b) is the correct answer.**

Source

- <https://www.worldatlas.com/seas/north-sea.html>



Oceanography

Total No. of questions: 05

2015

What explains the eastward flow of the equatorial counter-current?

- (a) The Earth's rotation on its axis
- (b) Convergence of the two equatorial currents
- (c) Difference in salinity of water
- (d) Occurrence of the belt of calm near the equator

Answer: A



Explanation:

- As the Earth rotates from west to east, the trade winds push water westward, piling it up in the western part of the ocean basins. This creates a slope in the sea surface, with higher water levels in the west and creates an eastward flow of the equatorial counter-current

Source

- <http://www.britannica.com/science/equatorial-countercurrent>

2015

Tides occur in the oceans and seas due to which among the following

1. Gravitational force of the Sun
2. Gravitational force of the Moon
3. Centrifugal force of the Earth

Select the correct answer using the codes given below.

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Answer: D



Explanation:

- The periodical rise and fall of the sea level, once or twice a day, is mainly due to the attraction of the Sun and Moon. Another factor is centrifugal force, which is the force that acts to counterbalance gravity.

Source

- NCERT Class 11th Fundamental of Physical Geography Page 121

2018

Coral Reefs

Consider the following statements:

1. Most of the world's coral reefs are in tropical waters.
2. More than one-third of the world's coral reefs are located in the territories of Australia, Indonesia and Philippines.
3. Coral reefs host far more number of animal phyla than house hosted by tropical rainforests.

Which of the statements given above is/are correct?

- (a) 1 and 2 only
- (b) 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Answer: D



Explanation:

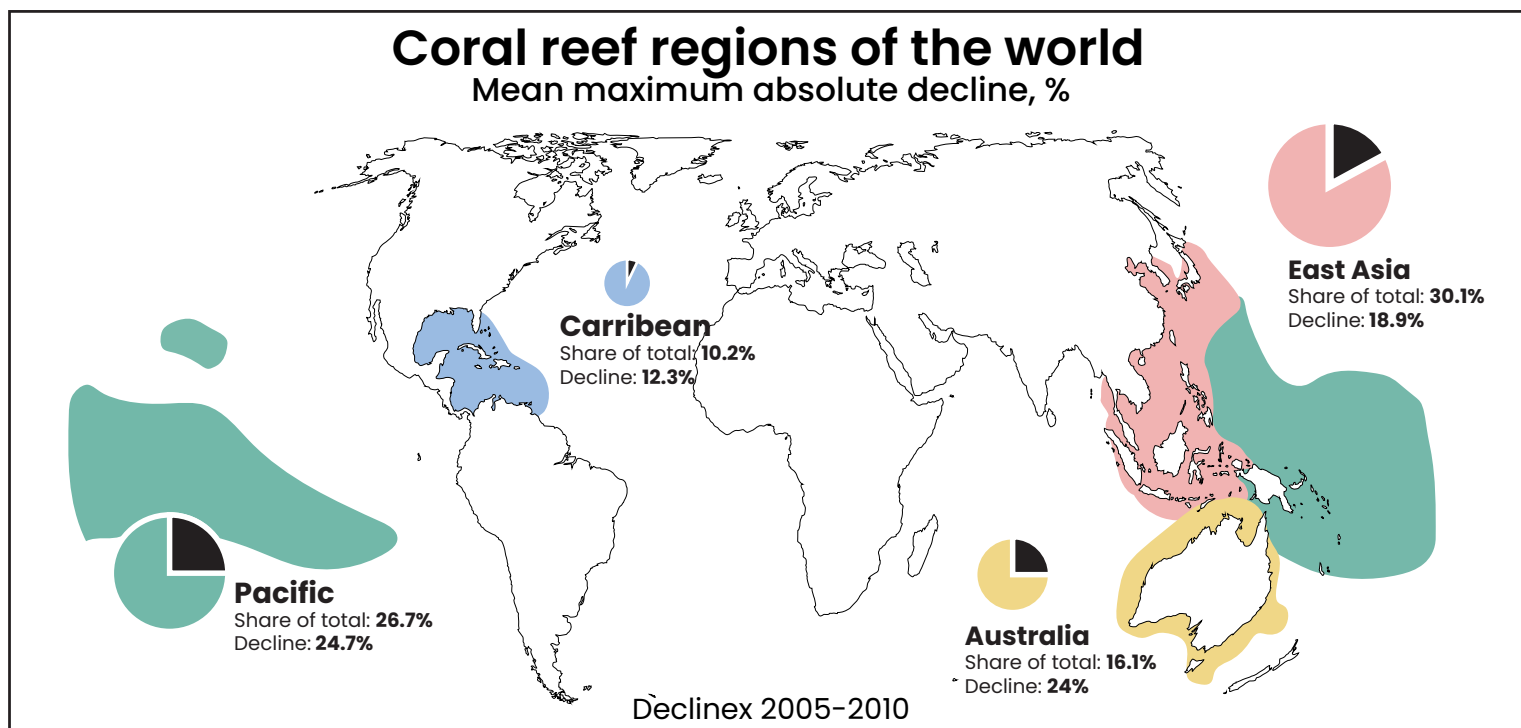
- Most coral reefs are located in tropical shallow waters less than 50 meters deep, in the western regions of the Indian ocean, Pacific Ocean and Greater Caribbean. **Hence statement 1 is correct.**
- Global distribution of corals:
 - Australia-17%
 - Indonesia-16%
 - Philippines-9%
 - So, Australia, Indonesia and Philippines together host more than one-third of the world's corals. **Hence statement 2 is correct.**
- 32 of the 34 recognised animal Phyla are found on coral reefs compared to only 9 Phyla in tropical rainforests. **Hence statement 3 is correct.**

Source

➤ Official Website of Global Coral Reef Alliance

Elimination Technique:

In statement 2 the territories of Australia, Indonesia and Philippines come under tropical areas. Hence, statements 1 and 2 are in congruence.



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LAUNCHING SOON

With reference to Ocean Mean Temperature (OMT), which of the following statements is/are correct?

1. OMT is measured up to a depth of 26°C isotherm which is 129 meters in the south-western Indian Ocean during January–March.
2. OMT collected during January–March can be used in assessing whether the amount of rainfall in monsoon will be less or more than a certain long-term mean.

Select the correct using the code given below:

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Answer: B



Explanation:

- Ocean heat content (OHC) and Ocean Mean Temperature (OMT) are important climatic parameters required for atmospheric and oceanic studies like cyclone and monsoon predictions and ocean heat transport estimations. Sea surface temperature (SST) is routinely used for predicting whether the total amount of rainfall that India receives during the monsoon season will be less or more than the long-term mean of 887.5 mm. Now, scientists from Pune's Indian Institute of Tropical Meteorology (IITM) find that ocean mean temperature (OMT) has a better ability to predict this than the sea surface temperature. Compared with SST which has a 60% success rate of predicting the Indian summer monsoon, OMT has an 80% success rate. **Hence statement 2 is correct.**
- The SST is restricted to a few millimetres of the top ocean layer and is largely influenced by strong winds, evaporation, or thick clouds. In contrast, OMT, which is measured up to a depth of 26 degree C isotherm, is more stable and consistent, and the spatial spread is also less. The 26 degree C isotherm is seen at depths varying from 50–100 metres. During January–March, the mean 26 degree C isotherm depth in the Southwestern Indian Ocean is 59 metres. **Hence statement 1 is not correct.**

Note:

More information on Warm and Cold Ocean Currents is mentioned in Annexure C of this document.

Source

- The Hindu Article titled "Ocean mean temperature can better predict Indian summer monsoon" dated 01 September, 2018



Motivation: Recently, scientists from Pune's Indian Institute of Tropical Meteorology (IITM) found that ocean mean temperature (OMT) has better ability to predict monsoons than the sea surface temperature.

2023

Consider the following statements :

Statement-I

The temperature contrast between continents and oceans is greater during summer than in winter.

Statement-II

The specific heat of water is more than that of land surface.

Which one of the following is correct in respect of the above statements?

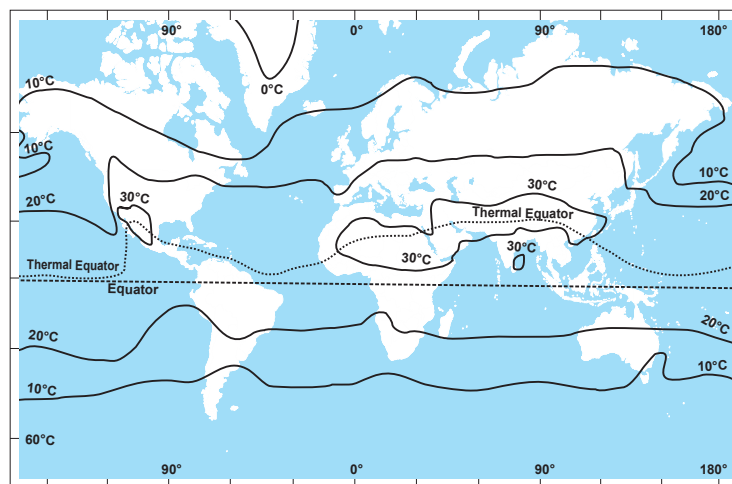
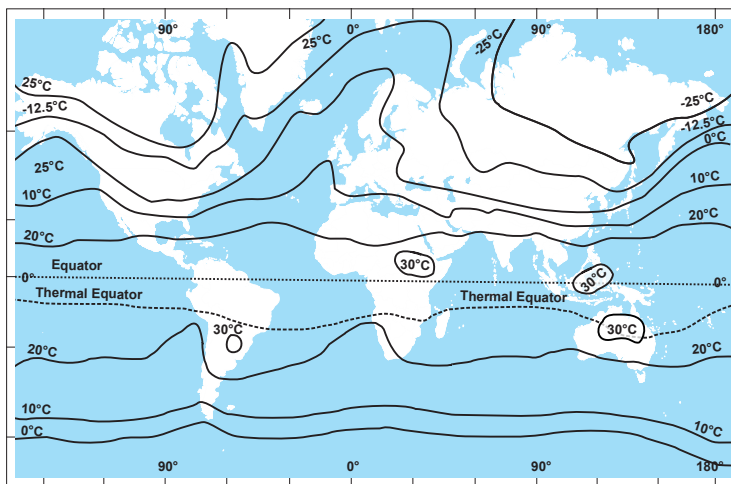
- (a) Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I
- (b) Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I
- (c) Statement-I is correct but Statement-II is incorrect
- (d) Statement-I is incorrect but Statement-II is correct

Answer: D



Explanation:

- The temperature contrast between continents and oceans is greater during winter than in summer. During winter, the land cools down faster and to a larger extent compared to oceans, leading to a significant temperature difference. In summer, while land heats up quickly, the difference in temperature between land and sea is less extreme compared to winter.
- The specific heat capacity is the amount of heat needed to raise the temperature of one gram of the substance by 1 degree. The specific heat Capacity of water is much greater than the land because the relative density of water is much lower than that of the land surface. It means more heat is required to raise the temperature of one gram of water by 1 degree than one gram of land.
- **Statement-I is incorrect and Statement-II is correct. Hence option (d) is the correct Answer**



The distribution of surface air temperature in different seasons

Source

➤ NCERT Class XI: Fundamentals of Physical geography



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Performance and Progress Analysis

Soil

Total No. of questions: 02

2021

The black cotton soil of India has been formed due to the weathering of

- (a) brown forest soil
- (b) fissure volcanic rock
- (c) granite and schist
- (d) shale and limestone

Answer: B



Explanation:

- Black soils are mineral soils which have a black surface horizon, enriched with organic carbon that is at least 25 cm deep. Black soil is formed due to the weathering or erosion of the basalt rocks. **Hence, option (b) is the correct Answer.**
- Basalt is an aphanitic extrusive igneous rock formed from the rapid cooling of low-viscosity lava rich in magnesium and iron exposed at or very near the surface.
- The black soil in India is found in the states like Maharashtra, Gujarat, Madhya Pradesh, and Karnataka.

Source

- Class X NCERT Chapter: Resources and Development

Additional Information



Consider the following statements :

Statement-I :

The soil in tropical rain forests is rich in nutrients.

Statement-II :

The high temperature and moisture of tropical rain forests cause dead organic matter in the soil to decompose quickly.

Which one of the following is correct in respect of the above statements?

- (a) Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I
- (b) Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I
- (c) Statement-I is correct but Statement-II is incorrect
- (d) Statement-I is incorrect but Statement-II is correct

Answer: D



Explanation:

- In the rainforest, most of the carbon and essential nutrients are locked up in the living vegetation, dead wood, and decaying leaves. As organic material decays, it is recycled so quickly that few nutrients ever reach the soil, leaving it nearly sterile. The high temperature and moisture of tropical rainforests cause dead organic matter in the soil to decompose quickly.
- Decaying matter (dead wood and leaf litter) is processed so efficiently because of the abundance of decomposers including bacteria, fungi, and termites. These organisms take up nutrients, which are released as wastes when organisms die. Virtually all organic matter is rapidly processed, even fecal matter and perspiration. It is only a matter of minutes, in many rainforests, before dung is discovered and utilized by various insects. **Hence statement 1 is not correct and 2 is correct.**

Source

- NCERTs Geography and Biology of Class XII

Vegetation

Total No. of questions: 04

2015

In India, in which one of the following types of forests is teak a dominant tree species?

- (a) Tropical moist deciduous forest
- (b) Tropical rainforest
- (c) Tropical thorn scrub forest
- (d) Temperate forest with grasslands

Answer: A



Explanation:

- The Moist deciduous forests are more pronounced in the regions which record rainfall between 100–200 cm. These forests are found in the northeastern states along the foothills of Himalayas, eastern slopes of the Western Ghats and Odisha. Teak, sal, shisham, hurra, mahua, amla, semul, kusum, and sandalwood etc. are the main species of these forests.

Source

- Ncert Class 11th India Physical Environment Page 59

2016

Recently, our scientists have discovered a new and distinct species of banana plant which attains a height of about 11 metres and has orange-coloured fruit pulp. In which part of India has it been discovered?

- (a) Andaman Islands
- (b) Anaimalai Forests
- (c) Maikala Hills
- (d) Tropical rain forests of northeast

Answer: A



Explanation:

- Scientists at the Botanical Survey of India (BSI) discovered a new species of banana from a remote tropical rain forest on the Little Andaman Islands. The new species has orange-coloured fruit pulp and is about 11 metres high, whereas the usual banana species is about three to four metres high.

Source

- <http://www.thehindu.com/news/national/other-states/andamans-yield-a-sweet-banana-with-orange-pulp/article7750316.ece>

2021

Leaf litter decomposes faster than in any other biome and as a result the soil surface is often almost bare. Apart from trees, the vegetation is largely composed of plant forms that reach up into the canopy vicariously, by climbing the trees or growing as epiphytes, rooted on the upper branches of trees." This is the most likely description of

- (a) coniferous forest
- (b) dry deciduous forest
- (c) mangrove forest
- (d) tropical rainforest

Answer: D



Explanation:

- The tropical rainforest is a hot, moist biome found near Earth's equator. The hot and humid conditions make tropical rainforests an ideal environment for bacteria and other microorganisms. Because these organisms remain active throughout the year, they quickly decompose matter on the forest floor. In other biomes, such as the deciduous forest, the decomposition of leaf litter adds nutrients to the soil. But in the tropical rainforest, plants grow so fast that they rapidly consume the nutrients from the decomposed leaf litter. As a result, most of the nutrients are contained in the trees and other plants rather than in the soil. Most nutrients that are absorbed into the soil are leached out by the abundant rainfall, which leaves the soil infertile and acidic.
- Epiphytes - these are plants which live on the branches of trees high up in the canopy. They get their nutrients from the air and water, not from the soil. Epiphytes are found throughout rainforests. Hence **option (d) is the correct answer.**

Source

- Certificate Physical and Human Geography by GC Leong

Consider the following trees:

1. Jackfruit (*Artocarpus heterophyllus*)
2. Mahua (*Madhuca indica*)
3. Teak (*Tectona grandis*)

How many of the above are deciduous trees?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

Answer: B



Explanation:

➤ Tropical Deciduous Forests:

- A deciduous forest is a biome dominated by deciduous trees which lose their leaves seasonally. These are the most widespread forests in India. They are also called the monsoon forests. They spread over regions which receive rainfall between 70–200 cm. On the basis of the availability of water, these forests are further divided into moist and dry deciduous.
- The Moist deciduous forests are more pronounced in the regions which record rainfall between 100–200 cm. These forests are found in the northeastern states along the foothills of Himalayas, eastern slopes of the Western Ghats and Odisha. Teak (*Tectona grandis*), sal, shisham, hurra, mahua (*Madhuca indica*), amla, semul, kusum, and sandalwood etc. are the main species of these forests. **Hence options 2 and 3 are correct.**
- The Dry deciduous forest covers vast areas of the country, where rainfall ranges between 70–100 cm. On the wetter margins, it has a transition to the moist deciduous, while on the drier margins to thorn forests. These forests are found in rainier areas of the Peninsula and the plains of Uttar Pradesh and Bihar. In the higher rainfall regions of the Peninsular plateau and the northern Indian plain, these forests have a parkland landscape with open stretches in which teak and other trees interspersed with patches of grass are common. As the dry season begins, the trees shed their leaves completely and the forest appears like a vast grassland with naked trees all around. Tendu, palas, amaltas, bel, khair, axlewood, etc. are the common trees of these forests. In the western and southern part of Rajasthan, vegetation cover is very scanty due to low rainfall and overgrazing.
- Jackfruit, (*Artocarpus heterophyllus*), evergreen tree (family Moraceae) native to tropical Asia and widely grown throughout the wetland tropics for its large fruits and durable wood. It is the largest tree-borne fruit in the world, reaching up to 60 cm (about 2 feet) long and weighing up to 18 kg (about 40 pounds). **Hence option 1 is not correct.**
- **Hence option (b) is the correct Answer.**

Source

- NCERT : India Physical Environment Class XI



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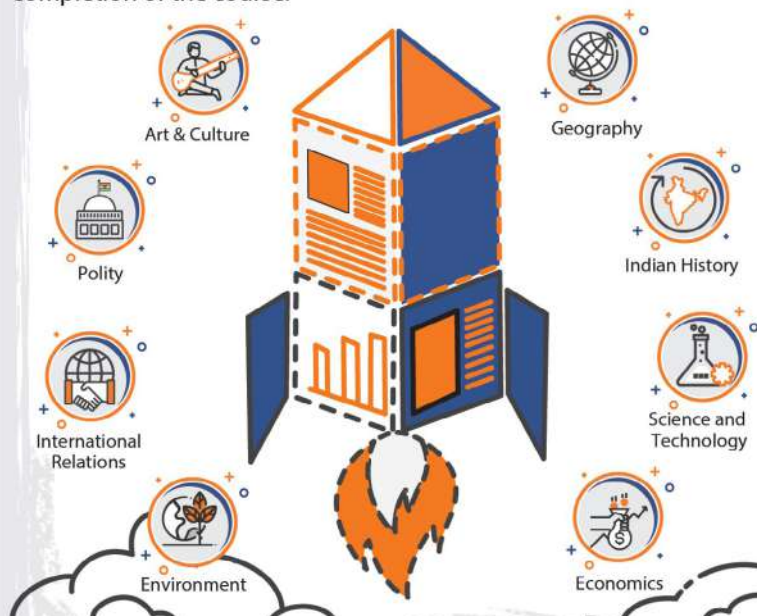
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Water Resources

Total No. of questions: 01

2021

With reference to the water on the planet Earth, consider the following statements:

1. The amount of water in the rivers and lakes is more than the amount of groundwater.
2. The amount of water in polar ice caps and glaciers is more than the amount of groundwater.

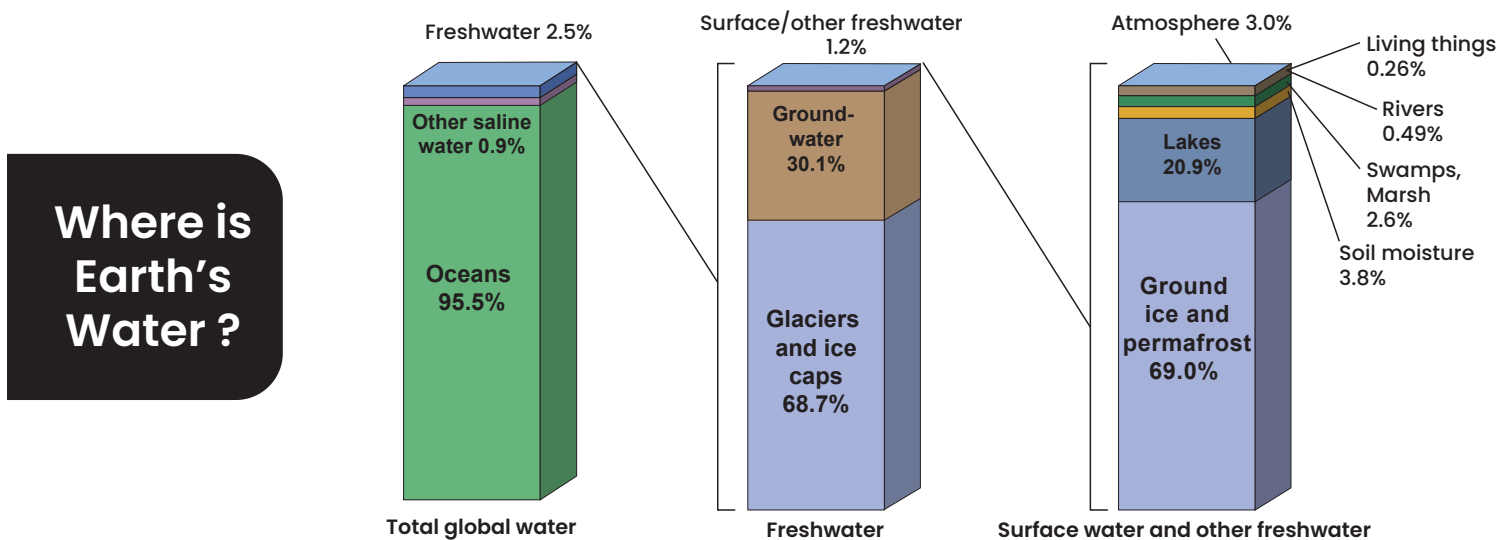
Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Answer: B



Explanation:



➤ Hence option (b) is the correct answer.

Source

➤ NCERT, Class XI, Fundamentals of Physical Geography

World Drainage System

Total No. of questions: 04

2018

Major Seas

Which of the following has/have shrunk immensely/dried up the recent past due to human activities?

1. Aral Sea
2. Black Sea
3. Lake Baikal

Select the correct answer using the code given below:

- (a) 1 only
- (b) 2 and 3
- (c) 2 only
- (d) 1 and 3

Answer: A



Explanation:

- As per NASA, over the last 50 years, Aral Sea has shrunk by about 75% of its original size mainly because of water diversion for agricultural usages in surrounding areas. Also from a depth of 68 meters in the 1960s, today it stands at less than 10 meters. In Lake Baikal, water level has gone below the critical mark of 456m eleven times since 1962. Thus, while Aral Sea has dried up immensely, Lake Baikal shows more of a fluctuating regime and shrinkage is not comparable to that of Aral Sea. **Hence option (a) is the correct answer.**

Source

- The hindu article titled "Aral Sea region in ecological tailspin" dated 01 October, 2017



Motivation: Drying up of lakes is constantly in news

Elimination Technique:

Lake Baikal is one of the deepest lakes in the world. Hence less chance of shrinking immensely. Also, Black is the largest inland body of water drained by major rivers of Europe. **Hence, option (a) is the correct answer.**

2020

Major Rivers

Consider the following pairs:

River		Flows into
1. Mekong	:	Andaman Sea
2. Thames	:	Irish Sea
3. Volga	:	Caspian Sea
4. Zambezi	:	Indian Ocean

Which of the pairs given above is/are correctly matched?

- (a) 1 and 2 only
- (b) 3 only
- (c) 3 and 4 only
- (d) 1, 2 and 4 only

Answer: C



Explanation:

- **Pair 1 is not correctly matched:** Originating in the icy headwaters of the Tibetan highlands, the Mekong River flows through the steep canyons of China, known as the upper basin, through lower basin countries Myanmar, Laos, Thailand, and Cambodia, before fanning across an expansive delta in Vietnam and emptying into the South China Sea.
- **Pair 2 is not correctly matched:** The River Thames is the longest river in England, flowing 215 miles from the Cotswolds to the North Sea.
- **Pair 3 is correctly matched:** The Volga is the longest river in Europe, and its catchment area is almost entirely inside Russia. It belongs to the closed basin of the Caspian Sea, being the longest river to flow into a closed basin.
- **Pair 4 is correctly matched:** The Zambezi is the fourth-longest river in Africa, the longest east-flowing river in Africa and the largest flowing into the Indian Ocean from Africa.

Source

- Standard Reference Atlas



Motivation: World map based basic questions are often covered by UPSC.

2023

Congo River Basin

Which one of the following is a part of the Congo Basin?

- (a) Cameroon
- (b) Nigeria
- (c) South Sudan
- (d) Uganda

Answer: A



Explanation:

- In February 2023, leaders from across Africa and Asia gathered in Gabon for an international summit on the state of the world's tropical forests, which experts say are disappearing at an alarming rate.
- Many of the discussions focused on the Congo Basin, which stores more planet-warming carbon than the Amazon but is disappearing.
- The Congo Basin is home to the world's largest tropical peatlands, along with Brazil and Indonesia. The peat swamp forest of the Congo Basin stores around 29 billion tons of carbon – approximately equivalent to three years' worth of global greenhouse gas emissions – while the Basin as a whole absorbs nearly 1.5 billion tons of carbon dioxide a year.
- The Basin stretches across six countries– Cameroon, Central African Republic, Democratic Republic of the Congo, Congo, Equatorial Guinea and Gabon. The Congo Basin is one of the world's last regions that absorbs more carbon than it emits. **Hence option (a) is the correct Answer.**



Source

➤ Standard Atlas



Motivation: In the news in Feb 2023, UNEP on Peatlands of Congo Basin.

Elimination Technique:

Nigeria is a part of Niger River Basin, South Sudan and Uganda are part of Nile River Basin.
Hence option (a) is correct.

2024

Consider the following statements:

1. The Red Sea receives very little precipitation in any form.
2. No water enters the Red Sea from rivers.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Answer: C



Explanation:

- The Red Sea is located in a region characterized by arid desert climates, resulting in very low levels of precipitation. High evaporation rates further reduce the water level, which contributes to its high salinity. The Red Sea receives minimal rainfall, often averaging less than 100 mm annually, which is negligible in comparison to the evaporation rate. **Hence statement 1 is correct.**
- Unlike many other seas, the Red Sea does not receive any freshwater input from rivers. There are no major rivers or streams flowing into it. This lack of freshwater input, combined with high evaporation, significantly increases the salinity of the water in the Red Sea. The lack of river inflow and freshwater sources helps maintain its unique marine ecosystem. **Hence statement 2 is correct.**

World Economic Geography

Total No. of questions: 04

2023

About three-fourths of world's cobalt, a metal required for the manufacture of batteries for electric motor vehicles, is produced by

- (a) Argentina
- (b) Botswana
- (c) The Democratic Republic of the Congo
- (d) Kazakhstan

Answer: C



Explanation:

- Cobalt has emerged as a vital ingredient of the shift to a lower-carbon energy system, but reserves of the metal are concentrated heavily in just a single country.
- The Democratic Republic of Congo (DRC) is home to half of the world's known resources, and currently accounts for around 70% of global production. **Hence option (c) is the correct Answer.**
- The metal's use in electronic goods – particularly the rechargeable batteries used in electric vehicles and other forms of energy storage – means interest in it is growing as the electrification of the global energy system continues to gather pace.

Source

- Economic Survey 2022–23: Page No. 229, Critical minerals– key to green transition.



Motivation: Critical Minerals and their sources are frequently seen in the news

Consider the following countries:

1. Italy
2. Japan
3. Nigeria
4. South Korea
5. South Africa

Which of the above countries are frequently mentioned in the media for their low birth rates, or ageing population of declining population?

- (a) 1, 2 and 4
- (b) 1, 3 and 5
- (c) 2 and 4 only
- (d) 3 and 5 only

Answer: A



Explanation:

- Japan's population will more than halve, from a peak of 128 million in 2017 to less than 53 million by the end of the century. Japan already has the world's oldest population and the highest rate of people over the age of 100.
- The Italian population is also expected to more than halve, from 61 million in 2017 to 28 million by the end of the century. Like Japan, Italy is known for its ageing population. More than 23% of people there were over the age of 65 in 2019, according to World Bank data.
- Nigeria and other countries in Africa are bucking the trend of population decline. Nigeria will become the world's second most populous country, with 791 million people. The new study predicts that Nigeria will have one of the largest working-age populations in the world by 2100 and see big increases in GDP.
- South Korea's demographic crisis has become the top risk to economic growth and the social welfare system, with the country's population of 51 million on track to halve by the end of this century. South Korea has previously projected its fertility rate is likely to fall further to 0.68 in 2024.
- Hence option (a) is the correct answer.

Source

- <https://www.bbc.com/news/world-53424726>



Motivation: News related to ageing population levels in many countries is frequently seen in the news.

2024

The total fertility rate in an economy is defined as:

- (a) the number of children born per 1000 people in the population in a year.
- (b) the number of children born to couple in their lifetime in a given population.
- (c) the birth rate minus death rate.
- (d) the average number of live births a woman would have by the end of her child-bearing age.

Answer: D



Explanation:

- The Total Fertility Rate (TFR) is the average number of children born to a woman over her lifetime. In India, the TFR has experienced a significant decline, from nearly 6.2 children per woman in 1950 to just under 2 children per woman in 2021.
- TFR of about 2.1 children per woman is called Replacement-level fertility. TFR lower than 2.1 children per woman — indicates that a generation is not producing enough children to replace itself, eventually leading to an outright reduction in population.
- **Hence option (d) is the correct answer.**

Source

- <https://ncert.nic.in/ncerts/l/lesy102.pdf>



Motivation: News related to Fertility levels drop below one in many Asian nations is frequently seen in the news.

2024

Which of the following countries are well known as the two largest cocoa producers in the world?

- (a) Algeria and Morocco
- (b) Botswana and Namibia
- (c) Cote d'Ivoire and Ghana
- (d) Madagascar and Mozambique

Answer: C



Explanation:

- Cocoa, a native to the Americas, was a valuable crop in the earliest South American cultures. The term cocoa originated from the Nahuatl word “cacahuatl”. Many believe that the plant first grew in the Amazon and upper Orinoco basins but the Mayans and the Aztecs eventually developed techniques to cultivate cocoa successfully. The plant was considered as a symbol of wealth for these civilizations and its beans were used as currency. Cocoa beans are the main ingredient for making chocolate.
- Cocoa beans are produced in tropical zones around the Equator, where climate conditions are well suited for growing cocoa trees. About 70 percent of the world’s cocoa beans come from four West African countries: Ivory Coast, Ghana, Nigeria and Cameroon. The Ivory Coast and Ghana are by far the two largest producers of cocoa, accounting for more than 50 percent of the world’s cocoa. In 2021, the Ivory Coast alone produced approximately 2.1 million metric tons of cocoa beans.
- **Hence option (c) is the correct answer.**

Source

- <https://www.statista.com/statistics/263855/cocoa-bean-production-worldwide-by-region/>

Sandhan Vision IAS Prelims Test Series Initiative

Sandhan Vision IAS Prelims Test Series Initiative

Consider the following pairs:

Plantation crop	Top producer
1. Banana	India
2. Rubber	Congo
3. Cocoa	Brazil

Which of the pairs given above is/are correctly matched?

- (a) 1 only
- (b) 1 and 2 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

Biogeography

Total No. of questions: 01

2016

Recently, for the first time in our country, which of the following States has declared a particular butterfly as 'State Butterfly'?

- (a) Arunachal Pradesh,
- (b) Himachal Pradesh
- (c) Karnataka
- (d) Maharashtra

Answer: D



Explanation:

- The Blue Mormon is a large, swallowtail butterfly found primarily in Sri Lanka and India, mainly restricted to the Western Ghats of Maharashtra, South India and coastal belts. It may occasionally be spotted in the Maharashtrian mainland between Vidarbha and Western Maharashtra. It is reportedly the second largest butterfly found in India, just smaller than the southern birdwing. Not uncommon. Not thought to be threatened. Occurs throughout the year but more common in the monsoon and immediately after it. Maharashtra has become the first State in the country to have a 'State butterfly'. **Hence, option d is the correct answer**

Source

- <http://www.thehindu.com/news/national/other-states/maharashtra-gets-state-butterfly/article7342955.ece>

Annexure

Annexure (A)

Name	Type	Climate	Area/Region	Temperature	Rainfall
Rice	Kharif	Hot & Humid Climate	North, NE India, coastal areas, deltaic region Upto- 2000m Altitude	>25°C Range:21-37°C Max: 40-42°C	Greater than 100 cm
Wheat	Rabi	Cool Growing season, Temperate, Bright sunshine at time of ripening	Ganga Satluj Plain and Black Soil Region of Deccan, Malwa Plateau	Winter Crop 20°C-25 °C	Less than 100 cm
Maize	Kharif , Rabi in Peninsular India & Bihar, and Spring in northern India.	Semi-arid regions Sown all over India	Gulf Type climate Regions	21°C-27°C	Below 100cm (50-75 cm)
Jowar	Kharif (Rabi in South India)	Warm weather (Tropical Climate)	Semi-arid areas of Central and Southern India	25°- 32°C, and temperature condition below 16°C is harmful for the crop.	40-50cm Rainfed crop, mostly grown in moist areas which hardly need irrigation
Bajra (Pearl Millet)	Kharif	Dry and Warm	North Western and Western part of India	20°- 30°C,	40-60 cm
Ragi	Kharif	Tropical & Sub-Tropical Climate	Karnataka, Maharashtra, Uttarakhand, Tamilnadu, Andhra Pradesh, Jharkhand, Odisha, Chattisgarh and Gujarat.	20°C- 40° C	50-100 cm
Barley	Rabi	Tropical & Sub-Tropical Climate	Uttar Pradesh, Rajasthan, Punjab, Madhya Pradesh, Haryana, Bihar, Himachal Pradesh, West Bengal and Jammu Kashmir	12°C-15°C during growing period and around 30 0C at maturity	40-50 cm
Gram	Rabi	Mild cool comparatively dry climate	Central, Western and North Western part (Subtropical)	20°C-25°C	60-80 cm
Tur (Red Gram or Pigeon Pea)	Kharif (In mild winters Rabi)	Mild winter	Uttar Pradesh, Madhya Pradesh, Maharashtra, Gujarat and Karnataka etc.	26°C-30°C for growth 15°C-18°C for maturity	60-65 cm
Pea	Rabi	Mild winter	Winter moderate temperature	13°C-18°C	

Lentils	Rabi	Cold/ Temperate	Uttar Pradesh, Madhya Pradesh, Bihar, Jharkhand, West Bengal, Maharashtra, Chattisgarh, Gujarat, Haryana, Punjab, Rajasthan, Uttarakhand, Jammu and Kashmir	18°C- 30°C	
Moong	Kharif and Rabi	Warm humid	High temperature, less humidity and moderate rainfall		60-80 cm no water logging
Groundnut	Kharif & Rabi	Tropical Climate	Gujarat, Andhra Pradesh, Tamil Nadu, Karnataka, Maharashtra, Madhya Pradesh, Uttar Pradesh, Rajasthan, Punjab and Orissa.	26°C-34°C	50 to 125 cm
Rapseed & Mustard	Rab	Cool and dry weathe	Rajasthan, Uttar Pradesh, Haryana, Madhya Pradesh, and Gujarat	18°C - 25°C	Low humidity
Cotton	Kharif	Tropical & Sub-Tropical Regions	Punjab, Haryana and Rajasthan, Gujarat, Maharashtra and Madhya Pradesh, Telangana, Andhra Pradesh, Karnataka and Tamil Nadu	20°C -30°C	50-100 cm
Jute & Mesta	-	Humid Tropical Climate	Jute: Eastern and North Eastern India Mesta: Almost throughout the country.	18°C-34°C	100-250 cm
Sugarcane	Kharif	Tropical & Sub-Tropical Climate	All over the country, except cold hilly areas like Kashmir valley, Himachal Pradesh and Arunachal Pradesh.	27° to 33°c	110-150 cm
Tea	-	Hot & Humid Climate, Winter is not too cold	Assam, Tripura, Arunachal Pradesh, Himachal Pradesh, Uttar Pradesh, Sikkim, Andhra Pradesh, Manipur, Nagaland, West Bengal, Tamilnadu, Kerala, Karnataka etc.	18 °C-30°C 80%-85%, annual relative humidity	150-250 cm
Coffee	-	Hot & Humid Climate	Traditional: Karnataka, Kerala, Tamil Nadu Non Traditional: Andhra Pradesh, Odisha, North East states	20°C-28°C	150-250 cm
Pea	Rabi	Mild winter	Winter moderate temperature	13°C-18°C	

Rubber	-	Tropical Climate (June – July is optimum for cultivation)	-	25°C–34°C with 80% relative humidity	A rainfall of 200 cm or more, evenly distributed without any marked dry season and with 125–150 rainy days per annum,
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Name	Soil	Top 3 Production states	Gross Area (MM Hec.)	Miscellaneous
Rice	Loamy, clayey and alluvial soils	West Bengal Uttar Pradesh Punjab	46.4 Kharif: 41.1 Rabi: 5.3	1. India contributes 22.07 per cent of rice production in the world and ranked second after China in 2018. 2. About one-fourth of the total cropped area in the country is under rice cultivation. West
Wheat	Alluvial, Loamy, Clayey and Black Soil	Uttar Pradesh Madhya Pradesh Punjab	30.5	1. 100 isohyet separates wheat from rice 2. India produces about 12.8% of total wheat production of world (2017). 3. About 85% of total area under this crop is concentrated in north and central regions of the country i.e. Indo-Gangetic Plain, Malwa Plateau and Himalayas up to 2,700 m altitude. 4. About 14 per cent of the total cropped area in the country is under wheat cultivation.
Maize	In variety of soils ranging from loamy sand to clay loam	Karnataka Madhya Pradesh Maharashtra	10	1. This crop occupies only about 3.6% of total cropped area. 2. Maize cultivation is not concentrated in any specific region. It is sown all over India. 3. Yield level of maize is higher than other coarse cereals. 4. Kharif maize represents around 83% of maize area in India, while rabi maize correspond to 17% maize area. 5. Over 70% of kharif maize area is grown under the rainfed condition 6. Madhya Pradesh and Karnataka has highest area under maize. 7. It accounts for around 10 percent of total food grain production in the country.
Jowar	Clayey deep regur and alluvium	Maharashtra Karnataka Madhya Pradesh Tamil Nadu	3.8 Kharif: 1.5 Rabi: 2.3	1. The coarse cereals together occupy about 16.50% of total cropped area in the country. Among these, jowar or sorghum alone accounts for about 5.3% of total cropped area. 2. It is main food crop in semi-arid areas of central and southern India. 3. Maharashtra alone produces more than half of the total jowar production of the country

Bajra (Pearl Millet)	Poor light sandy soils, black and red soils	Rajasthan Uttar Pradesh Haryana	6.7	1. Hardy crop resist frequent dry spells and drought in this region 2. Rain fed crop 3. The crop may tolerate drought but cannot withstand high rainfall of 90 cm or above. 4. Crop does not tolerate soil acidity 5. Rajasthan accounts for 45% of Pearl Millet (Bajra) produced in India.
Ragi	Red, Light black, Sandy and well drained alluvial loam	Karnataka	-	1. Rainfed kharif crop 2. Raised on variety of soils 3. Karnataka (73% of production) 4. The crop possesses good drought tolerance but is highly sensitive to frost.
Barley	Sandy Loam, Loam and Medium & Heavy Black Soils	Rajasthan Uttar Pradesh Madhya Pradesh	-	1. Can't tolerate high heat and humidity. 2. Rajasthan contributes to over 40% of total Barley production in India.
Gram	Loamy soils Silty clay loams Also in fine textured deep black soils	Maharashtra Madhya Pradesh Rajasthan	10.9	1. Most important of all pulses 2. Just one or two light showers 3. Rainfed crop
Tur (Red Gram or Pigeon Pea)	Black cotton soil	Maharashtra Karnataka Uttar Pradesh	5	1. Water logging, heavy rains, frost are very harmful 2. Grows as dry crop mixed with other kharif millets, cotton, groundnut etc. 3. India ranked first in area and production in the world.
Pea	Well drained loamy soils free from excessive soluble salts			Origin- Mediterranean region of Southern Europe and West asia
Lentils	Loamy soils (acidic soils are not fit)	Madhya Pradesh Uttar Pradesh Bihar		Origin- Turkey and South Iran Intercropping Mostly Micro-nutrient required (Sulphur, Boron, Nitrogen and Phosphorus)
Moong	Loamy soil with good drainage no alkaline or saline soil	Rajasthan Maharashtra Andhra Pradesh	5.5	Origin- India and Central Asia

Groundnut	Well drained, light textured, loose, friable and sandy and sandy loam soils most suitable. However, clay or heavy soils are not suitable for this crop	Gujarat Rajasthan Tamilnadu	5.7	1. Among all oilseed crops, groundnut accounts for more than 40-50 % in area and 60 to 70 % in production in the country.
Rapseed & Mustard	Sandy loam to clay loam soils but thrive best on light loam soils.	Rajasthan Madhya Pradesh Haryana	8.1	1. Do not tolerate water logging conditions or heavy soils. Soil having neutral pH is ideal. 2. The production of rapeseed and mustard in India accounts for about 30% of the total oilseed production of the country.
Cotton	Well drained deep alluvial soils in the north, black clayey soils of varying depth in central region and in black and mixed black and red soils in south zone.	Cotton Maharashtra Telangana	11.9	1. Nearly 65 percent cotton area is rainfed, mainly in the Central and Southern States. 2. Cotton crop is highly prone to pests and diseases. 3. Warm days of cool nights with large diurnal variations during the period of fruiting are conducive to good boll & fibre development. 4. India accounts for around 25% of the total global cotton production. 5. Due to its economic importance in India, it is also termed as "White-Gold". 6. A long growing period of at least 210 frost-free days is required.
Jute & Mesta	All kinds of soils from clay to sandy loam, but loamy alluvial are best suited	West Bengal Assam Bihar	0.6	1. Laterite and gravel soils are not suitable for Jute cultivation. 2. Jute cultivation is mainly concentrated in the eastern and north eastern India while that of mesta cultivation is spread almost throughout the country. 3. India is the 1st largest jute producing country in the world
Sugarcane	Loams & clay loams of Gangetic & other alluviums, brown or reddish loams, laterites and black cotton soils	Uttar Pradesh Maharashtra Karnataka	5.1	1. Tropical region shared about 45% and 55% of the total sugarcane area and production in the country, respectively. 2. Sub-tropical region accounted for about 55% and 45% of total area and production of sugarcane, respectively. 3. It is largely an irrigated crop in India. 4. In 2022, India emerged as the world's largest producer and consumer of sugar and world's 2nd largest exporter of sugar.

Tea	Deep & fertile well drained acidic soil, rich in humus and organic matter (4.5-5.5 pH)	Assam West Bengal Tamil Nadu	0.6	1. In North India tea plantations are located in plains while in South India tea is cultivated in hilly terrain. 2. China tea is grown on higher altitudes. 3. Frost causes adverse effect on tea bush, especially in the ravines. 4. India, China and Sri Lanka are the main tea producing countries in the world. India produces nearly 30 % of world tea production followed by China (22 %) and Sri Lanka (8%). 5. China is having maximum area under tea in the world(over 40 % of world area) but the maximum production is from India mainly because of higher productivity. 6. Darjeeling produces the finest quality of tea, its productivity is low. 7. The Largest state with area under Tea Plantations in India is Assam.
Coffee	Well drained, loams containing good amount of humus and slightly acidic	Karnataka Kerala Tamil Nadu	0.4	1. Frost, snowfall, high temperature above 30°C and strong sun shine is not good for coffee crops and is generally grown under shady trees. 2. Karnataka is the largest producer accounting for about 70% of the total coffee production. 3. Coffee can be grown on lots of soils but the ideal types are fertile volcanic red earth or deep sandy loam. 4. India mostly grows superior quality of arabica coffee. 5. India produces only about 3.17 per cent coffee of the world and ranks eighth after Brazil, Vietnam, Indonesia, Colombia, Honduras, Ethiopia and Peru.
Rubber	Deep and lateritic fertile soil with an acidic pH of 4.5 to 6.0 and highly deficient in available phosphorous.	Kerala Tripura Assam	0.8	1. It is indigenous to the rain forests of the Amazon basin. 2. The commercial source of natural rubber is Hevea brasiliensis, introduced in India in the year 1902 and planted in Kerala. Even though it is a tropical tree, it grows well in the soils of N.E. states. 3. The tyre industry consumer nearly 85% of the natural rubber produced in India. 4. It can be successfully grown up to the elevation of 450 metre from mean sea level. 5. The Rubber Board is a statutory body constituted by the Government of India, under the Rubber Act 1947. 6. Thailand is the leading Rubber producing country in the world, followed by Indonesia and China at the second and third place respectively. 7. India stands at fourth position.

EL Nino and LA Nina

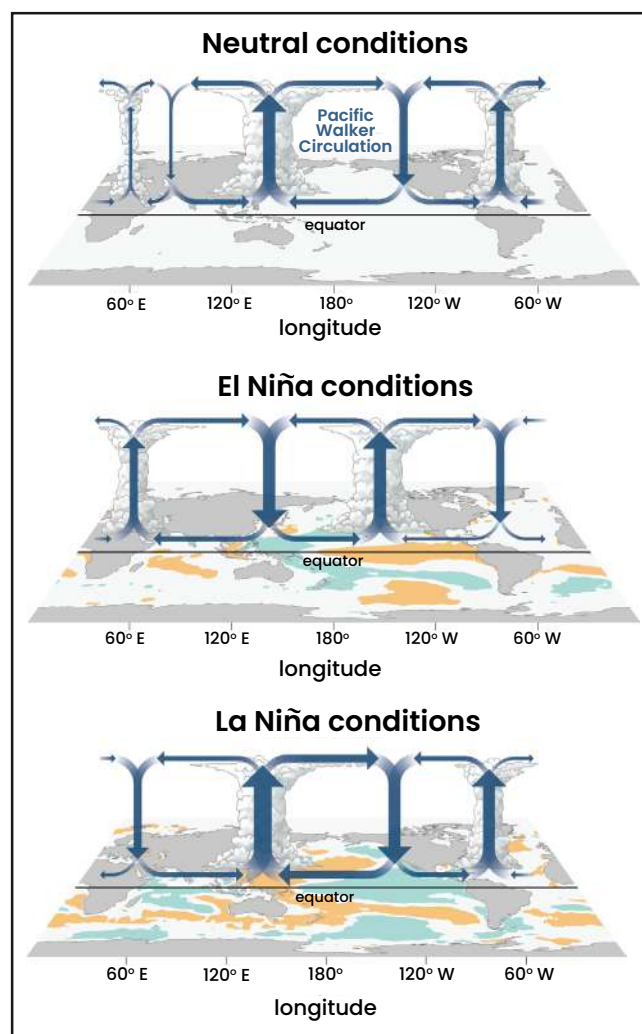
➤ During normal conditions in the Pacific ocean, **trade winds blow west along the equator, taking warm water from South America towards Asia.** To replace that warm water, cold water rises from the depths — a process called upwelling. El Niño and La Niña are two opposing climate patterns that break these normal conditions. Scientists call these phenomena the El Niño-Southern Oscillation (ENSO) cycle. El Niño and La Niña can both have global impacts on weather, wildfires, ecosystems, and economies. Episodes of El Niño and La Niña typically last nine to 12 months, but can sometimes last for years. El Niño and La Niña events occur every two to seven years, on average, but they don't occur on a regular schedule. Generally, El Niño occurs more frequently than La Niña.

➤ El Niño:

- In an El Niño year, air pressure drops over large areas of the central Pacific and along the coast of South America.
- The normal low pressure system is replaced by a weak high in the western Pacific. This change in pressure pattern causes the trade winds to be reduced.
- This reduction allows the equatorial counter current (current along doldrums) to accumulate warm ocean water along the coastlines of PThis accumulation of warm water causes the thermocline to drop in the eastern part of Pacific Ocean which cuts off the upwelling of cold deep ocean water along the coast of Peru.
- Climatically, the development of an El Niño brings drought to the western Pacific, rains to the equatorial coast of South America, and convective storms and hurricanes to the central Pacific. Severe droughts occur in Australia, Indonesia, India and southern Africa. Whereas heavy rains in California, Ecuador, and the Gulf of Mexico.

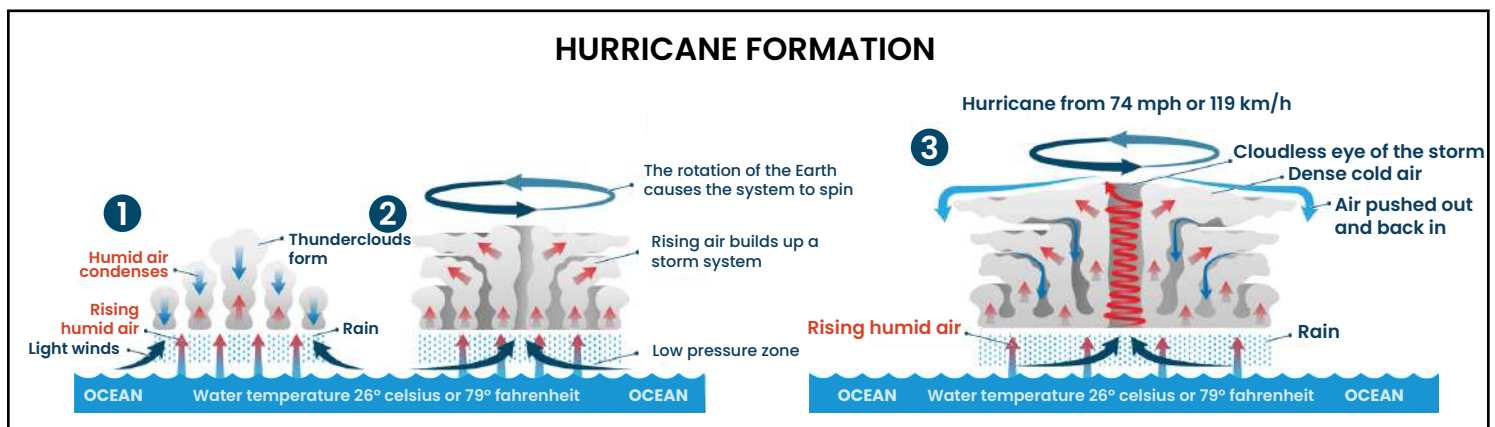
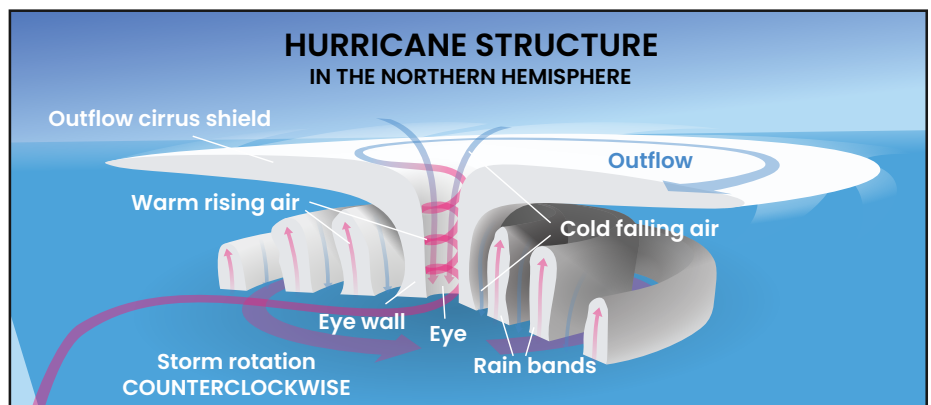
➤ La Niña:

- La Niña has the opposite effect of El Niño. During La Niña events, trade winds are even stronger than usual, pushing more warm water toward Asia. Off the west coast of the Americas, upwelling increases, bringing cold, nutrient-rich water to the surface.
- It causes abnormally heavy monsoons in India and Southeast Asia, cool and wet winter weather in southeastern Africa, and wet weather in eastern Australia.

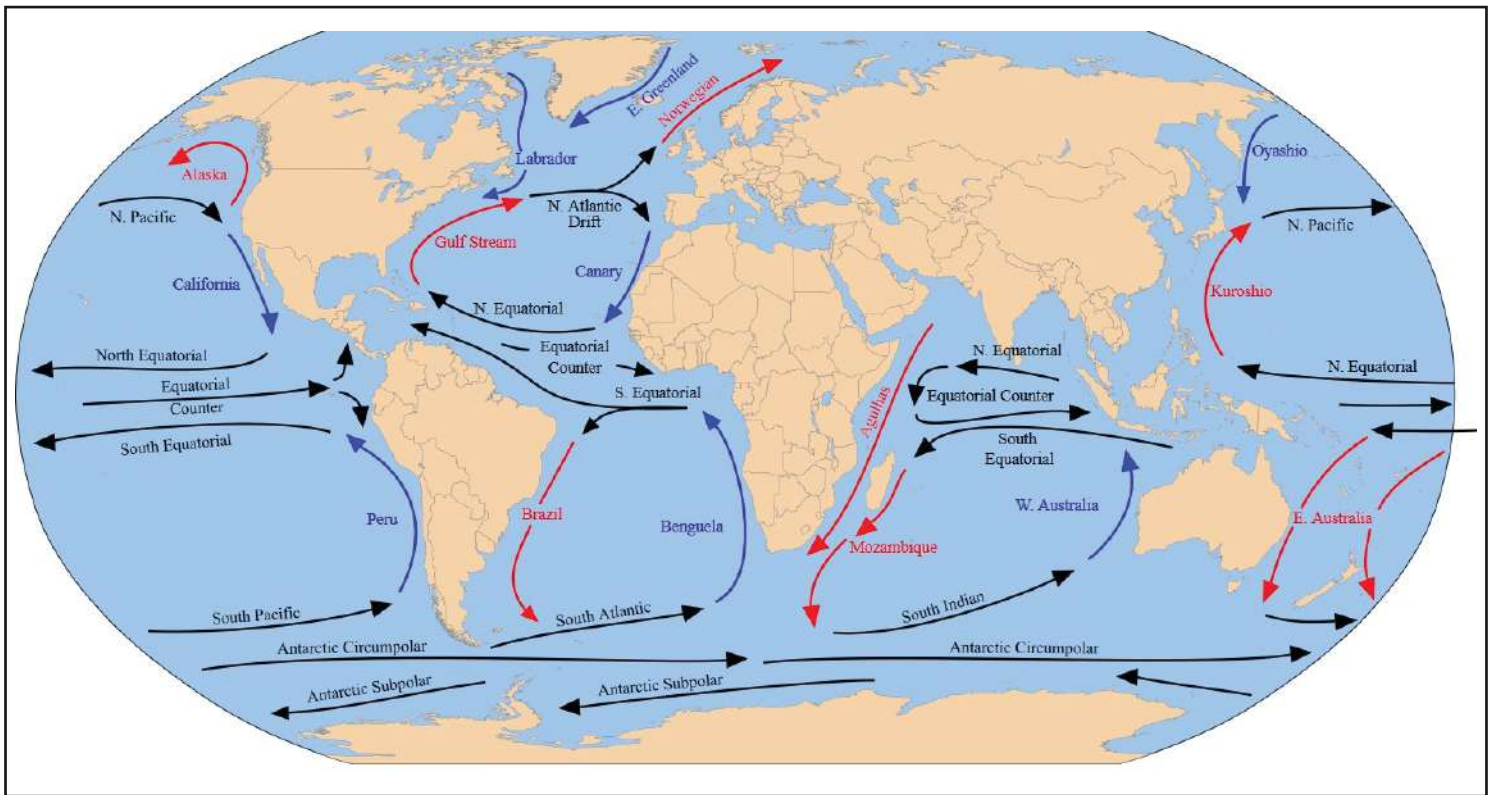


How do tropical cyclones form:

- They are like giant engines that use warm, moist air as fuel. That is why they form only over warm ocean waters near the equator. The warm, moist air over the ocean rises upward from near the surface. Because this air moves up and away from the surface, there is less air left near the surface. Another way to say the same thing is that the warm air rises, causing an area of lower air pressure below.
- Air from surrounding areas with higher air pressure pushes into the low pressure area. Then that "new" air becomes warm and moist and rises, too. As the warm air continues to rise, the surrounding air swirls in to take its place. As the warmed, moist air rises and cools off, the water in the air forms clouds. The whole system of clouds and wind spins and grows, fed by the ocean's heat and water evaporating from the surface.
- Storms that form north of the equator spin counterclockwise. Storms south of the equator spin clockwise. This difference is because of Earth's rotation on its axis.
- **As the storm system rotates faster and faster, an eye forms in the centre.** It is very calm and clear in the eye, with very low air pressure. Higher pressure air from above flows down into the eye.
- When the winds in the rotating storm reach 39 mph, the storm is called a "tropical storm." And when the wind speeds reach 74 mph, the storm is officially a "tropical cyclone," or hurricane.
- Tropical cyclones usually weaken when they hit land, because they are no longer being "fed" by the energy from the warm ocean waters.



Annexure (C)



Warm Ocean Currents

- North Equatorial Current
- Kuroshio Current
- North Pacific Drift
- Equatorial Counter Current
- South Equatorial Current
- East Australian Current
- Gulf Stream
- Irminger Current
- Brazilian Current
- Agulhas Current
- Southwest Monsoon Current

Cold Ocean Currents

- Humboldt or Peruvian Current
- Kurile or Oyashio Current
- California Current
- Antarctic Circumpolar Current or West Wind Drift
- Labrador Current
- Canary Current
- Eastern Greenland Current
- Benguela Current
- Falkland Current
- Western Australian Current

Annexure (D)

Types of Lakes	Names of Lakes	States
Artificial lakes	Hamirsar Lake	Gujrat
	Kankaria Lake	Gujrat
	Chembarambakkam	Tamil Nadu
	Hussain Sagar	Telangana
	Osman Sagar	Telangana
	Jaisamand Lake (2nd Largest in Asia)	Rajasthan
	Hirakud Lake	Odisha
	Bhojtal Lake	Madhya Pradesh
	Barapani Lake	Meghalaya
	Narayan Sarovar	Gujarat
	Thol Lake	Gujarat
Brackish Water	Pulicat Lake	Andhra Pradesh
	Ashtamudi Kayal	Kerala
	Kuttanad	Kerala
	Vembanad	Kerala
	Chilika lake	Odisha
	East Calcutta wetlands	West Bengal
Brackish Water	Lonar lake	Maharashtra
Freshwater Lakes	Kolleru Lake	Andhra Pradesh
	Nagarjuna Sagar	Andhra Pradesh
	Haflong Lake	Assam
	Son Beel	Assam
	Vastrapur	Gujarat
	Badkhal Lake	Haryana

	Blue Bird Lake	Haryana
	Brahma Sarovar	Haryana
	Damdama Lake	Haryana
	Karna	Haryana
	Maharana Pratap Sagar	Himachal Pradesh
	Chandra Taal	Himachal Pradesh
	Agara lake	Karnatka
	Bellandur Lake	Karnatka
	Kukarahalli lake	Karnatka
	Honnamana	Karnatka
	Pampa Sarovar	Karnatka
	Shashtamkotta	Kerala
	Loktak Lake	Manipur
	Salim Ali	Maharashtra
	Shivsagar	Maharashtra
	Umiam	Meghalaya
	Tam Dil	Mizoram
	Kanjia lake	Odisha
	Harike	Punjab
	Ropar	Punjab
	Kanjli	Punjab
	Wular Lake	Jammu & Kashmir
	Tsomgo Lake	Sikkim
	Khecheoplari	Sikkim
	Ooty Lake	Tamil Nadu
	Badrakali Lake	Telangana
	Belasagar	Uttar Pradesh
	Bhimtal	Uttar Pradesh
Freshwater (canal inflow)	Tilyar	Haryana

Freshwater (High Altitude)	Suraj Taal	Himachal Pradesh
Freshwater (Tectonically formed)	Deepor Beel	Assam
Holomitic (Freshwater)	Prashar lake	Himachal Pradesh
Warm Monomitic	Dal Lake	Jammu & Kashmir
Mixing Monomitic	Manasbal Lake	Jammu & Kashmir
Man-made lake	Govind Ballabh Pant Sagar	Uttar Pradesh
Oxbox (Freshwater)	Chandubi lake	Assam
Saltwater	Kanwar Lake	Bihar
	Sambhar lake	Rajasthan
	Pangong Tso	Ladakh
	Tso Moriri	Ladakh



PRELIMS MENTORING PROGRAM 2025

4 Month Expert Intervention

A Strategic Revision, Practice, and Mentoring Program for UPSC Prelims Examination

11 JANUARY

-  Highly experienced and qualified team of Mentors for continuous support and guidance
-  A structured plan of revision for GS Prelims, CSAT, and Current Affairs
-  Effective Utilization of learning resources, including PYQs, Quick Revision Modules (QRMs), and PT-365



PRELIMS & MAINS INTEGRATED MENTORING PROGRAM 2025

Lakshya Prelims & Mains Integrated Mentoring Program 2025

(A 7 Months Strategic Revision, Practice, and Mentoring Program for UPSC Prelims and Mains Examination 2025)

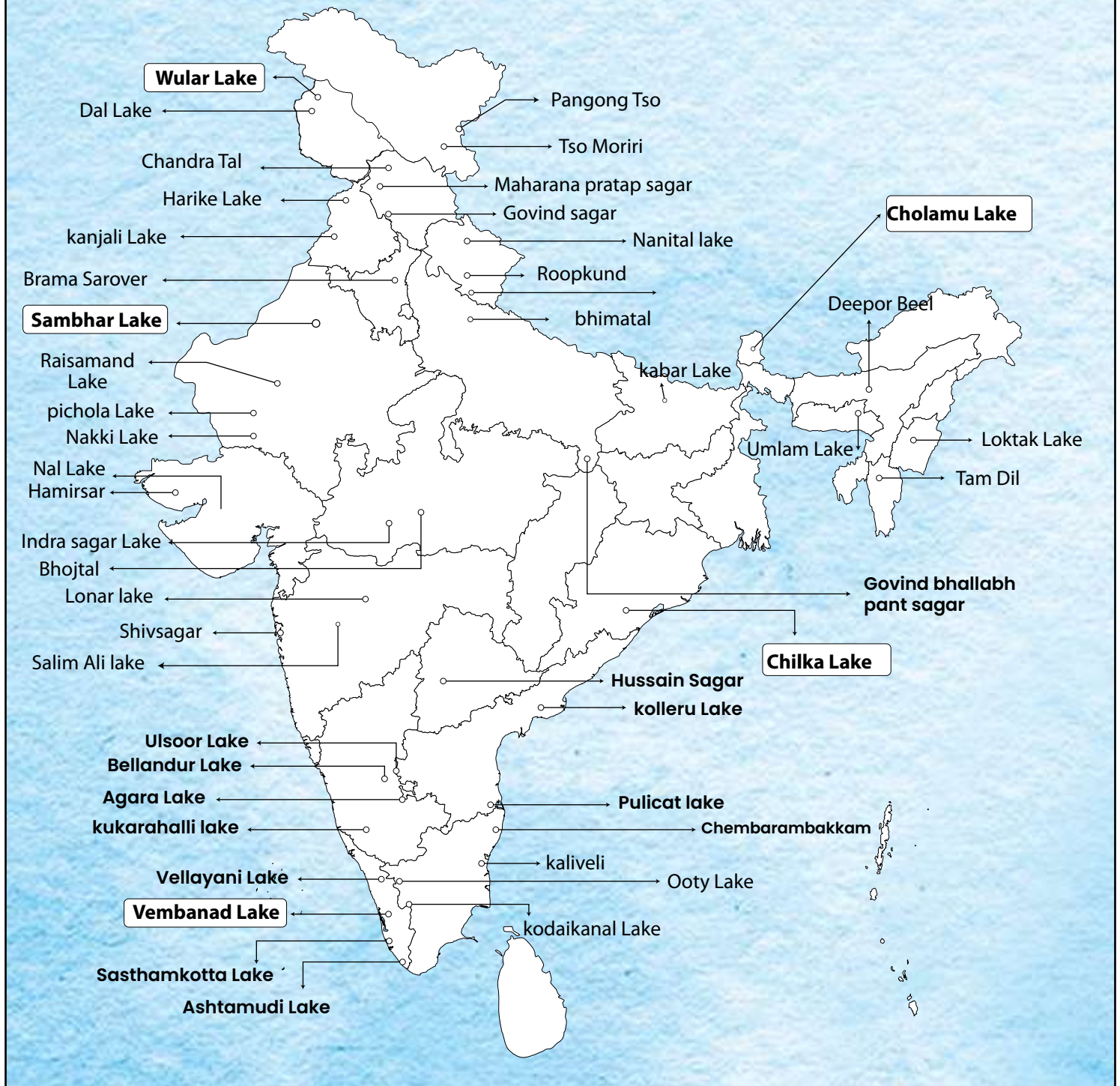
VisionIAS introduces the Lakshya Prelims & Mains Integrated Mentoring Programme 2025, offering unified guidance for UPSC aspirants across both stages, ensuring comprehensive support and strategic preparation for success

6 JANUARY

Highlights of the Program

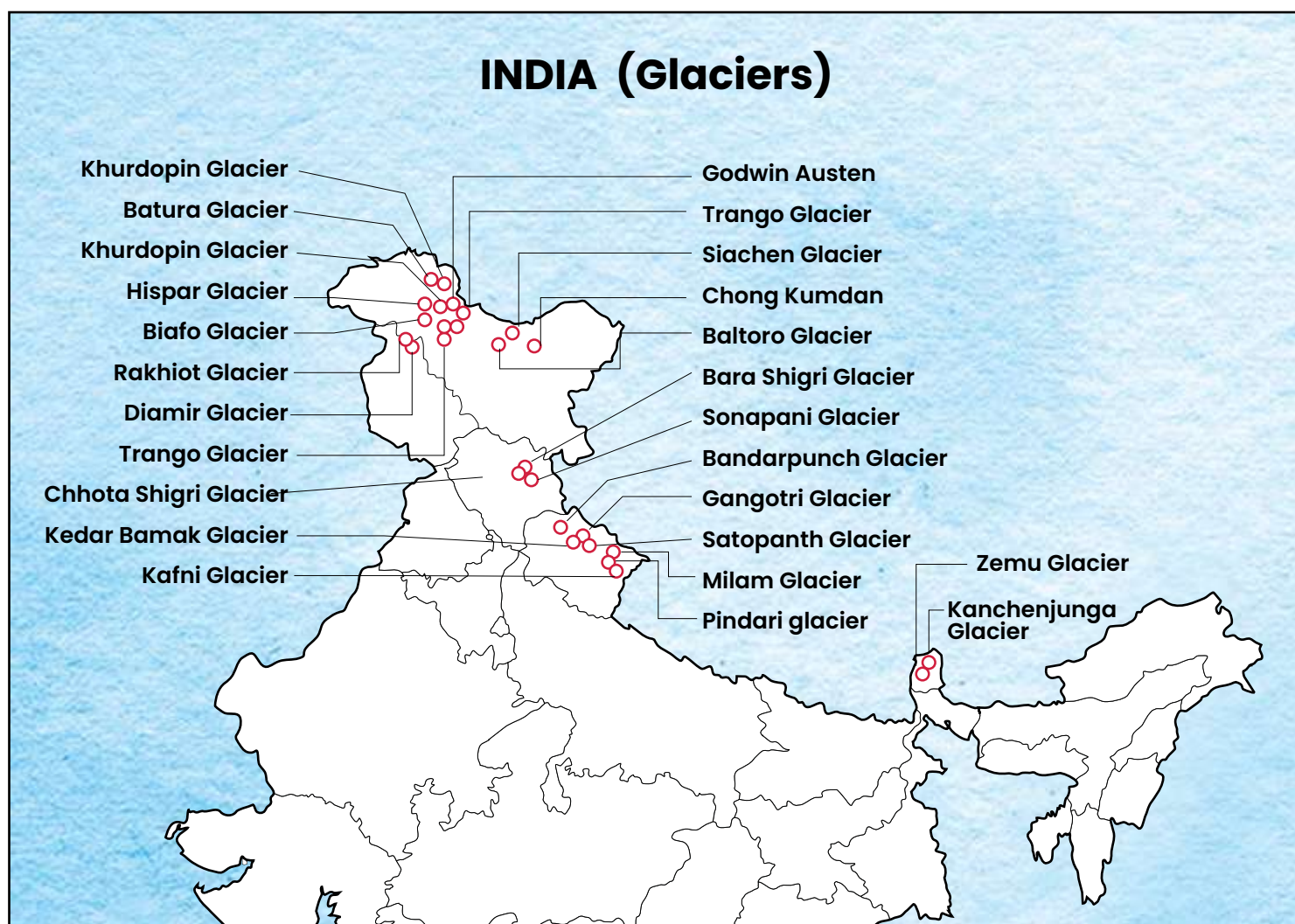
- Coverage of the entire UPSC Prelims and Mains Syllabus
- Development of Advanced answer writing skills
- Highly experienced and qualified team of senior mentors
- Special emphasis to Essay & Ethics

IMPORTANT LAKES IN INDIA



DAMS IN INDIA





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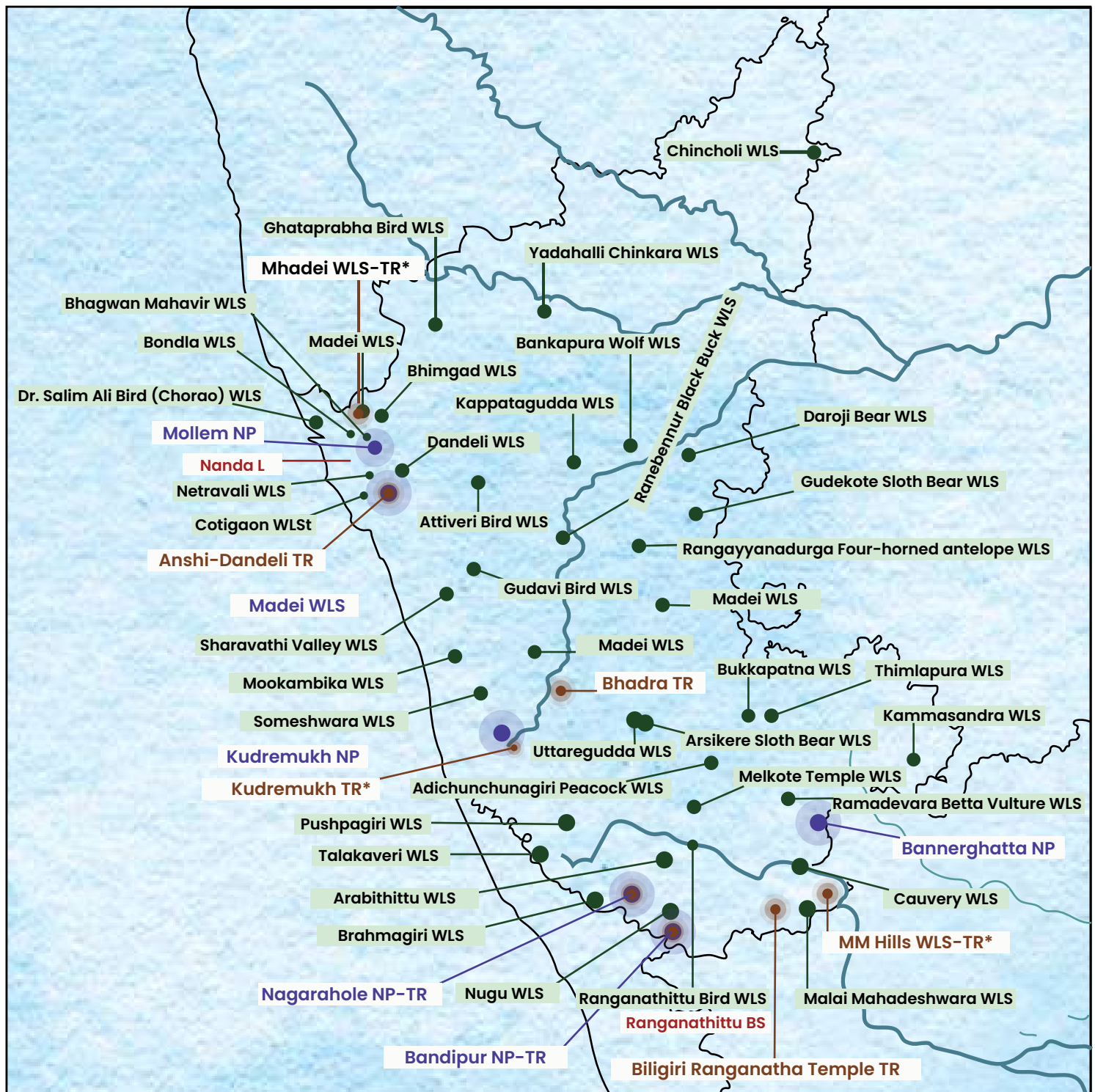
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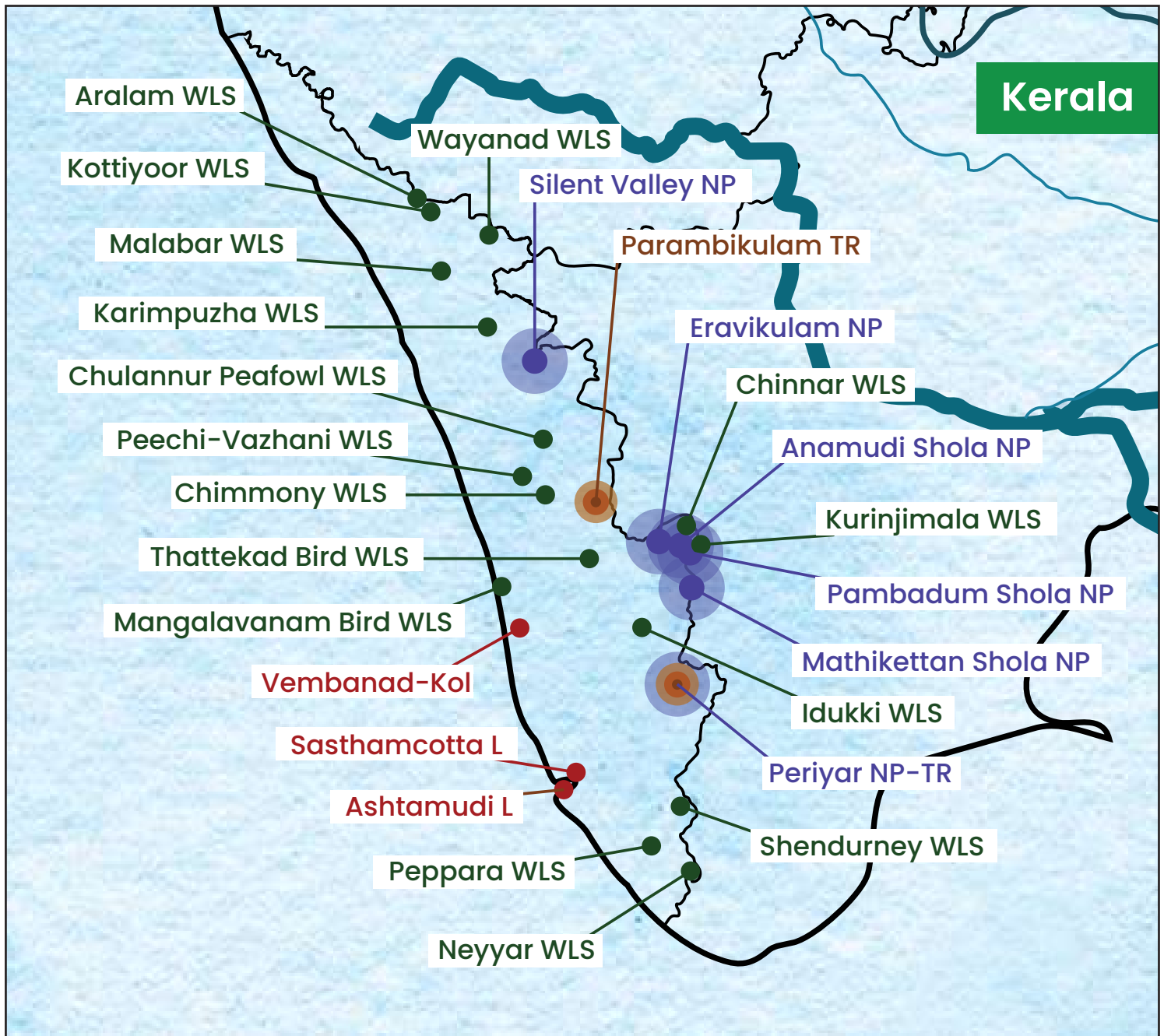
Glaciers of India

Name	State	Mountain Range	River
Batura Glacier	Jammu & Kashmir	Karakoram	Braldo/Shigar
Hispar Glacier	Jammu & Kashmir	Karakoram	Hunza
Biafo Glacier	Jammu & Kashmir	Karakoram	Tributary of Shigar
Baltoro Glacier	Jammu & Kashmir	Karakoram	Shigar
Chomolungma glacier	Jammu & Kashmir	Karakoram	Shyok
Diamir Glacier	Jammu & Kashmir	Karakoram	
Siachen Glacier	Jammu & Kashmir	Karakoram	Mutgah or Shaksgam Rive
Khurdapin glacier	Ladakh	Karakoram	
Nubra	Ladakh	Karakoram	Nubra
Godwin Austen	Ladakh	Karakoram	
Trango Glacier	Ladakh	Karakoram	
Chong Kumdan	Ladakh	Karakoram	
Saltoro	Ladakh	Karakoram	Saltoro/Shyok
Bara Shigri Glacier	Himachal Pradesh	Pir Panjal Range of the Inner Himalayas.	
Chhota Shigri Glacier	Himachal Pradesh	Pir Panjal	
Sonapani Glacier	Himachal Pradesh	Pir Panjal	
Rakhiot Glacier	Ladakh	Pir Panjal	
Gangotri Glacier	Uttarkashi, Uttarakhand	Himalayas	Bhagirathi
Bandarpunch Glacier	Uttarakhand	Western edge of the high Himalayan range	Yamuna
Milam Glacier	Uttarakhand	Trishul peak of Pithoragarh	Gori Ganga/Kali

Pindari glacier	Nanda Devi, Uttarakhand	Upper reaches of the Kumaon Himalayas	Pindar
Kafni Glacier	Uttarakhand	Kumaon-Garhwal	
Kalabaland Glacier	Uttarakhand	Kumaon-Garhwal	
Kedar Bamak Glacier	Uttarakhand	Kumaon-Garhwal	
Meola Glacier	Uttarakhand	Kumaon-Garhwal	
Namik Glacier	Uttarakhand	Kumaon-Garhwal	
Panchchuli Glacier	Uttarakhand	Kumaon-Garhwal	
Ralam Glacier	Uttarakhand	Kumaon-Garhwal	
Sona Glacier	Uttarakhand	Kumaon-Garhwal	
Satopanth Glacier	Uttarakhand	Kumaon-Garhwal	Alaknanda
Sunderdhunga Glacier	Uttarakhand	Kumaon-Garhwal	Mandakini
Dokriani Glacier	Uttarakhand	Kumaon-Garhwal	
Chorabari Glacier	Uttarakhand	Kumaon-Garhwal	
Zemu Glacier	Sikkim	Eastern Himalaya Located on Kanchenjunga peak	Teesta
Kanchenjunga Glacier	Sikkim	Eastern Himalaya	Arun & Koshi

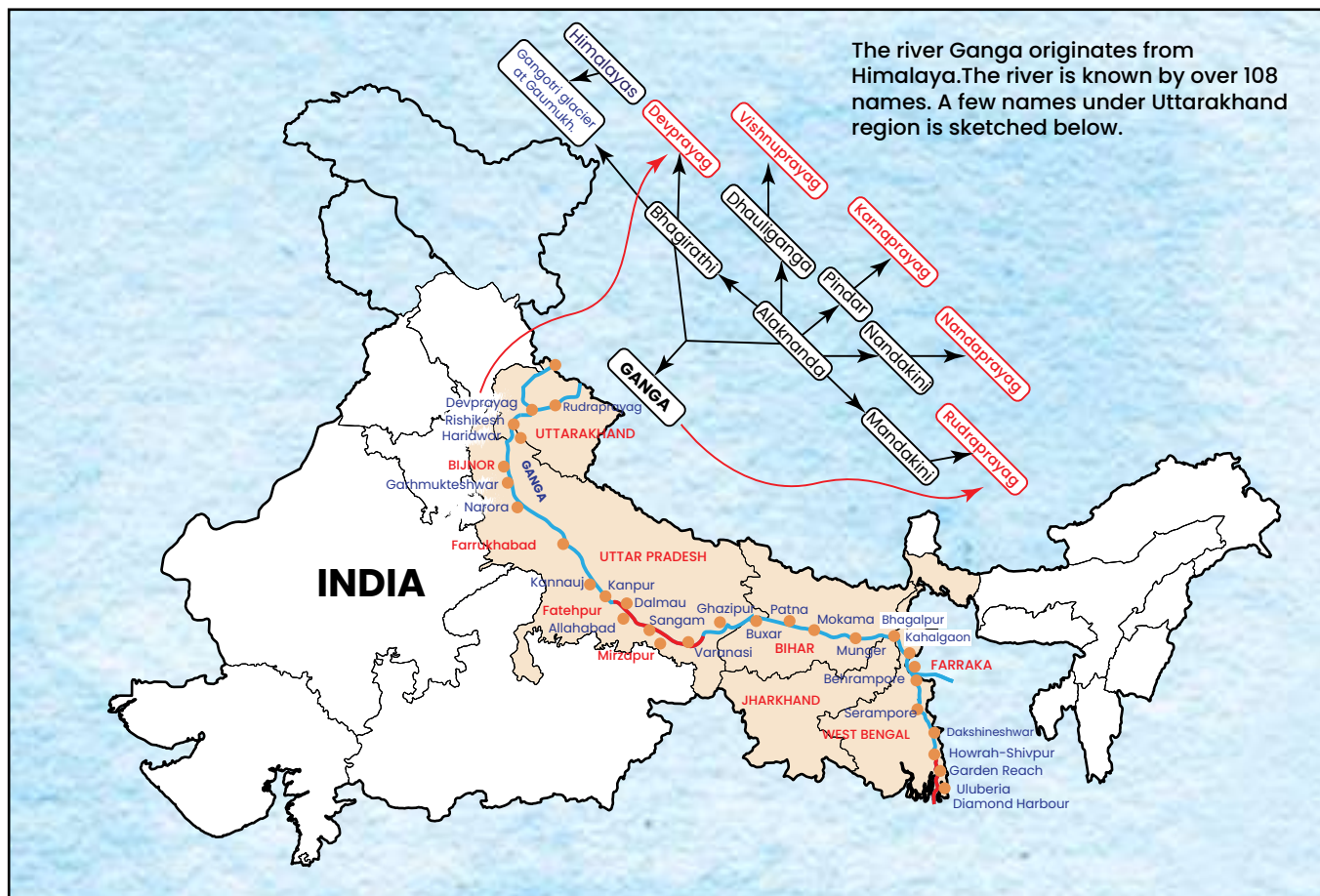
Annexure (G)





Annexure (G)





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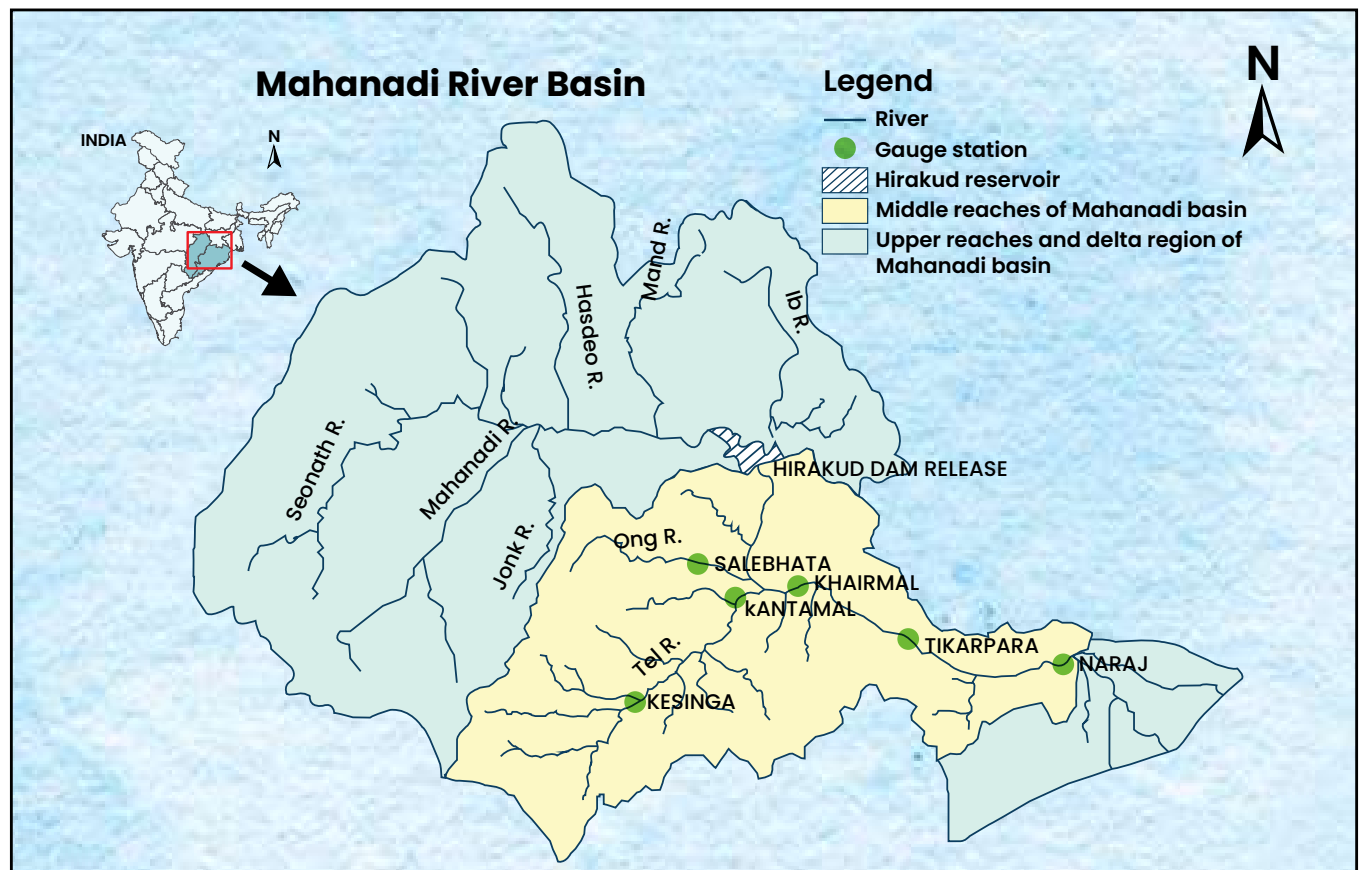
- **General Studies** (हिन्दी माध्यम में भी उपलब्ध)
- **CSAT** (हिन्दी माध्यम में भी उपलब्ध)
 - VISION IAS Post Test Analysis™
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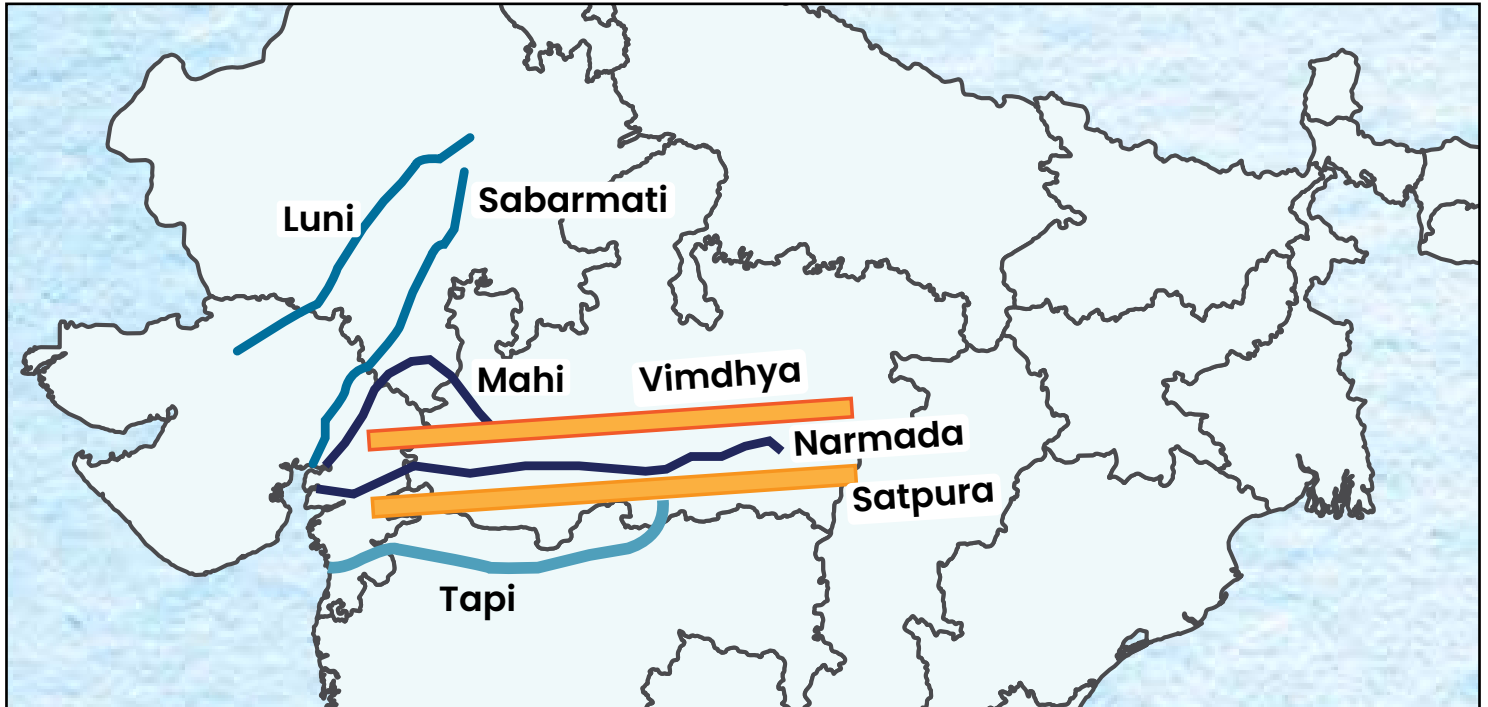
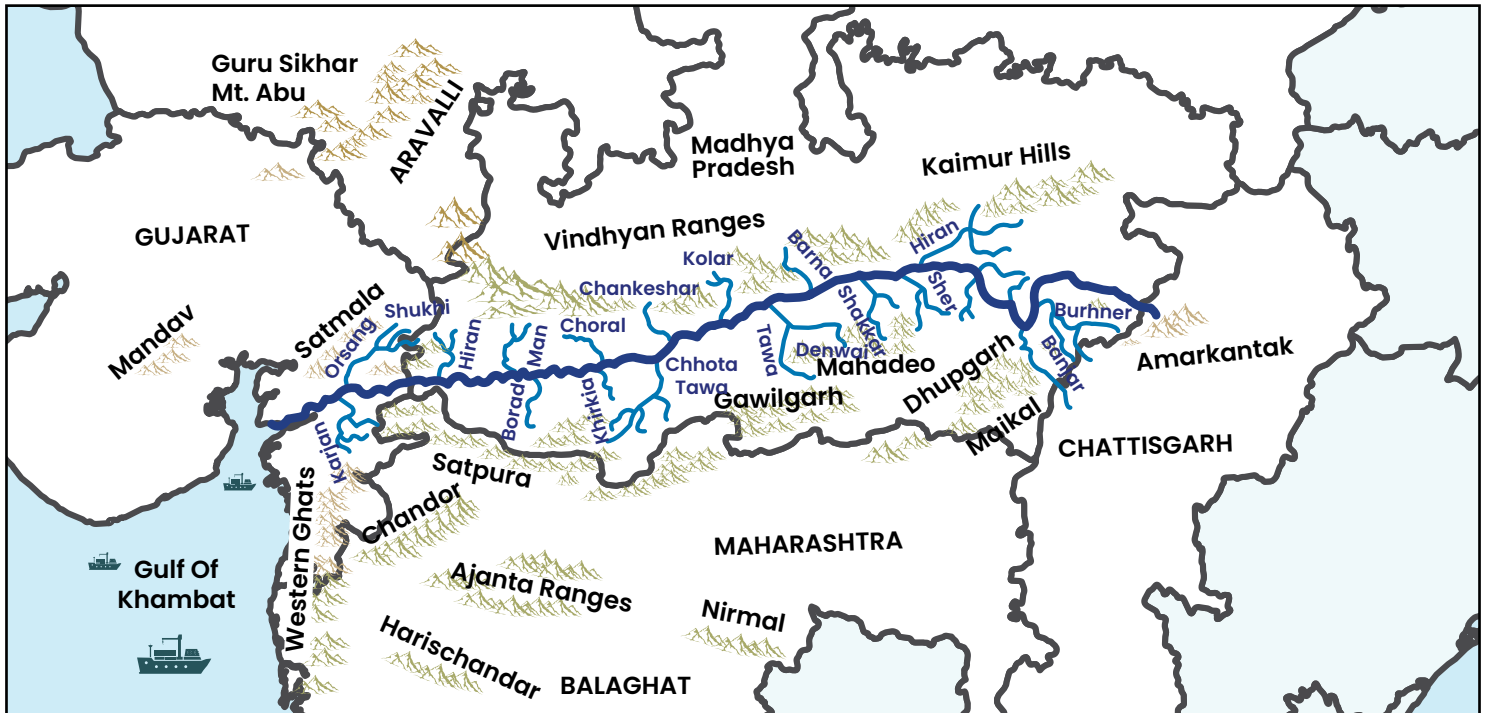
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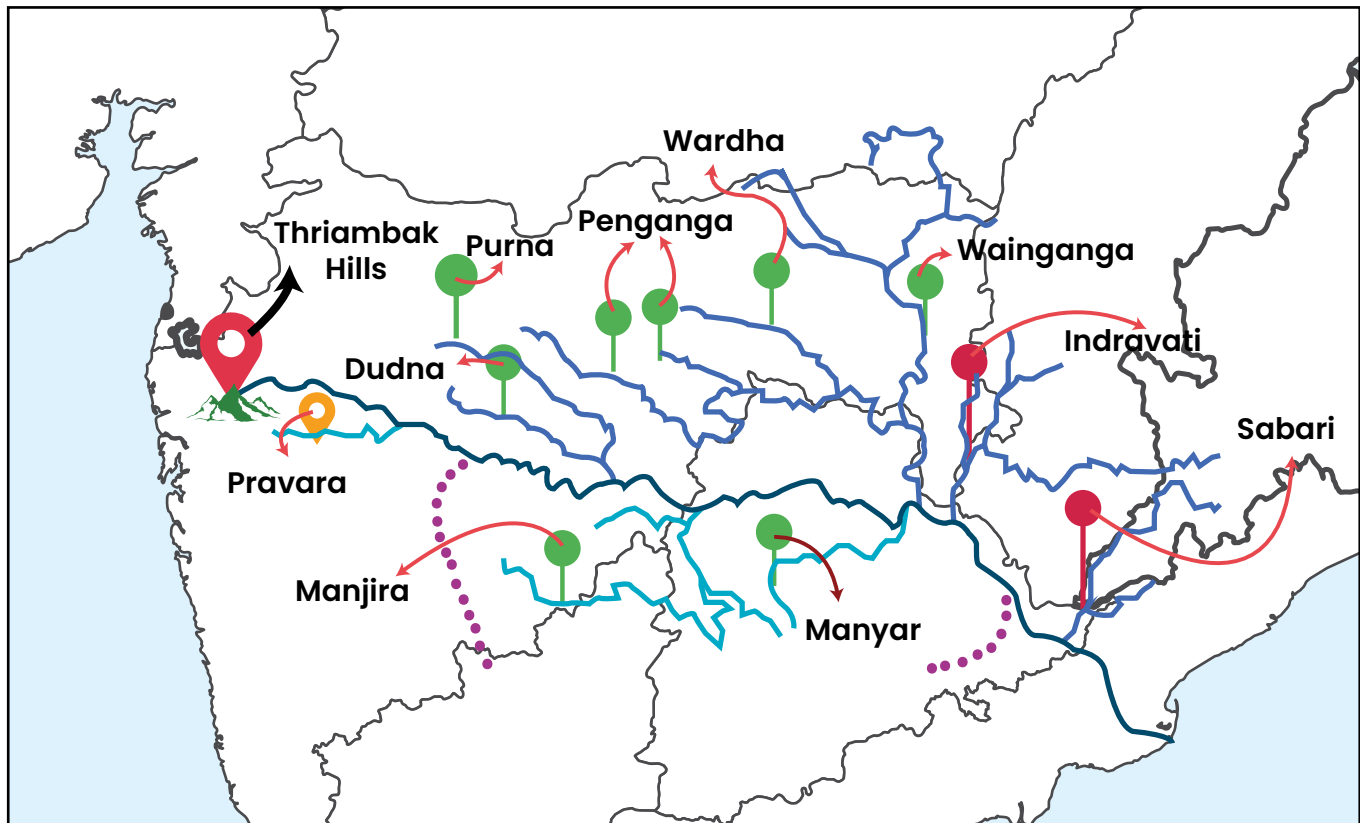
NARMADA RIVER & TRIBUTARIES



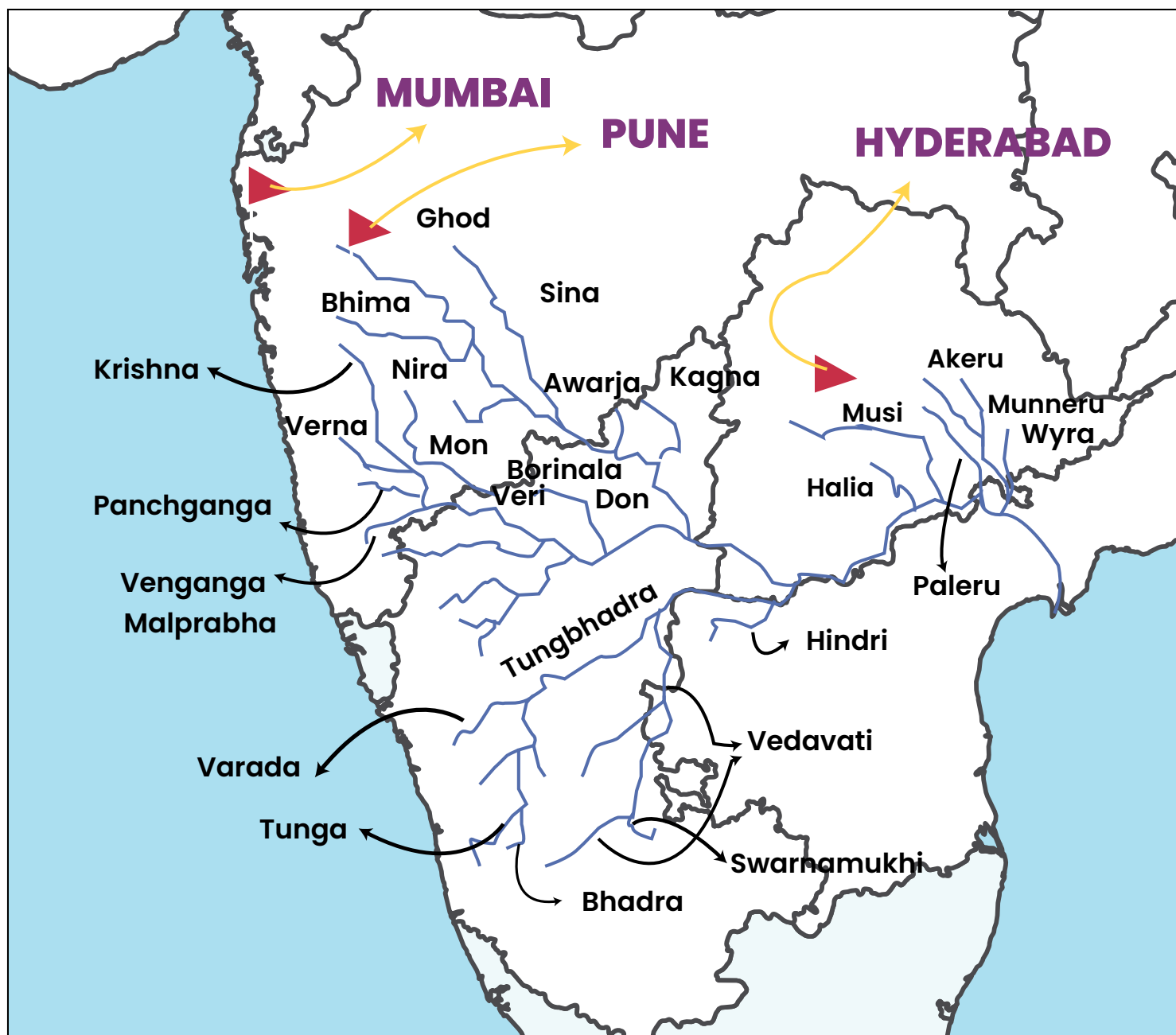
Rivers of Western Ghats



Godavari River Basin

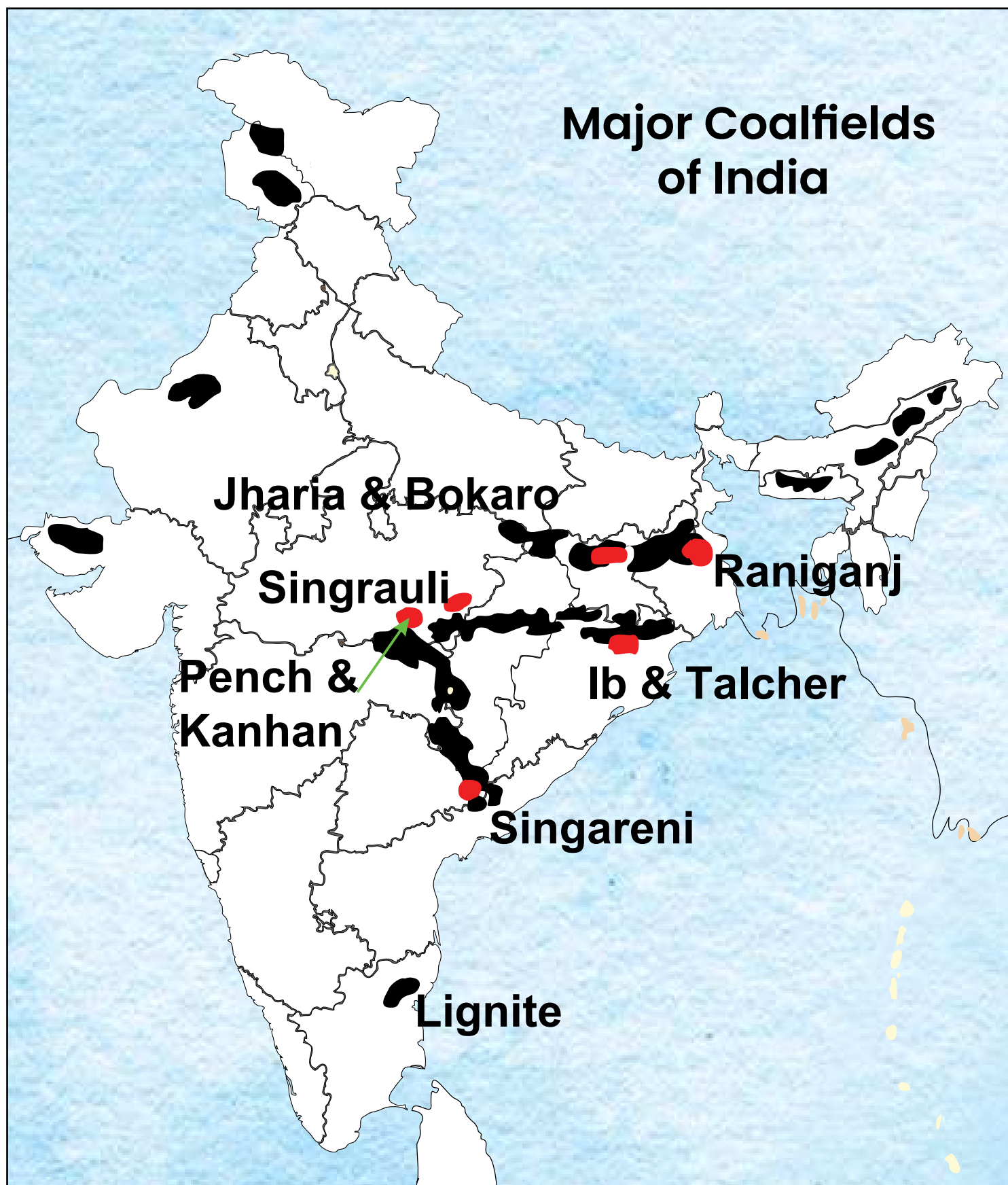


Krishna River Map



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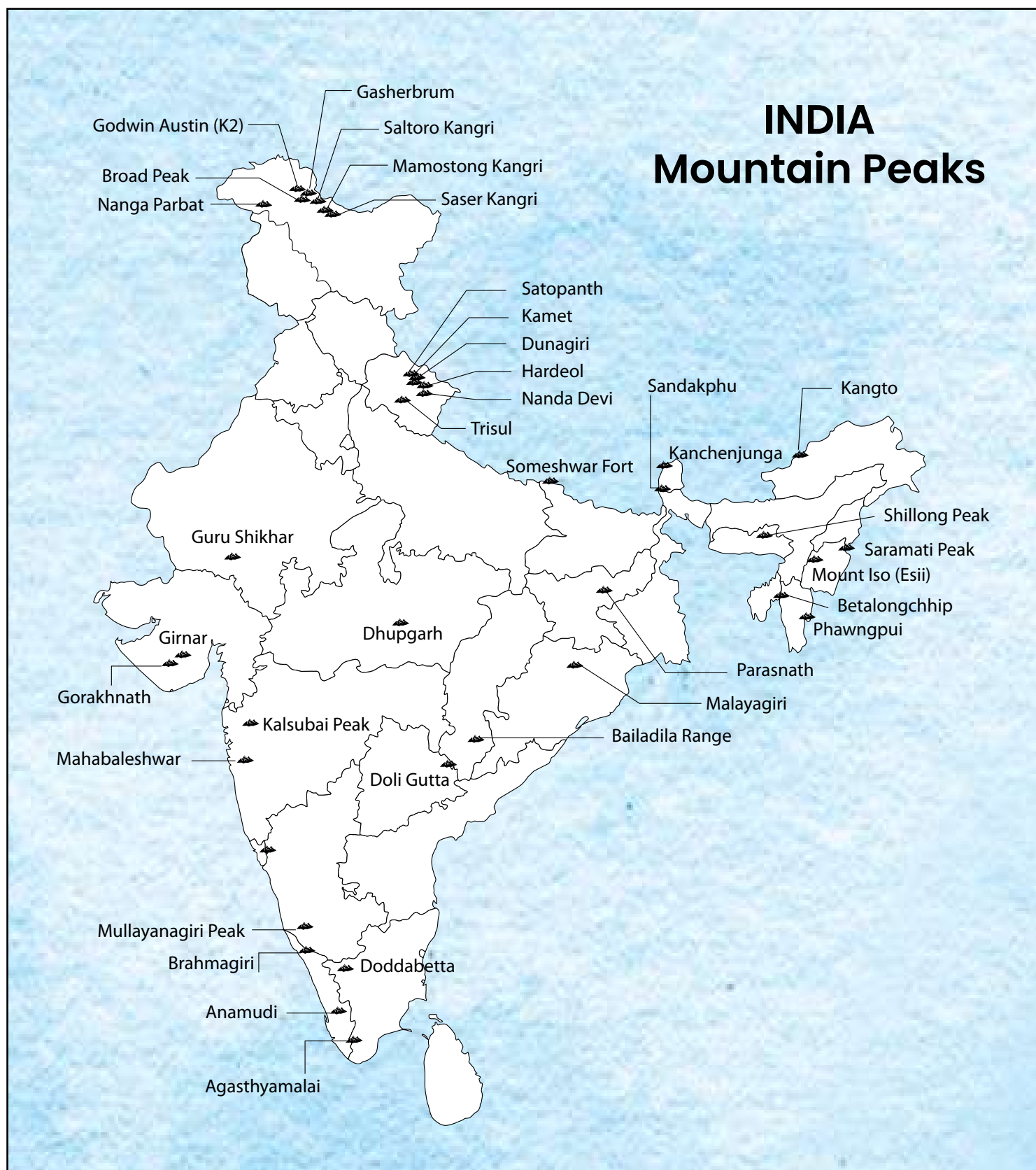
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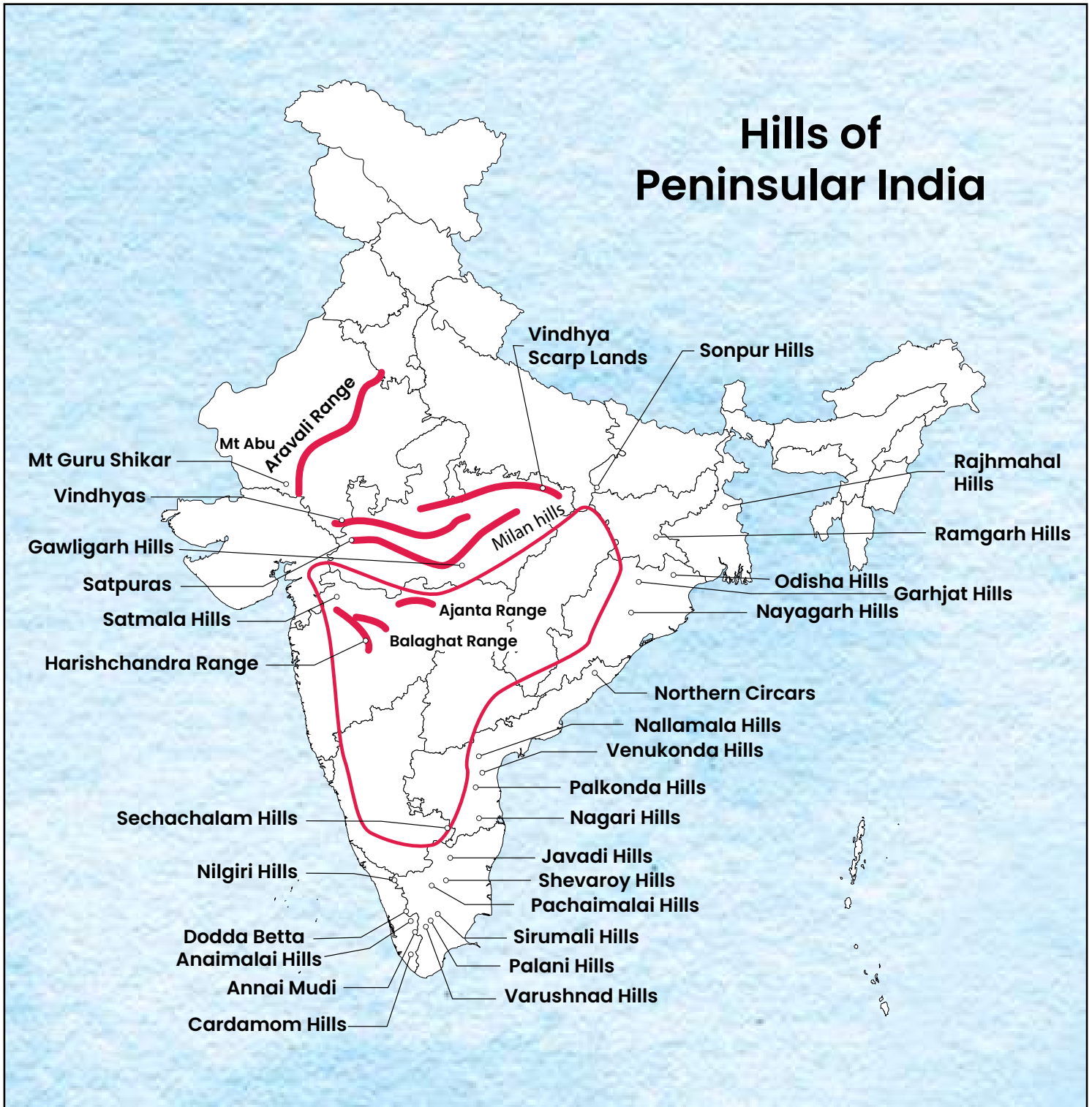
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GTB Nagar Metro (Mukherjee Nagar): 27 DEC, 6 PM | 6 JAN, 8 AM

हिन्दी माध्यम DELHI: 6 जनवरी, 11 AM | 13 दिसंबर, 8 AM

AHMEDABAD: 4 JAN BENGALURU: 5 DEC BHOPAL: 5 DEC HYDERABAD: 9 DEC JAIPUR: 20 JAN

JODHPUR: 3 DEC LUCKNOW: 5 DEC PUNE: 20 JAN ADMISSION OPEN CHANDIGARH

फाउंडेशन कोर्स सामान्य अध्ययन 2026

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JAIPUR: 20 जनवरी

JODHPUR: 3 दिसंबर

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1
AIR

Aditya Srivastava

16

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2
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**Animesh
Pradhan**



5
AIR

Ruhani



6
AIR

**Srishti
Dabas**



7
AIR

Anmol



9
AIR

Nausheen



10
AIR

**Aishwaryam
Prajapati**

39
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Kishore**



2
AIR

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